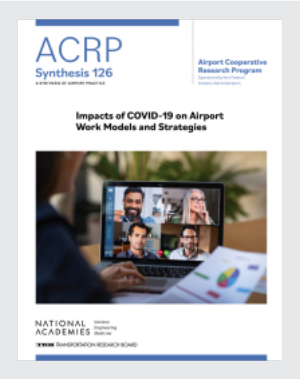
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AIRPORT COOPERATIVE RESEARCH PROGRAM

ACRP SYNTHESIS 126

Impacts of COVID-19 on Airport Work Models and Strategies

A Synthesis of Airport Practice

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Riverside, CA

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2023

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AIRPORT COOPERATIVE RESEARCH PROGRAM

Airports are vital national resources. They serve a key role in transportation of people and goods and in regional, national, and international commerce. They are where the nation's aviation system connects with other modes of transportation and where federal responsibility for managing and regulating air traffic operations intersects with the role of state and local governments that own and operate most airports. Research is necessary to solve common operating problems, to adapt appropriate new technologies from other industries, and to introduce innovations into the airport industry. The Airport Cooperative Research Program (ACRP) serves as one of the principal means by which the airport industry can develop innovative near-term solutions to meet demands placed on it.

The need for ACRP was identified in *TRB Special Report 272: Airport Research Needs: Cooperative Solutions* in 2003, based on a study sponsored by the Federal Aviation Administration (FAA). ACRP carries out applied research on problems that are shared by airport operating agencies and not being adequately addressed by existing federal research programs. ACRP is modeled after the successful National Cooperative Highway Research Program (NCHRP) and Transit Cooperative Research Program (TCRP). ACRP undertakes research and other technical activities in various airport subject areas, including design, construction, legal, maintenance, operations, safety, policy, planning, human resources, and administration. ACRP provides a forum where airport operators can cooperatively address common operational problems.

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Primary emphasis is placed on disseminating ACRP results to the intended users of the research: airport operating agencies, service providers, and academic institutions. ACRP produces a series of research reports for use by airport operators, local agencies, the FAA, and other interested parties; industry associations may arrange for workshops, training aids, field visits, webinars, and other activities to ensure that results are implemented by airport industry practitioners.

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Airport administrators, engineers, and researchers often face problems for which information already exists, either in documented form or as undocumented experience and practice. This information may be fragmented, scattered, and unevaluated. As a consequence, full knowledge of what has been learned about a problem may not be brought to bear on its solution. Costly research findings may go unused, valuable experience may be overlooked, and due consideration may not be given to recommended practices for solving or alleviating the problem.

There is information on nearly every subject of concern to the airport industry. Much of it derives from research or from the work of practitioners faced with problems in their day-to-day work. To provide a systematic means for assembling and evaluating such useful information and to make it available to the entire airport community, the Airport Cooperative Research Program authorized the Transportation Research Board to undertake a continuing project. This project, ACRP Project 11-03, "Synthesis of Information Related to Airport Practices," searches out and synthesizes useful knowledge from all available sources and prepares concise, documented reports on specific topics. Reports from this endeavor constitute an ACRP report series, Synthesis of Airport Practice.

This synthesis series reports on current knowledge and practice, in a compact format, without the detailed directions usually found in handbooks or design manuals. Each report in the series provides a compendium of the best knowledge available on those measures found to be the most successful in resolving specific problems.

FOREWORD

By Jordan Christensen
Staff Officer
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Due to the COVID-19 pandemic, airport operators had to develop strategies that maintained operations while ensuring employee safety and public health. Though not all airport-related tasks can be performed from remote worksites, many airports identified tasks that could be performed remotely. The objective of this synthesis is to provide information on those airports that experimented in remote work, options for airports that did not participate in remote work, and to identify emerging trends. Information used in this study was attained through a literature review and surveys of airport/human resources managers and airport employees. This report provides eight case examples from airports and provides sample work policies and agreements in the appendices.

Dr. Daniel Prather, DPrather Aviation Solutions, Riverside, CA, synthesized the information and wrote the report. The members of the topic panel are acknowledged on page iv. This synthesis is an immediately useful document that records the practices that were acceptable within the limitations of the knowledge available at the time of its preparation. As progress in research and practice continues, new knowledge will be added to that now at hand.

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SUMMARY

Impacts of COVID-19 on Airport Work Models and Strategies

The Coronavirus Disease 2019 (COVID-19) pandemic that was most pronounced during the March 2020–March 2022 time period drastically impacted the global economy through government-mandated stay-at-home orders, business shutdowns, and millions of COVID-19 infections and deaths. Until the first COVID-19 vaccine was available in December 2020, the most effective methods to reduce spread of the disease were to reduce travel, increase social distancing, and wear face coverings. Each of these measures specifically impacted the travel industry, with airports experiencing precipitous declines in passenger enplanements and aircraft operations, as well as the need to provide social distancing for employees and take other steps to minimize employee exposure to COVID-19.

Due to the COVID-19 pandemic, airport operators had to develop strategies that maintained operations while ensuring employee safety and public health. Even with the significant business slowdown (in areas such as aircraft operations, passenger screenings, concession activity, parking activity, etc.), managers of airports of all sizes had to ensure the safe and secure operation of the airport with "essential" or "operational" employees, while allowing "non-essential" or "non-operational" employees the ability to work remotely (from home, for example). Airports were somewhat unique in this regard. The airport needed to remain open and operational, and unlike some businesses, not all employees could perform their job functions remotely. This balance of requiring certain groups of employees to maintain an on-airport presence (with significant protections in place to prevent exposure to COVID-19), while other employees were either permitted or required by the airport to work remotely, is the focus of this synthesis.

This synthesis study was undertaken to address the evolution of airport work models and strategies. The objectives of this synthesis were to provide information on those airports who experimented in remote work, options for airports that did not participate in remote work, and to identify emerging trends in airport work models and strategies. To reach these objectives, a three-pronged approach was developed. First, a comprehensive literature review was conducted to review all relevant literature related to working arrangements during the COVID-19 pandemic. Due to the dearth of literature on this topic specifically related to airports, the literature review was expanded to include all segments of the United States (U.S.) economy. Second, an online survey was developed to collect data from airport managers and airport human resources (HR) managers on the topic of remote work and alternate work arrangements. Third, an online survey was developed to collect data from airport employees regarding their experiences with working remotely during the pandemic. By including both managers and employees, it was possible to compare managerial and employee views, which varied greatly at times, on airport work models and strategies.

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2 Impacts of COVID-19 on Airport Work Models and Strategies

The airport/HR manager survey gathered data from 151 individuals, 45.21% of whom represented Part 139-certificated airports. The employee survey gathered data from 76 individuals, 88.16% of whom were employed by Part 139-certificated airports. Although the country's largest airports are represented in the results, all sizes of airports are included. Indeed, all FAA National Plan of Integrated Airport System (NPIAS) airport categories and FAA regions (excluding Alaska) were represented in the manager survey. Likewise, all FAA NPIAS airport categories and FAA regions were represented in the employee survey.

In addition to the online surveys, phone calls to select airports with unique remote work practices or experiences resulted in the development of seven airport case examples and one employee case example for the report. These case examples highlight unique responses by these airports to the COVID-19 pandemic, as well as some insight into work models that have been (or will be) adopted moving beyond the pandemic.

Key conclusions of this synthesis are the following:

1. **Eligibility for remote work.** Airports commonly select employees eligible for remote work based upon the type of work performed by employees. Some employees are classified as essential and remain on-site and in-person due to the nature of their work, such as conducting airfield inspections, ensuring security, and responding to emergencies. Other employees are classified as non-essential since their tasks, such as processing accounts payable, managing contracts, and maintaining information technology resources, can be performed remotely (at home or at some other off-site location).
2. **Technological tools.** Rapid advances in cloud-computing and technological tools such as webcams enabled significant numbers of airport employees to work remotely during the pandemic.
3. **Business continuity.** Whether or not an airport had an existing business continuity or crisis plan, airports were confronted with a crisis caused by the COVID-19 pandemic. The pandemic created a sense of urgency at airports nationwide to consider how to continue operating in the midst of these challenges. Some airports have created a business continuity or crisis plan and have an emergency team structure in place to ensure they are equipped to handle future crises that may cause significant disruption of the business.
4. **Essential/Operational employees.** Airports benefit by determining a team of essential employees that need to remain at the airport during significant events, including a pandemic. Deciding this ahead of time minimizes the need to make these staffing decision "in the moment."
5. **Back to the office.** Now that the pandemic is officially over, numerous airports no longer allow employees to work remotely and are requiring all employees to be back in the office. Leadership at these airports generally see greater benefits in face-to-face interaction.
6. **Remaining remote.** Some airports, recognizing the benefits of remote work to both airports and employees, are continuing to allow "non-essential" employees to work remotely. These airports often have unique considerations for new employees, possibly requiring the new employees work on-site and in-person for the first 9 to 12 months before being eligible for remote work, for example.
7. **Long-term accommodation of remote employees.** As half of employees that have worked remotely at airports may like to continue working remotely, airports may consider how to accommodate these employees in a remote or hybrid fashion for the long term; otherwise, employee retention could be negatively impacted.
8. **Enhanced hiring practices.** Numerous airports have altered hiring practices for the foreseeable future to provide flexible options, such as conducting virtual interviews.

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9. **Generational similarities.** For the most part, there are no differences between people of different ages regarding their level of success with remote work.
10. **Sustainability of remote work.** Almost half of those airports experienced with remote work feel that remote work is sustainable for their airport years into the future. Almost half of employees with remote work experience feel that remote work is sustainable for the long-term. However, less than 20% of airports experienced with remote work actually have plans to support long-term remote work. Thus, the feasibility of long-term remote work does not correspond to the plans to maintain long-term remote work for employees.
11. **Employee training expectations.** Employees have high expectations for training delivered at a distance. Just over 40% of remote employees feel that training is more difficult while working remotely. Oftentimes, airports fell short in this regard, leaving employees to rate in-person training as more effective. Those airports receiving high marks from employees with distance training were intentional about delivering high-quality virtual training using motivated presenters, "live" webinars, and opportunities for employees to engage in the virtual environment.
12. **Benefits.** Generally, burnout was not experienced by airport employees working remotely. Employees with experience working remotely speak highly of numerous benefits, including less time commuting, more time with family, more time for hobbies, less stress, and in general, better work-life balance.
13. **Challenges.** Employees with experience working remotely also speak of challenges, including less face-to-face interaction with co-workers and supervisors, distractions at home, and difficulty in accessing "paper" files at work. More than eight out of ten airport employees with some remote work experience feel they socialize less with co-workers while working remotely. Just over one-third of airport employees with remote work experience feel that morale among their team has suffered while working remotely. Four out of ten feel their team's culture has suffered while working remotely.
14. **Innovation.** In the dynamic airport industry, innovation can be a source of competitive advantage. However, more than half of remote employees report that innovation is more difficult while working remotely. Although a remote employee may have fewer distractions and spend less time commuting, remote employees also report having less interaction with colleagues, making in-person collaboration all but impossible. Innovation may be more likely when in discussion and collaboration with colleagues.
15. **Career progression.** Generally, employees working remotely are not concerned about their future career progression, indicating that remote work has not hampered their careers.
16. **New employee integration.** Although remote work has numerous benefits, employees report that integrating new employees onto their team while working remotely is difficult. Indeed, three-quarters felt that it is difficult to add new members to their team while working remotely.

This report does not propose best practices or provide guidance on alternate work arrangements at airports but does offer a synthesis of practices from 227 total airport employees on this topic. Although practices vary and lessons learned differ, themes are identified that will be useful to airport and HR managers in debriefing on the immediate effects of the pandemic, as well as the future of alternate work arrangements to retain and attract a new generation of employees.

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CHAPTER 1

Introduction

The COVID-19 Pandemic

According to the U.S. National Intelligence Council (2021), the first-known clusters of COVID-19 cases appeared in Wuhan, China, in December 2019. Initially, Chinese government officials were unaware of the virus. By January 23, 2020, there were 17 deaths and more than 500 people sickened by the virus. As a result, Chinese officials made the strategic decision to close off Wuhan and its population of 11 million citizens, and also place a restricted access protocol on Huanggang, 30 miles to the east, where residents would not be able to leave without special permission. With these restrictions, 18 million people were soon under strict lockdown in China (*American Journal of Managed Care*, 2021).

Concern was expressed by leaders around the world as to the possibility of this contagious virus spreading worldwide. By early February, countries such as the United States, Australia, Germany, Italy, and New Zealand implemented restrictions on air travel in an effort to prevent spread of the virus. These efforts were generally unsuccessful in preventing worldwide spread of the virus. On January 21, 2020, the U.S. Centers for Disease Control (CDC) confirmed the first case of COVID-19 in the U.S. The infected individual had returned from Wuhan, China, on January 15. Several months later, on March 11, 2020, the World Health Organization (WHO) officially declared the COVID-19 disease a pandemic—an epidemic of an infectious disease that has spread across a large region, such as multiple continents or worldwide (National Intelligence Council, 2021).

Once COVID-19 was designated a pandemic, it required a global response. On March 13, 2020, President Trump declared COVID-19 a national emergency. Even though the response to the pandemic was on a global scale, by April 4, 2020, less than 4 months after the initial cluster of COVID-19 cases was discovered in China, the WHO reported more than 1 million cases of COVID-19 confirmed worldwide (*American Journal of Managed Care*, 2021; WHO, 2020).

In addition to federal-level efforts, further efforts to contain the virus and minimize its spread were implemented at the state and local levels. State governors and health departments across the country issued various types of orders and recommendations regarding the status of schools, businesses, and public services. For example, on March 19, 2020, California became the first state to issue a stay-at-home order, mandating all residents to stay at home except for those that must report to an essential job or shop for essential needs. In many states, non-essential businesses were required to cease in-person operations for extended periods. Even essential businesses were impacted, requiring strict cleaning, facial covering, and social distancing protocols.

Many employers were forced to maintain their business even with employees unable to report to work on-site. Thus, a great experiment began across the country (and world) with employees required to work from home, being quickly outfitted with resources necessary for

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remote work. With technology manufacturers and supply firms also being impacted by employees unable to work due to a positive COVID-19 diagnosis, obtaining needed equipment to support a remote workforce was not easy. Webcams, for example, were soon in short supply.

Worker unavailability (due to positive COVID-19 diagnoses), inability to obtain needed remote work tools (or inability to operate remotely, for businesses like coffee shops), and loss of in-person sales, was the fatal blow to many businesses. Unfortunately, many businesses permanently closed during the pandemic, as they were unable to financially sustain these extended periods of closure on reduced or non-existent sales/revenues (American Journal of Managed Care, 2021). An additional 200,000 businesses closed in the first year of COVID-19 (Simon, 2021).

Airports and the Pandemic

Airports, however, did not have the option to close their doors during this time. Airports continued to serve users even while many of the metrics related to the business dropped precipitously. For instance, the TSA screened a pandemic low of 87,534 passengers on April 12, 2020, compared to 2,208,688 on that day one year earlier. Annual airline passenger traffic was down 60% system-wide in 2020 compared to annual traffic levels in 2019 (Government Accountability Office, 2021). Further, passenger traffic in April 2020 was 96% lower than in April 2019 (pre-pandemic). These greatly reduced passenger screening numbers point to less airline activity, which then yields much less aeronautical and non-aeronautical revenue for airports. For example, landing fee revenue, parking revenue, and food and beverage and retail concession revenues all dropped significantly.

As explained by Airports Council International (ACI):

> Air traffic is the lifeblood of the airport business. Practically all aeronautical revenues are a direct function of traffic and include passenger-related charges and aircraft-related charges levied from aircraft operators. As traffic declines, revenues from charges decrease proportionally. Non-aeronautical revenues, which include such streams as retail concessions, duty free, car parking and food and beverage, are also very much linked to passenger traffic and throughput. As airports have little flexibility in operating expenditures but also have capital costs that are largely fixed, the current crisis represents an unprecedented challenge for the airport industry's financial viability. Compared to the projected baseline, the global airport annual revenue shortfall is projected to be US $83.1 billion in 2021 and an additional US $60.8 billion in 2022. (ACI, 2022, paragraph 17)

With the arrival of COVID-19 vaccines in December 2020, the closures, capacity limits, and large-scale mask orders that were characteristic of the previous 10 months shifted to vaccine and testing mandates, as well as specific face-covering rules. Even as new COVID-19 variants, including Delta and Omicron, appeared some states actively resisted restrictions, prohibited or limited mask and vaccine mandates, and allowed businesses to reopen. The response to the later phases of the COVID-19 pandemic varied from state-to-state.

As the passenger carriers suffered greatly during the pandemic, the air cargo carriers maintained operations and even increased activity—due in large part to vaccine delivery, as well as various types of personal protective equipment, such as masks and gloves. E-commerce was also a driver in air cargo activity. In 2020, freight forwarders generated a 4% profit, while air cargo carriers generated a 9% profit. In fact, the only five airlines that reaped profits in 2020—AirBridgeCargo, Atlas Air, Cargojet, Cargolux, and Kalitta—were cargo carriers (McKinsey & Company, 2022).

Passenger carriers generally welcomed an April 2022 decision by a federal judge that struck down the CDC mandate of face coverings in public transportation, including airlines. Several U.S. airlines, including American, Delta, Southwest, and United, quickly dropped their mask mandates, making them optional for travelers. As was typical with COVID-related policies, some travelers were ecstatic, while others were very disappointed.

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6 Impacts of COVID-19 on Airport Work Models and Strategies

By late spring 2022, pandemic-era restrictions were all but gone. After two years of unprecedented restrictions and serious concern for health and safety, Americans could finally return to some sense of normalcy—just in time for summer 2022.

Throughout the March 2020–March 2022 COVID-19 pandemic, U.S. airports had to confront a challenge never before experienced on such a grand scale: Maintain the operational integrity of the airport, even with minimal aircraft operations and passenger activity, with employees who could not work in close contact with one another without experiencing a significantly higher risk of potentially deadly COVID-19 exposure. How would the airport continue business as usual under such conditions? Short answer: It wouldn't.

One thing is clear: airports across the country (and world for that matter) had to confront a new reality. Deciding which employees would be required to perform their tasks in-person at the airport (termed "essential" workers) and which employees could perform their tasks remotely (termed "non-essential" workers).

Airports considered the work of "essential" employees to be "operational" in nature, meaning their job functions must be performed on-site at the airport, such as airfield inspections, building and airfield maintenance, and security. "Non-essential" employees had work that was considered "non-operational" in nature, meaning their job functions could be performed off-site, such as finance, marketing, and human resources (HR). It should be noted that "non-essential" does not mean these employees are not needed. It simply meant that it was not essential that these employees performed their necessary job functions at the airport.

For those essential workers that would remain at the airport, how could airport management ensure these employees did not face an increased risk of COVID-19 exposure? For those non-essential employees working remotely (often from home), how could airport management ensure these employees remained productive with adequate oversight, also contributing to support essential airport employees at the airport? What technological tools would be needed to support these two groups of employees? These were not easy questions to answer.

Synthesis Overview

This synthesis attempts to answer the difficult questions that faced airports as they managed remote work during the COVID-19 pandemic. Through a thorough literature review, two online surveys, and phone calls to several airports, this synthesis provides a resource that describes the tasks and missions that airports have been able to accomplish remotely, in response to the COVID-19 pandemic. The report describes the types of work and training options developed by airports to maintain continuity of operations during the COVID-19 pandemic, as well as societal trends toward hybrid and remote work models, specifically efforts by airports to accommodate employees during the pandemic. Challenges experienced by airports during the pandemic are explored, with an emphasis on innovative solutions adopted by airports to continue operations during the pandemic. The report is also forward-looking with an emphasis on remote and hybrid work and training arrangements designed to enhance efficiency and accommodate employee and airport needs moving forward.

The remaining chapters of the report will provide the reader with in-depth knowledge of the issues surrounding remote work, including those challenges and benefits specific to airports.

Chapter 2: Synthesis Methodology presents the methodology used in the project. Topics include recipients/participants, data collection instruments, survey and interview methodology, and data analysis techniques.

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Chapter 3: Literature Review, presents findings obtained from a thorough review of the literature, aiding the reader in more deeply understanding this issue, including relevant research. Topics include impacts of the COVID-19 pandemic, organizational responses to the COVID-19 pandemic, and considerations in supporting a remote workforce. An additional literature review appears in Appendix C.

Chapter 4: Survey Results, presents the findings from a survey of airport/HR managers and airport employees. Topics addressed include remote work models, employee productivity, benefits and challenges of remote work, maintaining a work-life balance, and cloud-based technology.

Chapter 5: Case Examples, presents seven in-depth case examples of airports, as well as one airport employee case example detailing this individual's experience with working remotely during the pandemic.

Chapter 6: Conclusions and Future Research, provides a summary of key findings of the synthesis, as well as comments on the state of research and opportunities for additional research related to this topic.

The following appendices can be found at the end of this synthesis:

- Appendix A: Airport Manager/HR Manager Survey
- Appendix B: Airport Employee Survey
- Appendix C: Additional Literature Review
- Appendix D: Remote Work Policy and Agreement Samples
- Appendix E: Verbatim Comments, Airport/HR Manager Survey
- Appendix F: Verbatim Comments, Airport Employee Survey

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CHAPTER 2

Synthesis Methodology

Literature Review

Prior to collecting data from airports on the topic of remote work arrangements, a comprehensive review of the literature was conducted. Due to the lack of literature specifically related to airports on this topic, literature highlighting studies in other professions was also reviewed as appropriate. Additionally, the general management/HR literature related to remote work in corporate America was included.

Sources were discovered through numerous platforms, including Google, Google Scholar, OneSearch (powered by EBSCO), TRID, TRB's database of worldwide transportation research, and the websites of the American Association of Airport Executives (AAAE) including the Association's Accreditation Papers library, and FAA.

Recipients and Participants

The population defined for this synthesis reflects the general idea that the COVID-19 pandemic impacted airports of all sizes, in all FAA regions (i.e., none were spared from the impacts of the pandemic). The project panel aimed to include airports from all National Plan of Integrated Airport System (NPIAS) airport categories and all nine FAA regions. Thus, every effort was made to cast a wide net so that each U.S. airport had the opportunity to participate, as long as the airport was a member of the AAAE or ACI-NA.

With such a large population, a streamlined approach to survey distribution was necessary. Based on topic panel feedback, both the employee and manager surveys were distributed to members of the AAAE. Additionally, the airport/HR manager survey was distributed to the ACI-NA Human Resources Committee.

To distribute the surveys, first, two separate informational posts to the AAAE Member Hub Community with a link to the online survey questionnaires were posted.

Second, the AAAE Member Directory was used to send emails to randomly selected members.

Third, and only for the airport/HR manager survey, the survey link was sent via email to the chair of the ACI-NA Human Resources Committee requesting survey distribution to members of the committee.

After several email reminders for both surveys, a total of 151 individuals responded to the airport/HR manager survey. A total of 76 individuals responded to the airport employee survey.

A response rate could not be determined for either survey because the total population of airports receiving the survey could not be accurately determined. The study included airports from each of the nine FAA regions (see Figures 1 and 2). Additionally, all NPIAS airport sizes—

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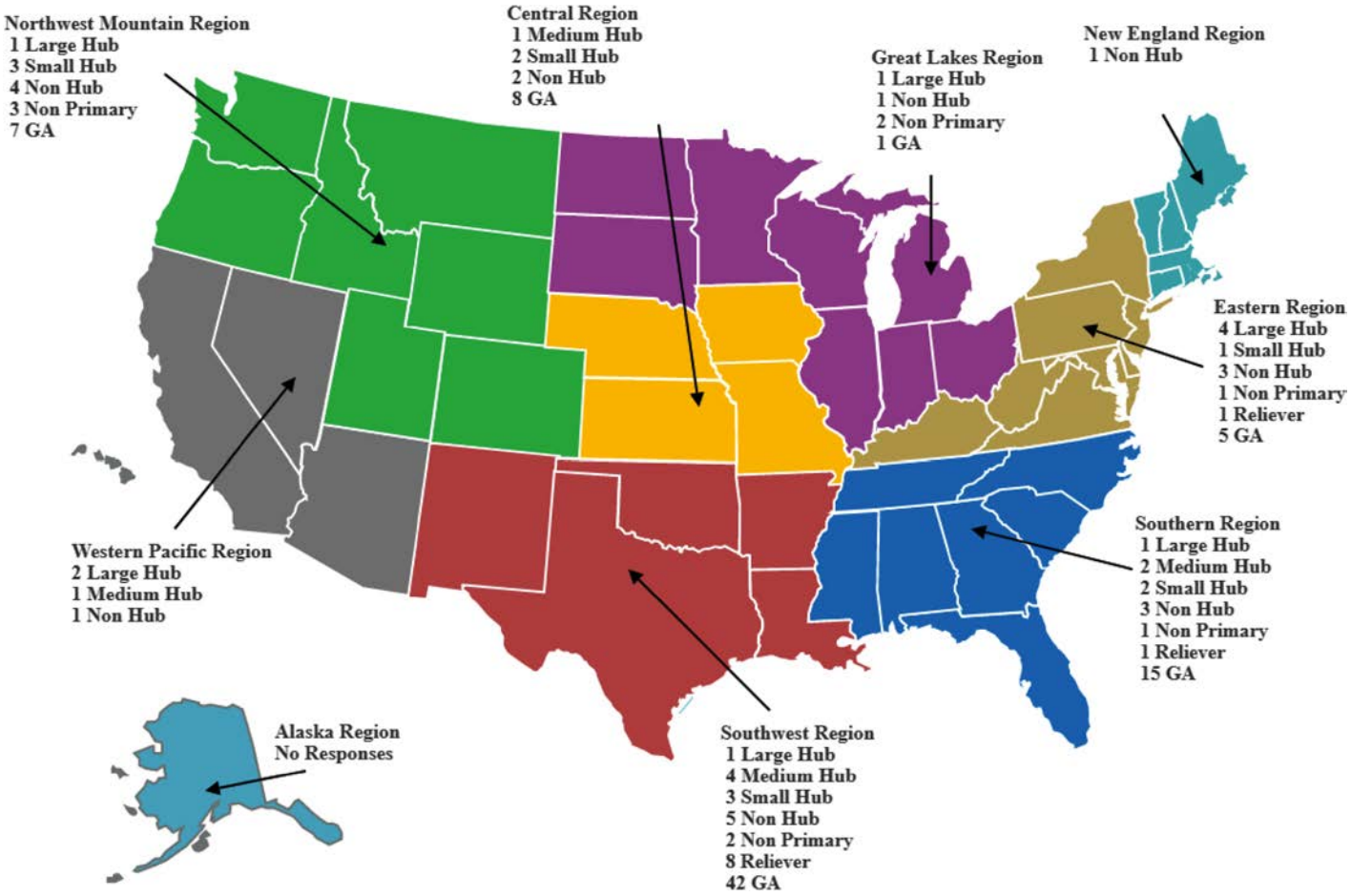


Figure 1. Participating airports for Airport/HR Manager survey by NPIAS category and FAA region.

including large hub, medium hub, small hub, non-hub, non-primary, general aviation (GA), and reliever—were represented. In fact, the only exclusion is the Alaskan region with the airport/HR manager survey. Even though a response rate could not be calculated, this number of responses, representing such diversity in geography and airport size, was deemed sufficient to synthesize nationwide information on remote work practices.

Survey and Interview Methodology

Two unique questionnaires were designed for this synthesis. The first questionnaire, "Airport Work Models—Airport Manager/HR Manager Survey—ACRP S06-07" consisted of 23 open-ended and close-ended items. (The complete survey can be found in Appendix A.) The second questionnaire, "Airport Work Models—Airport Employee Survey—ACRP S06-07" consisted of 28 open-ended and close-ended items. (The complete survey can be found in Appendix B.)

It was anticipated that an online survey protocol would result in a greater response rate than mailed surveys, garner faster response, and be more economical than telephone interviews. Both questionnaires were created using the web-based SurveyMonkey platform.

Both surveys were developed using the investigator's knowledge of remote work practices, effects of the COVID-19 pandemic, airport organizational structures, knowledge gained during the literature review, and feedback from the project panel. Following a project panel pretest, the survey instrument was further refined.

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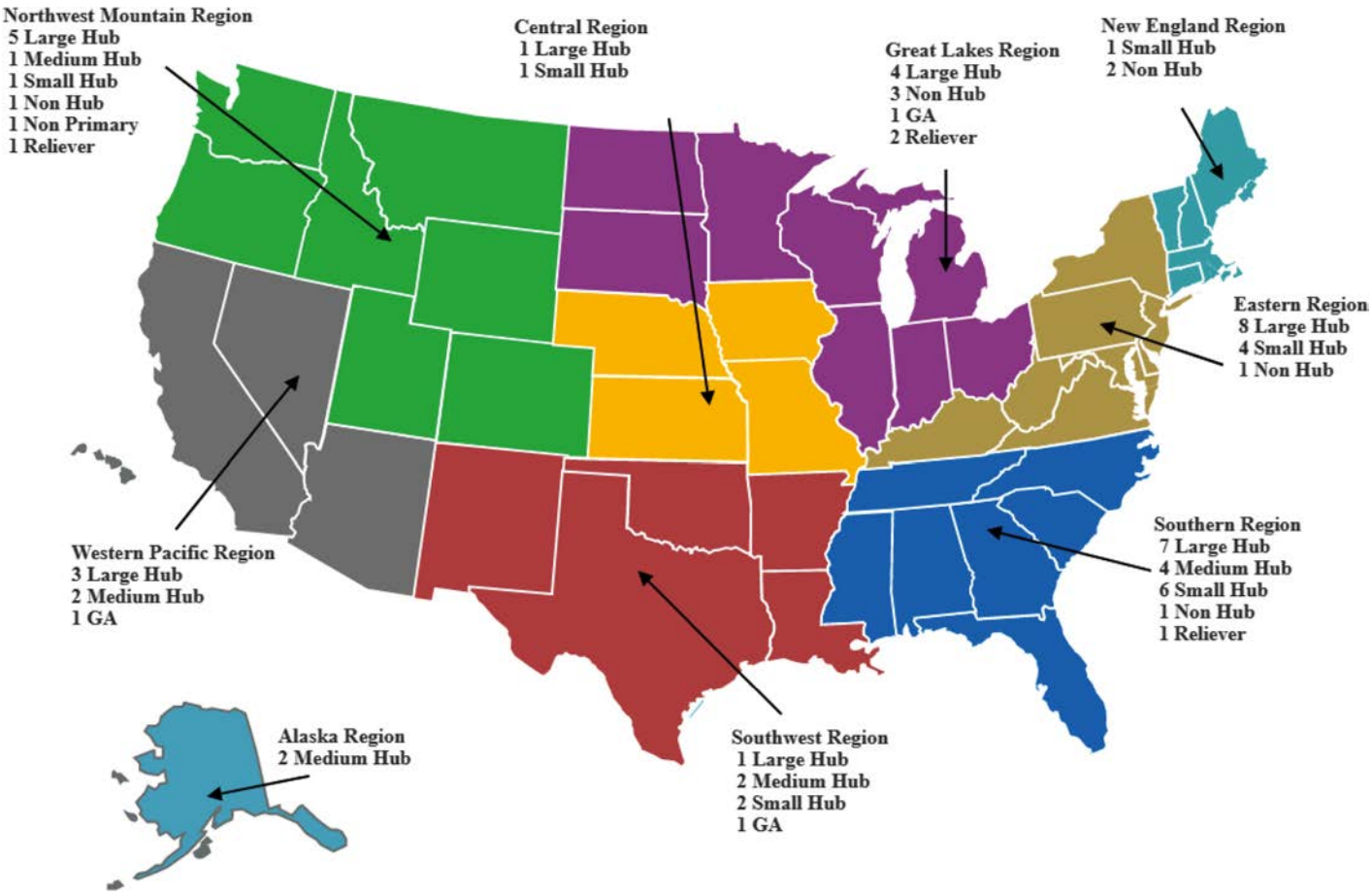


Figure 2. Participating airports for Airport Employee survey by NPIAS category and FAA region.

Both surveys were distributed via various channels from May to July 2022. Because of the extensive distribution over a period of multiple weeks with multiple follow-ups to enhance response, the survey remained open until July 31, 2022. Once the surveys closed, no further responses were possible.

On the basis of survey responses (including the willingness to be contacted for follow-up), seven airport managers and one airport employee were contacted via phone during August 2022. The intent of the phone calls was to better understand remote work practices, including challenges experienced, at the selected airports for the purpose of developing case examples for the report. A case example was developed for each of those airport managers and one airport employee (presented in Chapter 5).

The methods for analyzing survey data depended on the type of data collected:

- Quantitative data, mostly collected from close-ended items, was analyzed using basic Microsoft Excel statistical analysis tools, as well as the SurveyMonkey data analysis tool. Descriptive statistics were calculated. Charts were created using Excel.
- Qualitative data, collected from responses to open-ended items, were manually tabulated and analyzed using content analysis. Themes were uncovered from the verbatim responses and presented to better understand common perspectives among participating airports. If appropriate, word clouds were created to graphically portray those words most frequently used in response to open-ended questions.

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CHAPTER 3

Literature Review

Financial Impacts of the COVID-19 Pandemic on Airports

In an October 2021 report, the Government Accountability Office (GAO) recognized the severe hardship COVID-19 placed on the U.S. air travel industry, including airports. As the GAO points out, according to data from the Bureau of Transportation Statistics (BTS), "annual airline passenger traffic was down 60% system-wide in 2020 compared to annual traffic levels in 2019" (p. 12). Further, passenger traffic in April 2020 was 96% lower than the pre-pandemic levels of April 2019. The pandemic impacted not only airline companies, but airports as well, which experienced these significant reductions in passenger enplanements in the form of fewer flights (and significantly reduced landing fee revenue); reduced food, beverage, and retail sales (and significantly reduced concession revenue); etc. The GAO states, "We found that airports of all sizes experienced a decline in both aeronautical and non-aeronautical operating revenues as a result of the COVID-19 pandemic" (pp. 15–16). The GAO also discovered greater percentage declines in revenues for small-hub and non-hub airports, as contrasted to non-primary commercial service, GA, and reliever airports. This was due, as the GAO pointed out, to small-hub and non-hub airports' greater reliance on schedule passenger service.

Even with drastically reduced enplanements and aircraft operations, "Airfields had to be inspected; restrooms had to be cleaned [and] terminals had to be patrolled. While the pandemic put a drastic halt to the number of travelers visiting our airports, the responsibilities of airport workers did not cease" (Huckaby and Darras, 2021, p. 7). In other words, operational expenses continued to be incurred, while operational revenues were greatly reduced. Airports were in need of a lifeline.

On March 27, 2020, Congress and President Donald Trump provided this lifeline by passing the Coronavirus Aid, Relief, and Economic Security Act (CARES Act). This Act was a $2.2 trillion economic stimulus bill to provide much-needed economic relief to American businesses, including airports.

Airports of all sizes utilized federal COVID-19 relief grants to make needed debt and other payments, as well as to assist tenants, such as airlines, which were also facing extreme financial difficulty due to COVID-19 impacts. As the GAO (2021) explains:

> These federal grants of up to $20 billion in total allowed airports to respond to the COVID-19 pandemic, including funding their operations and meeting their ongoing debt payments, although the funding allocation and allowable uses for the grants differ under the CARES Act and subsequent COVID-19 relief laws. (p. 27)

The GAO (2021) found that a number of airports used CARES Act grant funding for routine, large expenses, including debt service payments and payroll. Some of the airports also used CARES Act grant funding to provide financial relief to concessionaires, including both food and

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beverage and retail. Some of the large- and medium-hub airports also used funding to avoid increasing airline rates and charges. The GAO found that smaller airports mostly used CARES Act grant funding for payroll and labor expenses, allowing these airports to avoid layoffs or furloughs of staff.

The GAO (2021) noted that:

> In response to the many effects of the COVID-19 pandemic, aviation stakeholders reported that throughout 2020, they quickly implemented measures to mitigate financial losses and position themselves to maintain business viability until demand recovers. These actions included managing costs – which often included reducing costs; using federal assistance provided through COVID-19 relief legislation; raising funds in the private market; and taking actions to mitigate the spread of COVID-19 among employees and customers. (p. 21)

Airports' Response to COVID-19

Airports of all sizes took actions in 2020 to manage non-labor operating costs and capital investments. Examples of these actions include closing airport parking lots, reducing airport shuttle operations, and altering the timeline of capital projects (or alternatively accelerating capital projects that could be more easily completed with reduced passenger traffic and aircraft operations). As Kim Becker, C. M., President and CEO of the San Diego Regional Airport Authority writes, "Core business strategies must be revisited with a laser focus on the passenger experience, all while maintaining our financial sustainability" (2021, p. 13). Of her airport's experience, she explains, "We are fortunate that we did not have layoffs but, like most airports, we implemented salary and hiring freezes, and deferred many capital projects in an effort to control costs" (p.13). In fact, the GAO (2021) estimates that one-third of smaller airports modified their infrastructure development timelines. Specifically, the GAO estimates that 15% of smaller airports canceled projects, 75% delayed projects, and 10% accelerated projects.

Airports also examined their retirement programs and hiring practices in the midst of COVID. The GAO (2021) shared that four of the airports it surveyed offered voluntary early retirement programs. One of the airports the GAO spoke to explained their reduction-in-force initiative that limited workforce size to 26 union and non-union employees, which reduced the airport's payroll by 5%. The GAO estimated that 5% of smaller airports implemented staff layoffs. The GAO also discovered that nearly all of the airports surveyed either suspended or significantly slowed down hiring. For example, new hiring was only permitted at some airports for critical positions, such as public safety. The GAO estimates that about 20% of smaller airports decreased staff hours to reduce operating costs.

In addition to cost control measures, airports had to implement a number of measures to protect the health of employees and customers. These include actions such as "requiring and providing masks, providing other personal protective equipment, implementing social distancing, and allowing employees to work from home when possible" (GAO, 2021, p. 33). Airports had to remain vigilant with COVID-19 protocols to ensure a safe work environment for on-site employees, as well as passengers and vendors. With large terminals and multiple facilities, commercial-service airports endured particularly significant costs associated with sanitization, installing signs and plexiglass barriers, etc. Even so, these efforts were necessary to provide a sense of safety for all involved.

Mental Health of Remote Workers

During the pandemic, the mental health of some employees suffered. "Living through a pandemic reduced people's sense of control, research has found, leading to greater feelings of helplessness and depression. Engaging in a leisure activity just for the pleasure of it can be

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a powerful antidote, as research finds that people who engage in such activities report greater satisfaction with their lives overall" (Zubernis, 2022, p. 56).

It is important for employers to regularly assess the mental health of remote workers. Employers may benefit by asking all employees, especially those working remotely, the following questions:

- Is your sleeping pattern disrupted by worries?
- Do you feel less effective at work?
- Are you fearful of starting your job?
- Do you feel overwhelmed by life?
- Do you see your friends and family less often?
- Do you experience more headaches and muscle cramps and are you sicklier?
- Do you check your watch more often?
- Are you resorting increasingly to bad habits such as drinking alcohol?
- Have you lost your sense of humor, and does negativity rule your life?
- Do your work and your life feel empty and meaningless?
- Do you feel overwhelmed by your work?
- Do you feel fragmented with all your different tasks?
- Are you busy isolating yourself from your colleagues and your friends?
- Do you feel that your situation is hopeless and without a solution?
- Are you angry, irritated, or disappointed in the people around you? (Esterhuizen, 2021, p. 42).

Answers to these questions will assist employers with gauging the mental well-being of employees. Although employees may be hesitant to answer some of these questions, it is incumbent upon management to know each employee well enough that red flags may be spotted early.

For those employees showing signs of burnout, employers can:

- Create healthy boundaries.
- Create a well-appointed work area that allows the employee to close the door at the end of the workday to "clock-out" (Esterhuizen, 2021, p. 42).

Additionally, employers can recommend that employees showing signs of burnout take the following measures:

- Live mindfully by selecting one pleasant activity you do each day, such as enjoying a cup of coffee each morning before you have a seat at your desk.
- Get into nature regularly.
- Get adequate sleep.
- Reward yourself for a job well done.
- Try breathing techniques for stress release.
- Make time to develop and maintain healthy relationships with friends and family.
- Talk to a supervisor or HR if you feel overwhelmed (Esterhuizen, 2021, p. 42).

Employers may also find it useful to assist employees with managing various responsibilities. Gaines (2021) recommends the following time management tips:

- Prioritize self-care time. Make time for 7–9 hours of sleep each night, eat three balanced meals, and factor in adequate time for family and leisure.
- If you are a leader, infuse plan into your employees' workday. Host themed virtual meetings, mail employees small care packages, host online birthday or holiday celebrations.
- If you are an employee, establish what the "workday" means to you and your boss. When will you be off duty?

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- Don't stay on your email all day. Constantly checking your inbox can be distracting and reduces your productivity.
- Step away from the Internet. This can be a huge time waster.
- Minimize distractions and the tendency to multitask.
- Set achievable goals each day.
- Listen up! Active listening requires being present and engaged.
- Stop shuffling papers. Plan to touch each piece of paper only once. Act on it, file it, or toss it.
- Recapture your throwaway moments. Achieve small goals and accomplish small tasks in brief five-minute intervals, rather than wasting this time.
- Do a priority tune-up. Ask yourself (a) What should I keep doing? (b) What should I eliminate or stop doing? (c) What's missing that I might want to add?
- Practice mindfulness and breathe.
- Learn to say no more often.
- Stop owning other people's stuff. Hold others accountable for their responsibilities, rather than doing it yourself.

Work-Life Policies

During the early stages of the pandemic, as stay-at-home orders and various restrictions were enacted, many employers sent workers to remote locations (or instructed them to work from home) without a clear remote work policy. This set the stage for "discrimination claims, confusion regarding expenses, misunderstandings regarding work expectations, lack of clarity regarding the applicability of other policies, and increased liability for workplace injuries" (Mutz, 2021).

Although there are certainly airport employees defined as civil servants with due process rights, in the general economy, most employees are at-will, meaning "they have no contractual rights to continued employment, let alone to particular working conditions" (Arnow-Richman, 2020, p. 1). In essence, if an employer is not willing to accommodate a remote work arrangement, the employee's recourse is to quit. "Absent a statutory disability, an individual employee who refuses to come to work is unlikely to be protected, at least where the employer is following basic sanitation and social distancing protocols that reduce the risk of [COVID-19] transmission" (Arnow-Richman, 2020, p. 3).

Developing and maintaining a remote work policy can establish clear expectations between employees and management and provide support for employees in alternative work arrangements. See Appendix D for sample remote work policies and forms.

Vetter (2021) recommends three tips for building remote-work policies that last:

- Start with your people. Do your team members need to work in close personal collaboration? Do they have long commutes? Do they prefer working from home all the time? Would they like the option of going into the office? In many cases, you'll find your teams don't adhere to a one-size-fits-all approach.
- Don't compromise on real estate. Explore workshare arrangements, nontraditional spaces, and other innovative options. Design the space that works for your business rather than tailoring your operations to suit a physical space.
- Solidify your tech. Select cloud-based platforms that support your teams and enable business success. Additionally, finalize tech policies that address, for example, means of communication.

Once developed and formally approved, employers commonly distribute the remote work policy to all employees (both remote and in-person). It is generally included in the employer

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handbook, which employees sign to acknowledge receipt and understanding of the remote work policy.

To address accountability and productivity concerns, Mutz (2021) recommends limiting remote work to only "long-standing, trusted employees (i.e., full-time employees with at least one year of employment, demonstrated good work habits, history of timely deliverables, and good evaluations)" (paragraph 3). Mutz goes on to write that:

> One of the biggest fears regarding remote work is accountability and productivity since employees work with all of the temptations of being at home. It is not outside the realm of possibility that an employee could choose to nap during work hours, take breaks to perform household chores, or run errands. Employers must compensate employees for all hours spent advancing the employer's interest under the Fair Labor Standards Act (FLSA), including work that may be unauthorized or performed off-the-clock. Accordingly, any remote work policy should include timekeeping practices (clock-in/clock-out), accountability expectations (work hours), and prohibit unauthorized or off-the-clock work. Employers generally may discipline employees for performing unauthorized or off-the-clock work while also paying employees for work time to ensure compliance with FLSA. (paragraph 7)

The work-life policy is generally developed with several employee productivity assurances. However, according to Vaillancourt (2021), employee productivity concerns may be overinflated.

Although micromanagers generally don't consider themselves micromanagers, for those willing to self-reflect, "two questions are useful: (a) If employees are not trustworthy, why are they still employees? (b) If managers have employees who cannot be trusted, why are they still managers?" (Vaillancourt, 2021, p. 48). This direct line of questions is important to consider, as Vaillancourt encourages managers to support employees and allow them to be productive in their own right without burdensome management.

"I can attest that it is not hard to tell who is doing work and who is not. In fact, it's fairly easy as long as performance objectives are clear, work tasks are well defined, and administrators are engaged in regular and meaningful conversations with the people they oversee. While ineffective managers believe in hovering, micromanagement, and seat time, strategic managers know that physical presence and the appearance of being busy is no substitute for actual productivity" (Vaillancourt, 2021, p. 48).

Setting communications expectations for remote employees serves multiple functions.

- They help align employees with organizational goals during remote work and how those goals might be best accomplished.
- They help employees establish new routines, which can facilitate effective communication and sensemaking of new work processes.
- They can serve as a resource in helping employees cope with uncertainty and resulting stress (Shockley et al., 2021, p. 1469).

A realistic work-life policy recognizes that remote work is not appropriate for all employees. Numerous researchers (Wang et al., 2021; Pinsonneault and Boisvert, 2001; and Golden and Veiga, 2005) agree that a remote work policy that requires all employees to work remotely may not be appropriate. Researchers agree that managers should provide remote work options for appropriate jobs and workers, meaning that a one-size-fits-all remote work policy is not advisable.

"Communication is often cited as critical to remote worker success. That is, in order to be effective, remote workers must be able to communicate with colleagues through means other than face-to-face, such as by email, phone, texting, instant messaging, and video conferencing" (Shockley et al., 2021, p. 1466).

According to the WHO and the International Labour Organization, to ensure employee work-life balance, protections should be in place to help remote workers switch off from work and also limit the ways employers can invade an employee's private spaces (Skelton, 2022). This is important, as it addresses a primary concern with remote work—the 24/7 home office.

Moving forward, setting clear guidance and expectations for employees regarding remote work is important. A comprehensive remote work policy will include some or all of the following:

- Defined employee eligibility to work remotely.
- Specific procedures to request approval.
- Reference to applicable reasonable accommodation procedures for employees with disabilities.
- Employee expectations (such as work hours, timekeeping, accessibility, secure remote access procedures, and work-related expenses.)

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- Employer responsibilities (such as technical support, equipment provided, and expense reimbursements.)
- Designated work areas and break times to avoid liability for injuries that are not work-related.
- Reminders that the employee remains subject to all employer policies.
- Explanation of the temporary nature and conditions of the remote work arrangement (for example, a note explaining that remote work arrangement is only available during governmental orders necessitating remote work) (Mutz, 2021, paragraph 2).

Even with a well-developed and refined remote work policy, evolution can be expected. According to Colvin (2022), "No remote-work policy will be invulnerable to post-pandemic reality" (p. 44). In other words, in unstable times, "all policies are experiments" (p. 44).

Support for Remote Employees

"To a large extent, your personality will determine how you handle this [remote work]. The introvert will probably be more at ease at home, while the extrovert, who gets his energy through interaction with other people, could . . . [experience] more stress at home" (Esterhuizen, 2021, p. 41).

There are a number of considerations to support successful remote employees. Organizations asking or requiring certain employees to work remotely are responsible for providing support for these employees. The degree of success with remote work depends on a number of factors, and it is important for employers to remember that employees are different. "Individuals vary significantly in their ability to stay on task, in their tendency to be distracted (or not) by what is happening at home, and in how well they respond to unusual pressures and demands" (Manko and Rosinski, 2021, p. 53). As Manko and Rosinski (2021) explain, "There is no one-size-fits-all approach to managing remote work" (p. 53). As Johnson and Suskewicz (2020) share, "To know what's 'best' for your organization's future when it comes to remote work, you have to put it in the context of all the things that you are looking to achieve" (p. 3). Considerations include needed technologies, needed resources, and the metrics used to evaluate success and measure productivity.

Success for remote employees is enhanced when management and employees create a partnership. Both have responsibilities. According to Manko and Rosinski (2021), "Ultimately, it would seem that a line should be drawn between what managers can do to make remote work go well, and what the individual must do for themselves" (p. 45). In essence, both employers and employees have responsibility for successful remote work arrangements. Managers are responsible for:

- Providing the necessary equipment and training for remote work.
- Giving remote workers access to the information they need for their jobs.
- Giving remote workers transparent information on the company's progress and management decisions.
- Providing mechanisms for questions, suggestions, and feedback.
- Being mindful of remote workers' emotional issues and needs and devising ways to address them.
- Finding the right balance between autonomy and oversight.
- Designing online interactions with an eye to the particular issues raised by virtual media (Manko and Rosinski, p. 44).

The research shows that some employees are better suited to remote work. In essence, the job type, as well as the individual's personality type, are both taken into consideration when making decisions about approving remote work arrangements. As stated by Toscano and Zappala (2021):

> Managers should consider that people who work less well in the office are likely to work poorly even at home, especially if they have children to care for. Where and when conditions (e.g., health emergency) allow it, therefore, it is possible to prioritize the return to the office of these employees who do not benefit from, and might suffer from, remote work. In contrast, employees who are usually productive may find in remote work a solution that improves their effectiveness at work and probably also their skills to cope with parenting commitments. This aspect may be very relevant for professionals and managers, who may be skeptical about granting remote work to their high-performing employees with children to take care of, so as not to alter their already positive results. (p. 140)

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Ergonomic Considerations

It is helpful to develop ergonomic guidelines for employees working remotely. Ergonomics, which is about reducing strain and fatigue by fitting tasks to the capabilities of the human performing them, is an important component in worker productivity. In other words, if someone is uncomfortable at the workstation, with lower back pain, eye strain, etc., productivity will suffer. While this may only be a minor concern at the office (with ergonomically correct desks and chairs in place), it is a greater concern for those working from home, as a kitchen table or bed may double as a workstation—creating less than ideal ergonomics.

Subramanian, Miller, and Fernandez (2020) offer the following ergonomic considerations:

- **Neutral positive.** Attain a proper posture while performing sitting or standing work; a neutral seated posture should include sitting with the neck straight, shoulders straight down loosely at the sides, elbows at a right angle, wrists straight, low back supported on the back rest of the chair, 90 degrees at the hips, 90 degrees at the knees, and feet flat on the floor or on a footrest.
- **Eye and elbow height.** Whether seated or standing, ensure that the keyboard (ASDF home row) and mouse are positioned at the elbow level. The top of the monitor should be at or slightly below eye height.
- **Work area.** Keep items that are used often in the primary work zone (the area when elbows are at the sides and the hands are moved side to side); keep items that are used less often in the secondary work zone (area within the outstretched arms). In the office, the keyboard and mouse should be in the primary work zone, centered with the user and the monitors. (p. 20)

By addressing ergonomic considerations of the home office, remote employees will feel less fatigue, less back and eye strain, and will generally be more productive. As management strives for maintaining productivity from remote employees, ergonomics is an important consideration.

Organizational Support

In developing a workplace that will benefit both remote workers and on-site workers, "The goal is to offer a workplace experience that is flexible, energizing, and inspiring. Talented people have options, and they will choose organizations that make it possible for them to have both meaningful jobs and meaningful lives" (Vaillancourt, 2021, p 48).

It is important to note that "even employees who do not work remotely must work in ways that are remote-friendly" (Limoncelli, 2020, p. 108). Employees working on-site may need to interact with remote employees throughout the day. Thus, it is important to consider the needs of these remote employees, to include communication tools, etc.

Tips to support remote work are provided by Limoncelli (2020):

- If anyone is remote, we're all remote. Meetings should be either 100% in-person or 100% remote. There are no mixed meetings.
- Set your chat status to away when you are away. Set it to available when you are available.
- Establish a low-overhead way to start a quick conversation.
- Work silently together in a videoconference room. Many teams hang out in a videoconference even when not having a meeting.
- Create social events specifically for remote workers.

For those organizations that desire to continue supporting a remote or hybrid workforce post-pandemic, the literature recommends a number of practices (Landry-Bourne and James, 2021, pp. 6 and 28–29):

- **Involvement.** Actively involve staff in creating their workplace plan. Staff needs to be actively involved in creating a schedule or workplace plan that is desirable for them, created with their input, and with agreement and support from their supervisor and manager.

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- **Togetherness.** Create opportunities for staff to be together. Create connectedness with virtual all-staff meetings and coffee hours, for example.
- **Lead by example.** Serve as a role model for remote or hybrid work. If employees act in a contradictory fashion, particularly those in leadership, this sends a mixed message to staff, which ultimately leads to confusion and uncertainty.
- **Resources.** Provide the needed technology and resources. When staff feels prepared and equipped to get their jobs done, productivity and staff morale are more likely to remain steady and positive.
- **Connectedness.** Be intentional in connecting with and celebrating staff. Staff who have a feeling of connectedness to others in the workplace have a better sense of well-being.
- **Transparency.** Communicate clear expectations that exhibit transparency. Employees will feel safer and more secure when they know what is expected of them in any given situation.
- **Work-life balance.** Keep employee mental health and well-being at the forefront. Work-life balance might adjust for employees in ways they did not need to think about before.
- **Strengths.** Seek opportunities to utilize personal strengths. The opportunity to utilize strengths on a regular basis contributes to job satisfaction.
- **Managing change.** Demonstrate patience and grace with change management. Allow staff to feel heard and have some control over the new working model.
- **Sharing.** Encourage staff to share successful strategies and outcomes. Leaders should encourage staff to be involved in strategic planning and sharing outcomes.

According to Zhang et al. (2021):

> Managers can provide structure for employees' daily work (e.g., delegate specific tasks and set clear timelines), help employees maintain work-life boundaries (e.g., keep work-related communications within business hours), ensure that employees have the home office setup and virtual technology they need, support training and development (e.g., online courses, webinars, and virtual conferences), encourage frequent breaks to combat risk of fatigue or burnout, and facilitate interpersonal connections among remote workers (e.g., periodic group meetings and virtual team happy team hours.) (p. 805)

In a vote for long-term remote work, companies are now hiring for a new executive position: head of remote work. "This individual provides support to off-site workers, championing their needs, ensuring they have the right tools to perform their jobs, and keeping them engaged" (Association for Talent Development, 2021, paragraph 2).

Summary

Although remote work is not a new concept for some industries, the prospect of significant numbers of employees working off-site and from home was generally a new concept for airports as the COVID-19 pandemic impacted the globe. Some airports required many employees to work remotely, while others did not. As with companies across the United States, this decision was often driven by management that either placed value in maintaining operations by sending employees home, or placed value in maintaining operations by continuing to serve patrons in-person and on-site. While the literature does not necessarily agree that one method was more successful than the other, there is agreement that a "one-size-fits-all" approach to work models during the pandemic does not exist. As this report presents findings based on data collected from airports across the United States, the reader may consider alternative ideas presented, but adopting these ideas (or any variation thereof) is best done with a consideration of the unique staffing and operational realities at the reader's airport.

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CHAPTER 4

Survey Results

Introduction

Each section of this chapter presents results from the two surveys administered during this project. First, responses from 151 airport managers or HR managers are presented. These responses provide the employer perspective. Second, responses from 76 airport employees are presented. These responses present the employee perspective. At times, these two perspectives are similar; at other times, these perspectives are quite different. Responses are presented in graphical format, along with narrative interpretation of the results.

It should be noted that, in general, not all participants answered each question; thus, percentages are based on the actual number of responses to each question, which may or may not match the total number of survey participants.

Airport Manager/HR Manager Survey

Participating Airports

The majority of participating airports (54.79%, n = 80) are not Part 139-certificated. Even so, 45.21% (n = 66) of participating airports are Part 139-certificated. Although the study gathered data from airports representing all FAA NPIAS categories, the majority of participating airports (53.38%, n = 79) are GA. This coincides with the fact that most of the participating airports are not Part 139-certificated. Non-hub airports are also well-represented at 13.51% (n = 20). The remaining NPIAS categories were similarly represented: large hub (7.43%, n = 11), medium hub (5.41%, n = 8), small hub (7.43%, n = 11), non-primary (6.08%, n = 9), and reliever (6.76%, n = 10).

The study also had wide geographic reach, with eight of nine FAA regions represented. There were no responses from the Alaskan region. The Southwest region was most represented, with 44.97% (n = 67) of participants. The Southern region was second highest with 16.78% (n = 25) of participants. Other regions include the Central (8.72%, n = 13), Eastern (10.07%, n = 15), Great Lakes (3.36%, n = 5), New England (0.67%, n = 1), Northwest Mountain (12.75%, n = 19), and Western Pacific (2.68%, n = 4). The Southwest region and Southern region are most represented with 61.75% of participating airports. These regions are similar and different in numerous ways, including climate, topography, wildlife challenges, and remote work considerations.

Remote Work

To initially determine the degree to which airports provided for remote work, participants were asked, "At any time during the COVID-19 pandemic (March 2020–March 2022) did

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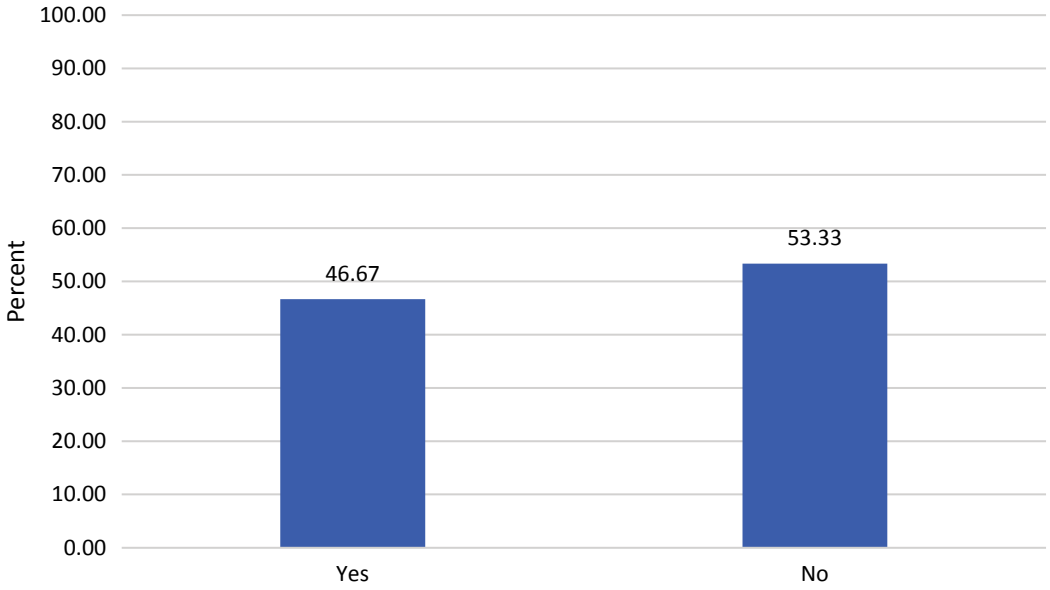


Figure 3. At any time during the COVID-19 pandemic (March 2020–March 2022) did your airport allow (or require) any airport employees to work remotely for any period of time?

your airport allow (or require) any airport employees to work remotely for any period of time?" Just under half of participating airports (46.67%, n = 70) provided for remote work during the pandemic. Interestingly, more than half of participating airports (53.33%, n = 80) did not allow for any employee groups to work remotely during the pandemic (see Figure 3). Remote work options tend to be more common at larger airports with more staff (see Figure 4). Generally, as the size of airport staff decreases, the opportunity for flexible working arrangements also decreases.

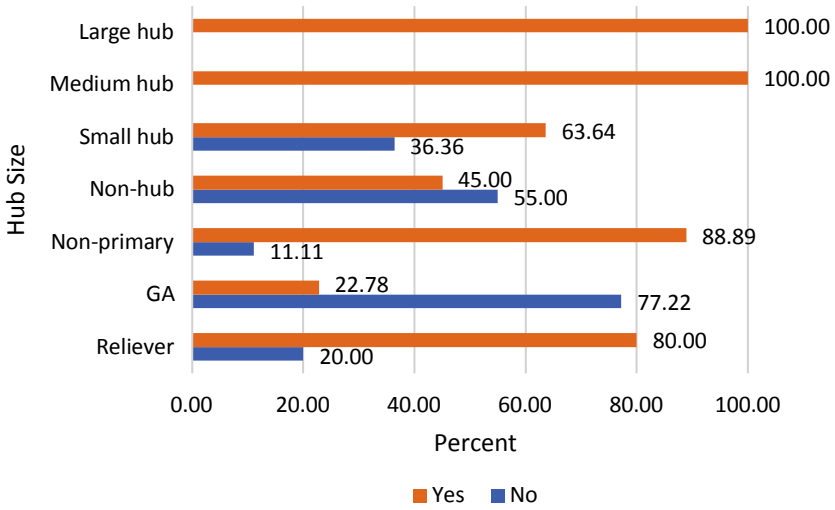


Figure 4. At any time during the COVID-19 pandemic (March 2020–March 2022) did your airport allow (or require) any airport employees to work remotely for any period of time?

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Those airports that did not allow for any employees to work remotely during the pandemic explained their rationale. Themes included:

- Very few employees
- Work could not be performed from home
- All employees are considered essential
- Employees chose to work on-site

Some airports instituted split shifts to rotate employees and minimize the number of staff in the office at any one time. As one participant explained: "Small staff of 1-5 employees. Instituted an every other day work schedule to split staff up. This ensured qualified people were available at all times for coverage. If one person got sick then the whole staff was not affected." As another participant explained, "With little to no traffic during COVID, the airport decided to rotate office staff. Our front desk person worked three days per week and the office manager worked two days per week. We paid them for 40 hours, but it was important to us to minimize possible contamination of both office staff at the same time. This schedule allowed for the office staff to not be here at the same time during COVID."

Of those 70 participating airports that did provide remote work arrangements, the survey queried participants as to which departments these remote employees were assigned. Participants were allowed to select more than one department. Finance was the most common department (66.67%, n = 36) with remote employees. Properties/Contracts/Commercial Development was the second most common department (48.15%, n = 26) with remote employees. Other departments selected by at least one-third of participating airports include Procurement, Public Relations, Planning and Design, Marketing, and Information Technology. The "Other" category was selected by 44.44% (n = 24) of participants, although most of these responses were Administration, which may have included HR, for example (see Figure 5).

Those participating airports with remote employees were also asked, "How did you choose the employee groups eligible for remote work?" Most responses indicate that "non-essential"

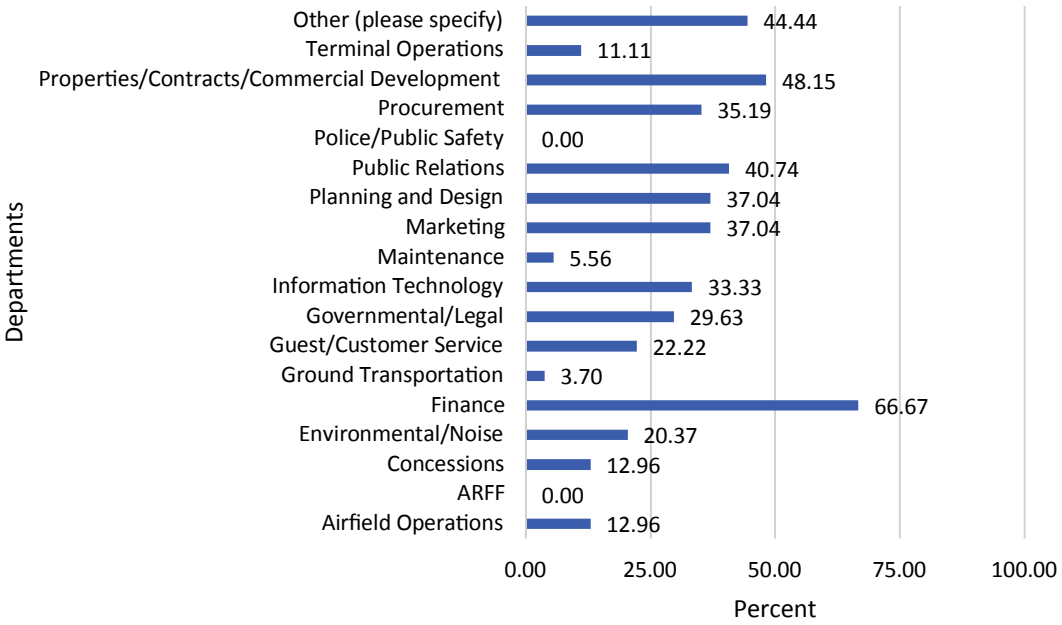


Figure 5. Which employee groups were required to work remotely at any time during the pandemic?

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employees who could perform tasks away from the airport were likely to be eligible for remote work. The following selected responses from airport managers give a sense of how they chose which groups of employees would be eligible for remote work:

- "It was simple. Can your work be done from a computer at home?"
- "The decision was based on the operational need to be physically located at the airport. Those employees who complete the majority of work by computer or phone were eligible for remote work."
- "Employees were eligible for remote work based upon their technical capabilities to work remotely. The employees were largely administrative functions."
- "If it was possible for them to do the work remotely, we were agreeable. Most jobs – custodians, parking lot attendants, maintenance personnel, and public safety could not, but most administration staff could."
- "Employees in jobs that cannot be performed remotely (maintenance, operations) were generally not eligible for remote work. All other jobs were allowed to perform work remotely at times."
- "Individuals that had childcare problems relating to school closures and those living with people in high-risk health groups."

It seems therefore, that among airports allowing remote work, only employees with job functions that could be performed off-site were eligible for remote work.

In a follow-up to the previous question, participants were asked, "Which employee groups did your airport consider essential, and thus in-eligible for remote work?" In other words, which employee groups were required to work in-person and on-site? The five most common departments are Airfield Operations (88.68%, n = 47), Maintenance (83.02%, n = 44), Airport Rescue and Fire Fighting (ARFF) (67.92%, n = 36), Terminal Operations (52.83%, n = 28), and Police/Public Safety (50.94%, n = 27) (see Figure 6). This is expected, as employees in each of these departments are generally required to perform their job functions at the airport. In other words, it is impossible to perform an airfield inspection or respond to an aircraft emergency from home.

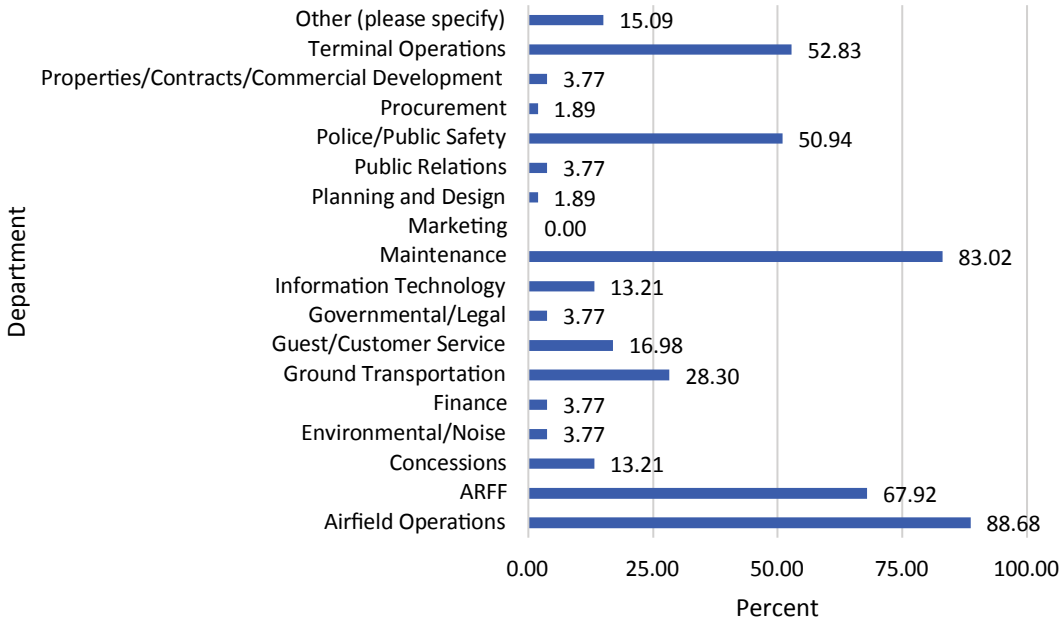


Figure 6. Which employee groups did your airport consider essential, and thus in-eligible for remote work?

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For those participating airports allowing remote work, participants were queried about the technology tools that enabled remote work. Although there was wide variation in specific tools and platforms in use, common themes emerged (see Figure 7). Airports with successful remote work arrangements provided needed technology tools to employees. Some participating airports utilized expensive cloud-based products, while others utilized more affordable or free platforms. IT infrastructure upgrades were also beneficial.

To determine how newly hired employees were accommodated via remote work arrangements, participating airports were asked, "What considerations are there for new employees regarding possible remote work arrangements?" One common theme was that no additional considerations were afforded newly hired employees in this regard. That being said, some participating airports were quite innovative in their approach.

Some participating airports have ceased all remote work arrangements as the pandemic ended. As one participant shared, "Remote work is not standard and is no longer allowed. Employees can request ADA alternate work arrangements but that typically involves workplace modifications, not locations." That being said, some airports are recognizing remote work as a legitimate ADA accommodation. Some airports are also considering how remote work arrangements may be considered in the future and are developing remote work policies as a result.

Hybrid arrangements were common among those continuing to offer remote work. As one participant shared, "Since COVID and prompted by the competitive employment environment and gas prices, we instituted a permanent limited remote work policy allowing certain regular employees to work up to 2 days per week remotely while managers and directors can do so no more than 1 day per week."

Another participant shared, "We currently have returned to a hybrid schedule. Those that can work from home are able to do so on M and F. They must be in the office T, W and Th. Some departments are already deviating from that schedule. Some people are back full time, others are staying at home up to four days. New hires are working up to five days a week depending on training needs."

Another participant shared, "Managers designate prior to hiring whether the position can work remotely. We have policies and procedures in place for remote work arrangements, how to manage in a remote environment, and how often. We also have hybrid work arrangements – employees work part time in the office and part time remotely (primarily those in office environments)."

Some airports are considering how such a policy can assist with HR challenges, including employee attraction and retention. As one participant shared, "Our airport authority ended all remote work June 2021; we are revisiting it to enhance employee attraction and retention and have drafted a remote work policy that we expect to implement in the next couple of months."

What technology tools enabled employees to perform their jobs in a remote manner?

"Employees who worked from home were provided with laptops, additional computer monitors, and in some instances cellular devices. In addition, employees were provided with access to VPN into the Airport's shared computer drives and other software."

"The airport network servers had previously been upgraded to allow for remote connections while staff travels or is otherwise off site."

"Employees were issued laptops, iPads, chromebooks, computer screens, connection cables to their TV, wireless keyboards, scanners/ printers . . ."

What considerations are there for new employees regarding possible remote work arrangements?

"For new employees, considerations regarding remote work arrangements are given in providing equipment while also ensuring they are in compliance and receive onsite training and work requirements."

"We offer a flexible work arrangement to all staff. They can share how and when they'd like to work and if it can be done, we do it."

"Usually have a mandatory on boarding period (e.g., 3 months) where one must be in the office, then can be considered to telework."



Figure 7. What technology tools enabled employees to perform their jobs in a remote manner?

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What are some methods your airport used to manage employees working remotely?

"We use Google Meets and Hangout. I am sure there are other tools departments are using. I keep a Hangout chat open with each of my staff and a group chat. This way I can see when they are away from their desk or I can ask them an individual question. I also liked to have a Google Chat to see each other face to face rather than just a phone call. I am not sure what worked for all others."

To gauge the structures put in place to manage remote work, participating airports were asked, "What are some methods your airport used to manage employees working remotely?" The goal was to understand how remote employee oversight was accomplished to ensure productivity in the remote environment. Basic methods such as virtual meetings, email, phone check-ins, etc., were used by many airports. Use of collaborative platforms also supported the management of remote employees. As one participant shared, "Office 365 allows for collaboration in a team environment. Daily meetings of the entire remote staff with those on site. Other meetings as needed to link staff on specific projects." One participating manager simply kept a chat window open during business hours and would converse with remote employees throughout the day.

One participating airport required remote employees to come into the office once weekly to check-in and pick up work assignments. Another airport utilized various metrics to track employee productivity. As this participant shared, "Daily check in with supervisor; our work group developed a daily staffing report which identified staff attendance, as well as who was on site and who was teleworking. This helped us track when call off rates were escalating and continuity of operations planning needed to be implemented to meet staffing requirements." Another participating airport required detailed work logs that could be audited. As this participant shared, "Employees were required to document daily work (by hour) and submit weekly to supervisor. These were tabulated and held by Finance for payroll and audit purposes."

One participating airport that adopted a hybrid work arrangement used performance standards to determine if employees would be allowed to remain in a hybrid arrangement. As this participant shared, "Aviation Department's management meetings occur on Wednesdays, so we have all personnel working in person on that day. The other weekdays, remote work employees rotate. Typically, all remote work employees have 2 days in office and 3 days working remotely. This proves to be a big incentive–if the tasks gets done, remote work can continue. If things start to slip, in-person work might return."

Another participant explains quite simply, "If you hire the right employees, trust your employees to do their work well, and base their work on results and not time in a chair, everyone wins."

Participating airport/HR managers were also asked, "What percentage of your airport employees would like to continue working remotely?" On average, 27% of airport employees would like to continue working remotely, according to participating airport/HR managers, with responses ranging from a low of 0% and a high of 100%. Clearly, there is wide variation among airports and likely, specific employee groups.

How have your hiring practices changed since, and as a result of, the COVID-19 pandemic?

"Many of our hiring practices have gone virtual due to the pandemic. For example (interviewing)."

"Many of our tools for hiring are now digital, including remote interviews. This allows us to recruit and hire more quickly if the hiring team is not all on site at the same time."

To gauge the degree that hiring practices may have changed at airports since the pandemic, participating airports were asked, "How have your hiring practices changed since, and as a result of, the COVID-19 pandemic?" This open-ended question was answered by participants in a varied manner. At many participating airports, nothing has changed. The most common theme was airports conducting virtual interviews. At some airports, a remote work policy has been developed. At others, drastic changes have taken place. Another airport has completed a thorough review of all job descriptions, eliminating some job requirements to recruit from a larger applicant pool. As this participant shared, "We've reevaluated a lot of the job descriptions and removed anything that wasn't truly required and had likely been rolled over year after year (Degrees, driver's license, certifications . . .) in hopes of having a bigger pool to choose from." Numerous airports mentioned how challenging it has been to fill vacant positions with qualified candidates. As one participant shared, "Our requirements for education and experience have been lowered, our wages are increasing and we are trying to be as flexible as possible about shifts and wfh [work from home]." Another participant shared, "To hire quality workers in such a competitive market, flexibility is necessary." Another participant shared, "We haven't changed our hiring yet, and it shows, we are getting 10% of the applications we used to get for Airport

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Operations positions. When I've contacted universities, I've been told that new graduates want jobs where they can work from home."

To further explore the role COVID-19 has had on the airport HR function, participating airport/HR managers were asked, "How has your overall HR function changed since, and as a result of, the COVID-19 pandemic?" A number of participating airport/HR managers mentioned COVID-19 cleaning protocols, COVID-19 vaccine requirements, and expedited/virtual interviewing/hiring practices. As one participant shared, "Interviews have been conducted remotely. Hiring decisions have been accelerated in an effort not to lose good candidates to competing opportunities."

One participant summed up these changes by sharing, "The economic impacts of the pandemic on budgeting, recruitment and retention have impacted human capital significantly and will have a lasting effect. HR is recruiting for many more positions and the pool of applicants is shallow which creates greater demand on training resources, some of have had limited availability until recently."

One participating airport manager expressed frustration with their airport HR department, by sharing, "From the HR customer perspective, they have not adapted fast enough nor innovatively enough to meet the changing conditions of job/hiring market, responding with policies and procedures to support the field in a timely manner, etc." Yet another participating airport manager expressed satisfaction with their airport HR department by sharing, "HR was great providing the information regarding public health orders and being creative with work schedules and identifying challenges of remote work at the onset." These two contrasting views are evidence of the wide variation of perspectives on this issue among airports.

To discern whether some age groups (or generations) were more successful with remote work, participants were asked, "During your experience with remote work, did you notice one generation of employees were more efficient/productive when working remotely? (If so, which?)" (see Figure 8). The results indicate that, overall, there is no difference among workers of different generations in their success as remote employees. Indeed, 81.13% (n = 43) of participants indicate

How has your overall HR function changed since, and as a result of, the COVID-19 pandemic?

"The great resignation has impacted our organization, hiring, attrition—all has increased. HR burnout is high. How we communicate with our employees has changed. How we provide training has changed (learning opportunities are hybrid or remote). This is in addition to most HR jobs being primarily remote now - almost 100% of the HR staff works remotely part or full time. Customer service to the organization is impacted. More reliance on data. At the same time, some learning opportunities and organization-wide meetings have more attendees by offering virtual options."

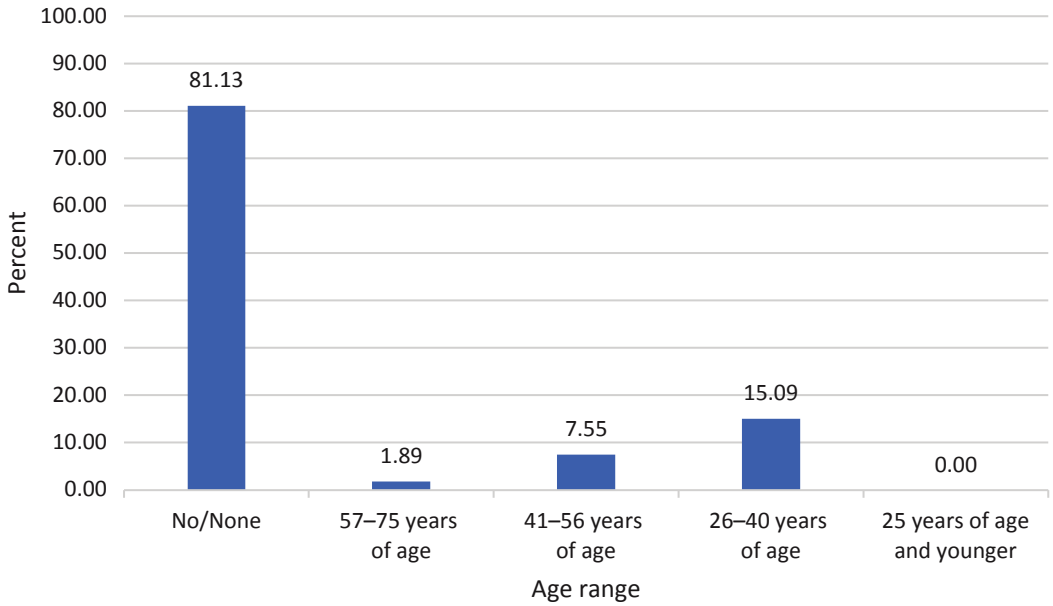


Figure 8. During your experience with remote work, did you notice one generation of employees were more efficient/productive when working remotely? (If so, which?)

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there was no difference. Among those few airport/HR managers indicating there was a difference, employees aged 26 to 40 years old seem to have greater success in the remote work environment. This is possibly due to this generation being considered "digital natives," making for an easy transition to a more technologically rich work environment.

To determine how airport/HR managers feel about the future of remote work, now that the COVID-19 pandemic is over, participants were asked, "Do you feel a remote work model is sustainable for your airport, even years into the future?" (see Figure 9). The responses to this question were almost evenly split. Although a slim majority (51.92%, n = 27) do not feel remote work is sustainable for the long-term, 48.08% (n = 25) do feel it is sustainable.

To follow-up on the previous question, participants were asked, "Does your airport have a long-term plan for remote work?" (see Figure 10). The vast majority of participating airports (80.77%, n = 42) do not have a long-term plan for remote work. Although 48.08% (n = 25) of participating airport/HR managers believe that remote work is sustainable for the long term, a much smaller percentage, 19.23%, (n = 10) actually have a long-term plan for remote work. Thus, not all airports that believe in the long-term viability of remote work actually plan to participate in this trend.

Airports were also forced to consider training for employees in innovative ways, especially in states with more restrictive stay-at-home orders. Participants were asked, "Were employees trained (either initially or recurrently) via either fully online asynchronous or hybrid/blended approach at any time during the March 2020–March 2022 time period?" The majority of participating airports (65.35%, n = 83) did not conduct employee training in a fully online or blended approach during the COVID-19 pandemic. However, 34.65% (n = 44) airports did (see Figure 11).

As a follow-up to the previous question, participants were asked to explain how training was conducted during the COVID-19 pandemic. Some airports utilized Microsoft Teams, AAAE online training, webinars, Veoci, and Zoom. One airport participant mentioned developing "new SIDA training through an online tool so candidates could take the training independently."

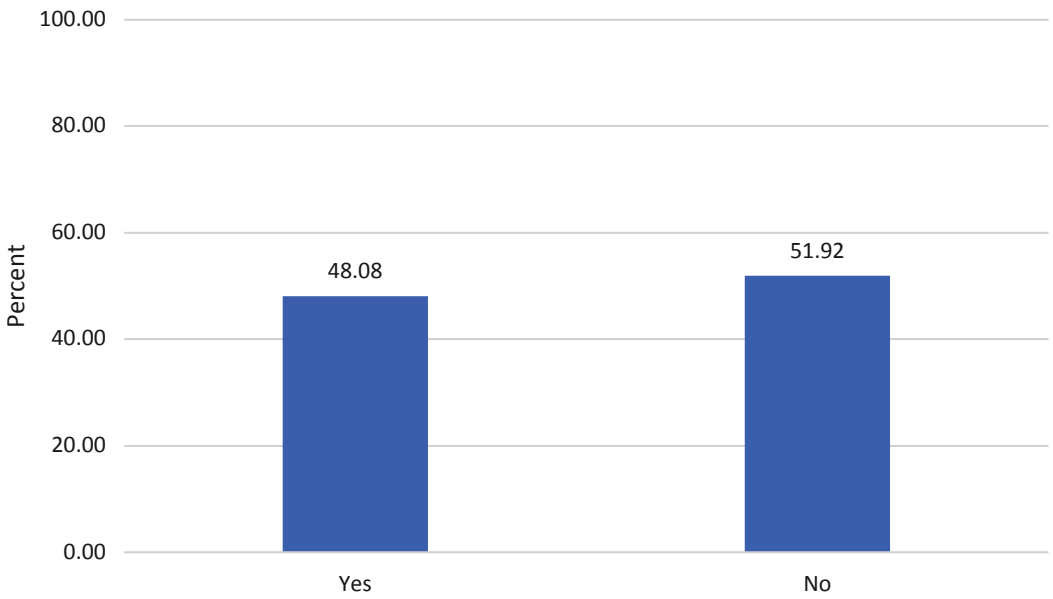


Figure 9. Do you feel a remote work model is sustainable for your airport, even years into the future?

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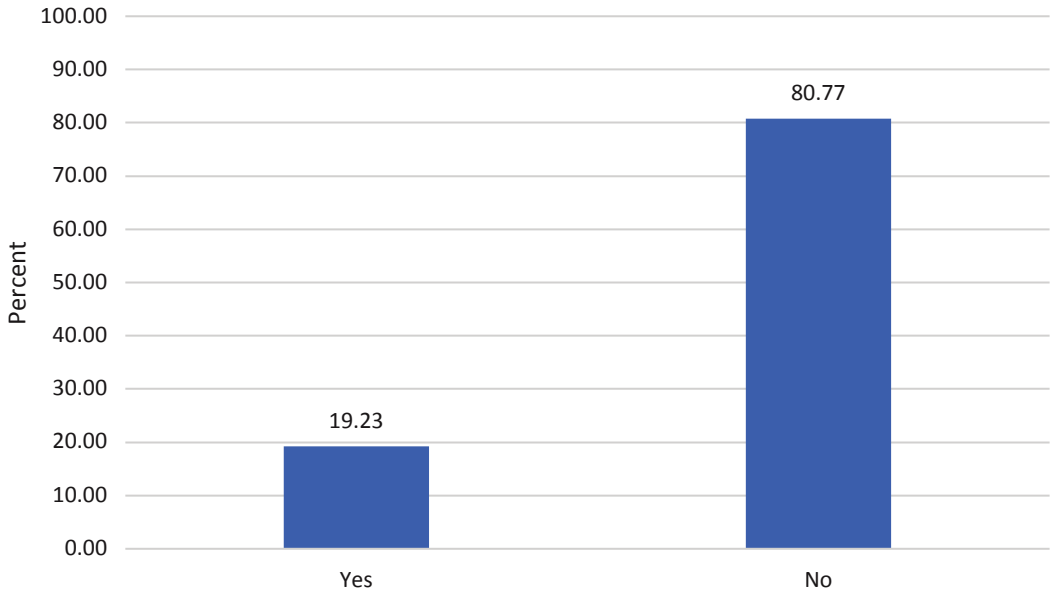


Figure 10. Does your airport have a long-term plan for remote work?

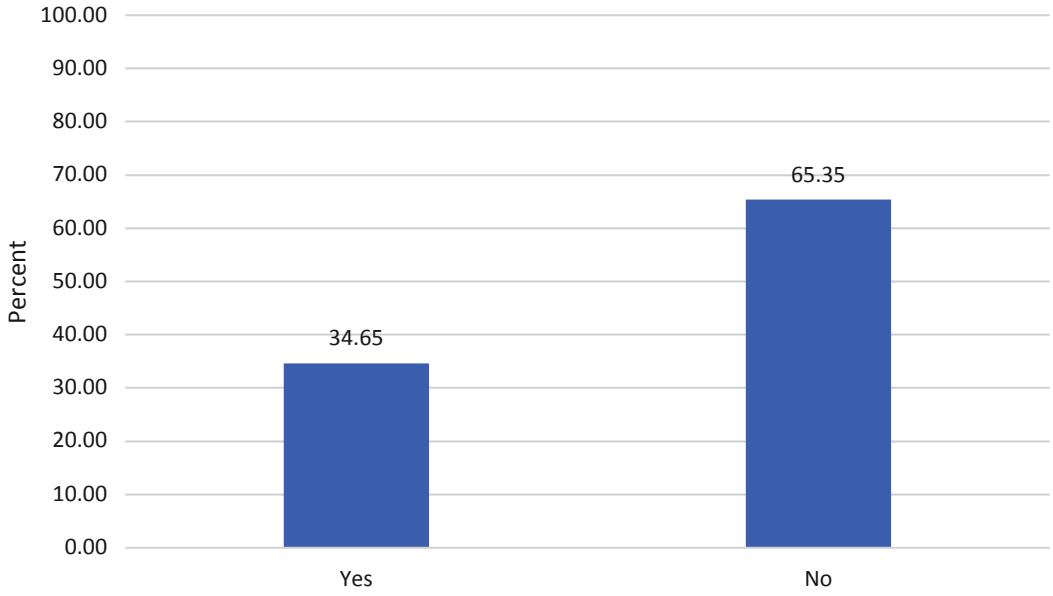


Figure 11. Were employees trained (either initially or recurrently) via either fully online asynchronous or hybrid/blended approach at any time during the March 2020–March 2022 time period?

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Another participant mentioned how they had to juggle in-person training for on-site employees and virtual training for remote employees. As this participant shared, "Virtually. We used MSFT Teams and shared our screen. – This was for remote workers. On-site continued to be trained in-person."

For those airports that did not adopt online or blended training, in-person training was conducted with some COVID-19 protections in place. For example, as one participant shared, "In-person training with social distancing in place, limited number of attendees."

Morale took a hit during the pandemic, what "fun" thing did your airport do that you can share?

"Nothing. There was a very strict crack down on essential employee interactions meaning no more team meals, birthday celebrations or even in-person shift briefings. It really isolated staff and made a sometimes stressful job harder."

"For our onsite workers we had an appreciation day where we provided them with food and snacks. In addition, we offered a number of online offerings including virtual wellness, and Thursday Water Cooler Conversations."

Obviously, the COVID-19 pandemic placed a severe strain on organizations of all types and sizes. It also placed great strain on the emotional well-being of employees (including those in management). In that regard, participants were asked, "Morale took a hit during the pandemic, what 'fun' thing did your airport do that you can share?" Responses were quite varied, with some airports reporting that they did nothing in this regard. Other airports were more intentional in supporting employee morale through fun activities. Innovation often created the most fun. As one participant shared, "Online team building with scavenger hunt app. We competed against other departments and all the items were found outdoors or something you could do at home." Another participant shared, "I produced videos to take the place of regular 'All Hands' meetings. When able, we met (socially distanced and with the doors open) in a hangar to create a sense of community and physical togetherness." Experience has shown that intentional "fun" events have significant positive impacts on employees during a very stressful time.

Participants were also afforded the opportunity to share any additional thoughts on this topic. Although lessons learned varied among participating airports, several themes were common.

- **Communication.** Communication proved highly important during the pandemic. As one participant shared, "The biggest lesson learned is that communication is key in an organization and that our essential staff are the heart of the organization."
- **Challenges.** Remote work presented new, and in some cases, insurmountable challenges. As one participant shared, "Remote work is not ideal at all for the airport industry. The work being produced at home with other distractions abound [sic] are not conducive for a steady output of work. That coupled with the true need to have that face-to-face interaction with the many tenants and all of their facilities make me believe there is not a place for remote work in this industry."
- **Desire.** It cannot be assumed that all employees want to work remotely. As one participant shared, "Not all employees desire to work remotely. It can be accomplished with a little planning and employers that are flexible will prevail in the competitive labor market."
- **Benefits.** Alternate work arrangements can benefit airports. According to one participant, "It took a pandemic for us to realize that work can get done in a different way. Employees feel like they get part of their lives back, not having to spend as much time and money commuting to/from work every weekday. If the organization remains intentional to connect and communicate with the employees, we expect to see continued positive results."

Airport Employee Survey

Participating Airports

Unlike the airport/HR manager participants, the majority of participating airport employees (88.16%, n = 67) work at Part 139-certificated airports. A minority (11.84%, n = 9) of participating airport employees work at airports that are not Part 139-certificated. Although the study gathered data from airports representing all FAA NPIAS categories, most of the participating airport employees (38.16%, n = 29) work at large-hub airports. The remaining NPIAS categories were also

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represented: medium hub (14.47%, n = 11), small hub (19.74%, n = 15), non-hub (11.84%, n = 9), non-primary (1.32%, n = 1), GA (9.21%, n = 7), and reliever (5.26%, n = 4).

The study also had wide geographic reach, with all nine FAA regions represented. The Southern region was most represented, with 27.63% (n = 21) of participants. The Eastern and Northwest Mountain regions were the second most represented regions with 17.11% (n = 13) each. Other regions include Alaskan (2.63%, n = 2), Central (2.63%, n = 2), Great Lakes (13.16%, n = 10), New England (3.95%, n = 3), Southwest (7.89%, n = 6), and Western Pacific (7.89%, n = 6).

All of the participating airport employees were at least 25 years of age. The age group with the most responses was 35–44 (36.84%, n = 28). A number of participating airport employees were also 45–54 (22.37%, n = 17) and 55–64 (22.37%, n = 17). The least represented age groups are 25–34 (17.11%, n = 13) and 65+ (1.32%, n = 1). This indicates that all age groups older than 25 years of age participated in the study.

Airport Departments

Participating airport employees work in a number of different airport departments (see Figure 12). The Airfield Operations department is the most common home department of participants, representing 31.58% (n = 24) of participants. The Planning and Design department was the next most common, representing 21.05% (n = 16) of participants. Other departments representing at least 10% of participants include Finance (14.47%, n = 11), Maintenance (10.53%, n = 8), and Properties/Contracts/Commercial Development (11.84%, n = 9).

Remote Work

Participants were asked "Were you trained via distance learning at any time during the pandemic (March 2020–March 2022)?" This question was asked to ascertain how much airport personnel training was conducted via distance learning for employees who were unable to be trained in-person during the pandemic (see Figure 13). The majority of participating airport employees (69.74%, n = 53) were trained via distance learning at some point during the pandemic.

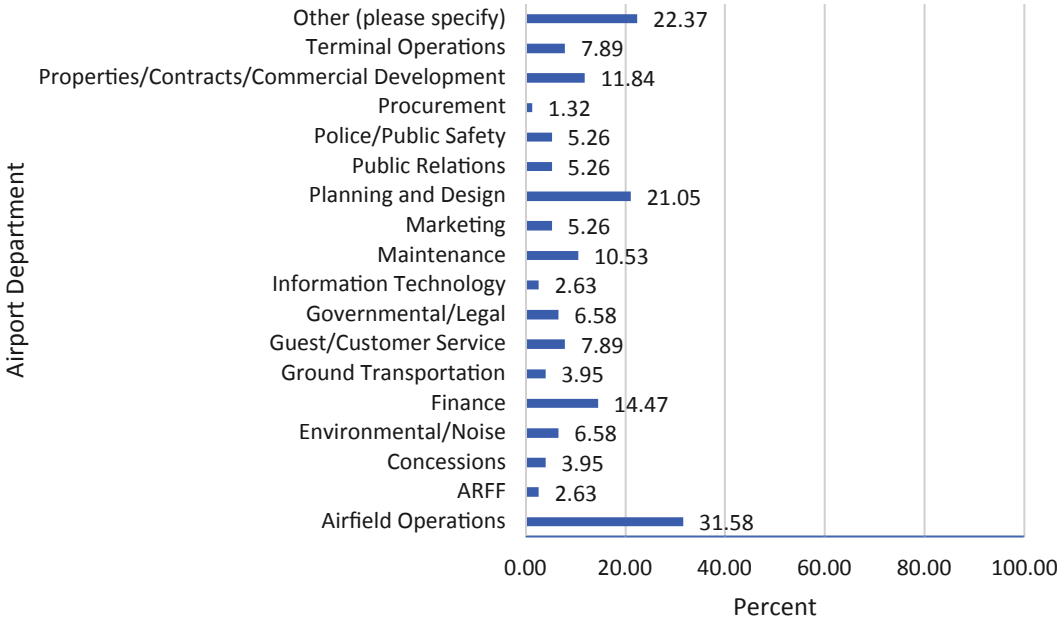


Figure 12. Which airport department do you work in?

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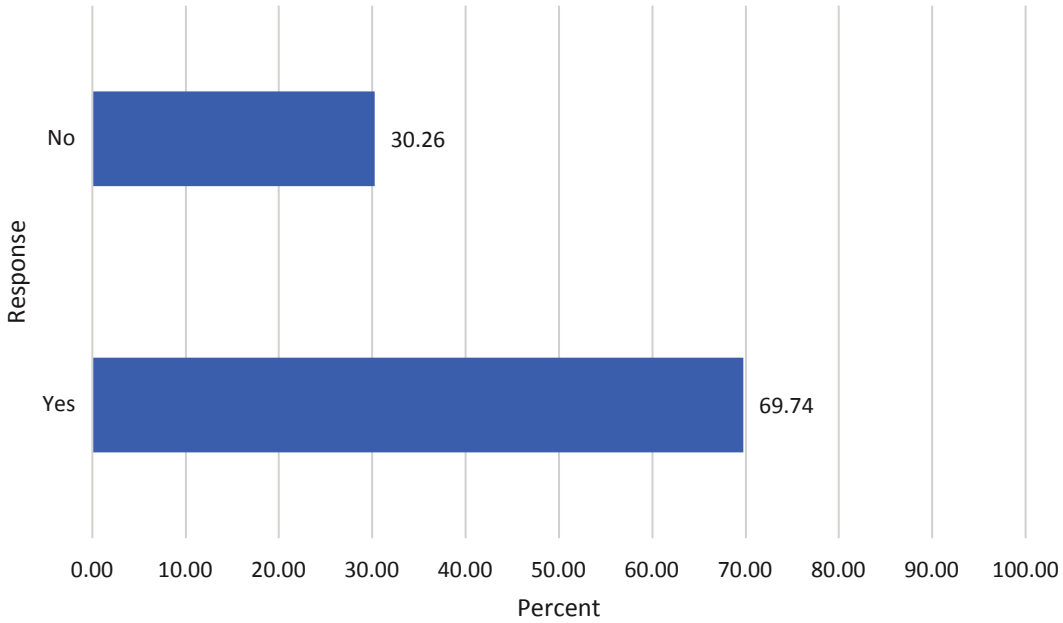


Figure 13. Were you trained via distance learning at any time during the pandemic (March 2020–March 2022)?

What is your perception of the quality of distance learning training versus in-person training?

"In-person is superior training. There is value in distance learning, however, the collaboration and learning discussions that take place in-person are critical to me for internalization of the knowledge gained."

"In-person is always better, but I felt the virtual training was of good quality and valuable."

For those participating airport employees who were trained via distance learning, the survey asked, "If so, what is your perception of the quality of distance learning training versus in-person training?" Although responses were varied, several common themes emerged. Numerous participants believed in-person training to be more effective than training via distance learning.

At least one participant referenced the quality of training material and ability of the presenter. As one participant shared, "It's good, although highly reliant on the quality of the presentation material and energy of the presenter while being highly susceptible to being ineffective based on the trainee's degree of attention."

Other participating employees referenced the difficulty in staying focused and remaining engaged via distance learning. As one participant shared, "While instructors tried to be engaging, I felt that the 'human element' that was missing from the training was a determent to my overall learning. I felt it was very difficult to remain focused and engaged."

As one participant summarized, "Some training was not as effective as in person training, but was able to meet certification standards. The quality of training is dependent upon how much time was spent preparing for training sessions. For example: distance learning is more restrictive for preparations but is heavily dependent on presentation media (i.e., how well put together a PowerPoint is); vs in person training where there is more engagement with procedures, processes, and how to handle situations/equipment/systems."

To qualify candidates for the study, a qualifying question was asked: "Did you work remotely at any point during the COVID-19 pandemic (March 2020–March 2022)?" (see Figure 14). The vast majority of candidates completing the survey (78.08%, n = 57) did work remotely at some point during the COVID-19 pandemic. Only 16 (21.92%) of those receiving the survey and deciding to complete the survey did not work remotely at any time during the pandemic. This is evidence of the survey's intended reach.

Of those who did not work remotely, almost half (46.67%, n = 7) would have liked to, although, interestingly, more than half (53.33%, n = 8) would not have liked to work remotely.

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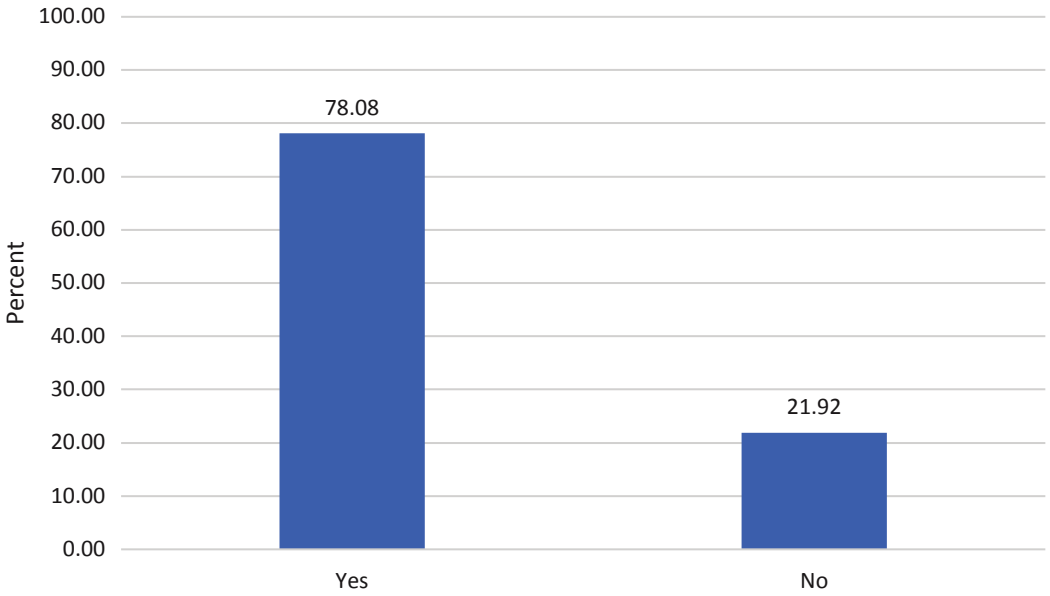


Figure 14. Did you work remotely at any point during the COVID-19 pandemic (March 2020–March 2022)?

(One of the participants who did not work remotely did not respond to the follow-up question about whether they would have liked to have worked remotely.) Among those employees who did work remotely at some point during the pandemic, 46.30% (n = 25) would prefer to continue working remotely. However, more than half (53.70%, n = 29) would like to return to the office (see Figure 15).

As the literature references "burnout" as a disadvantage of remote work, survey participants were asked, "To what degree have you experienced 'burnout' (physical or mental breakdown)

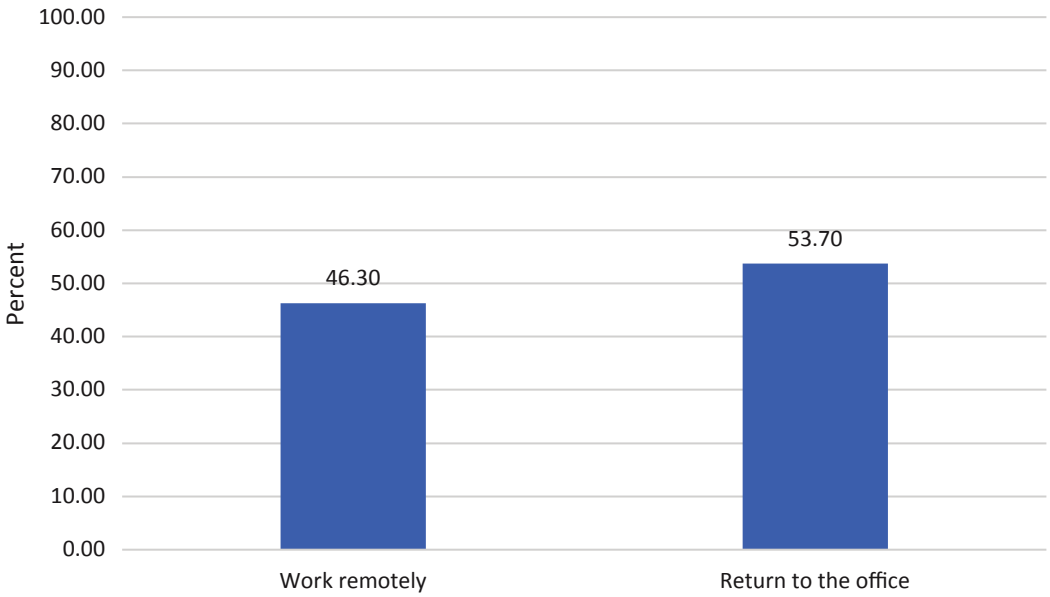


Figure 15. Which would you prefer?

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while working remotely?" Using a five-star rating system, the average response was 2.2 stars. Almost half of participants (46.15%, n = 24) responded "None." Unfortunately, 23.08% (n = 12) of participants reported "significant" or "extreme burnout."

To better understand benefits to remote working from the employee's perspective, participants were asked "What are the benefits to working remotely?" Although this open-ended question captured varied responses, several themes emerged:

- **Time Savings.** By avoiding the daily commute, additional time allowed time for hobbies, spending time with family, and other tasks. As one participant shared, "I get more done as I usually start when I wake up. I get to spend a lot more time with my family and mix my hobbies in my daily routine." As another participant shared, "Gained five hours from not having to commute."
- **Fewer Distractions.** For some, working from home was less distracting than the office. This resulted in greater productivity. As one participant shared, "I get more 'quiet work' done more quickly at home due to no interruptions and no 'get ready for work' time."
- **Reduced Expenses.** By driving less, remote employees saved money on fuel and incurred less wear and tear on their vehicle. As one participant shared, "No driving expenses."
- **Work-Life Balance.** With more available time, self-care became a priority, resulting in better work-life balance. As one participant shared, "More time for self-care like working out and walking the dog."

What are the challenges to working remotely?

"More difficult to engage with others. Loss of informal work discussions. Harder to monitor the work of subordinates."

"Some work physically can't be done remotely. More self-discipline is needed to stay on-task. Proper environment is needed to reduce distractions."

Realizing that remote work presents challenges (in addition to benefits), survey participants were asked, "What are the challenges to working remotely?" Responses were plentiful, with 53 participants responding to this question. Common responses include:

- Less community, less social interaction, feelings of isolation
- Difficulty in feeling productive, as well as colleagues and supervisors who question if you're productive
- No dedicated workspace
- Difficulty in unplugging from work
- Computer/wi-fi problems
- Distractions, including household chores and children

Because the literature discussed the challenge of maintaining a proper work-life balance with remote employees, the survey asked participants, "How were you able to maintain work-life balance while working remotely?" Common themes include:

- Leave work at a certain time
- Adhere to a workday routine
- Dedicated office space
- Setting boundaries for family not to disturb during work

One participant summed it up by sharing a formula: "More time at home = better work life balance."

To gauge employee perspectives on whether remote work is here to stay, survey participants were asked, "Do you feel working remotely is sustainable for the long-term?" (see Figure 16). Almost half (45.28%, n = 24) feel that remote work is sustainable for the long term. Although 24.53% (n = 13) are not sure, 30.19% (n = 16) do not feel that remote work is sustainable for the long term. This perspective closely mirrors the airport/HR manager perspective, with (as previously discussed), 48.08% (n = 25) of managers believing that remote work is sustainable for the long-term.

The literature referenced the challenge in training remote employees during the pandemic. To understand the degree to which airports experienced this challenge, survey participants

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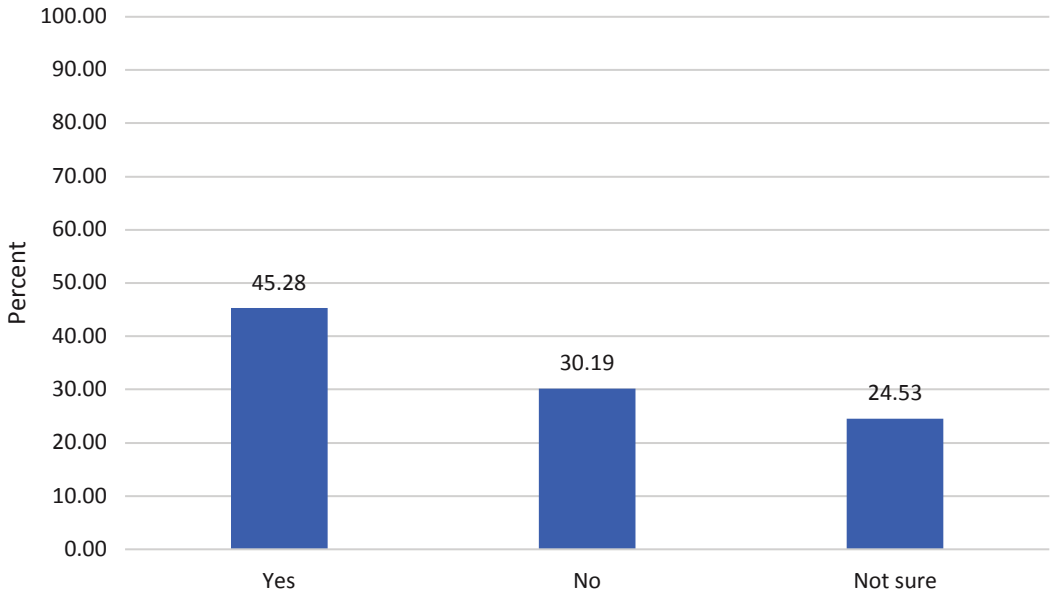


Figure 16. Do you feel working remotely is sustainable for the long-term?

were asked, "Do you feel that training is more difficult while working remotely?" (see Figure 17). Most of the participants (43.40%, n = 23) feel that training is more difficult while working remotely. However, 30.19% (n = 16) feel that training was not more difficult while working remotely. Just over one-quarter (26.42%, n = 14) were unsure.

Participating airport employees were also asked, "Do you feel that innovation is more difficult while working remotely?" Just over half (50.94%, n = 27) of participating airport employees feel that innovation is more difficult while working remotely (see Figure 18).

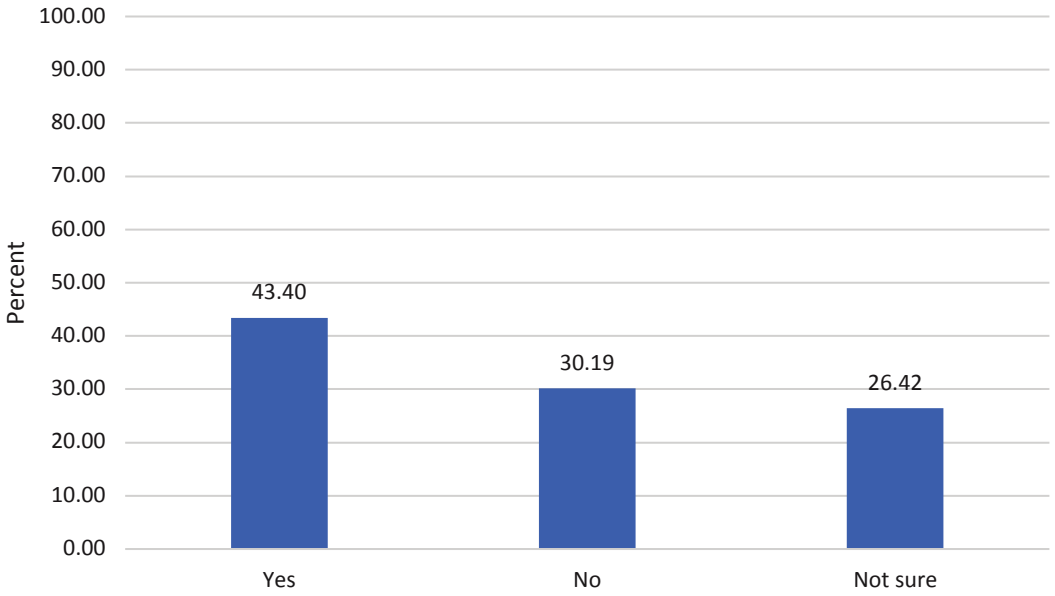


Figure 17. Do you feel that training is more difficult while working remotely?

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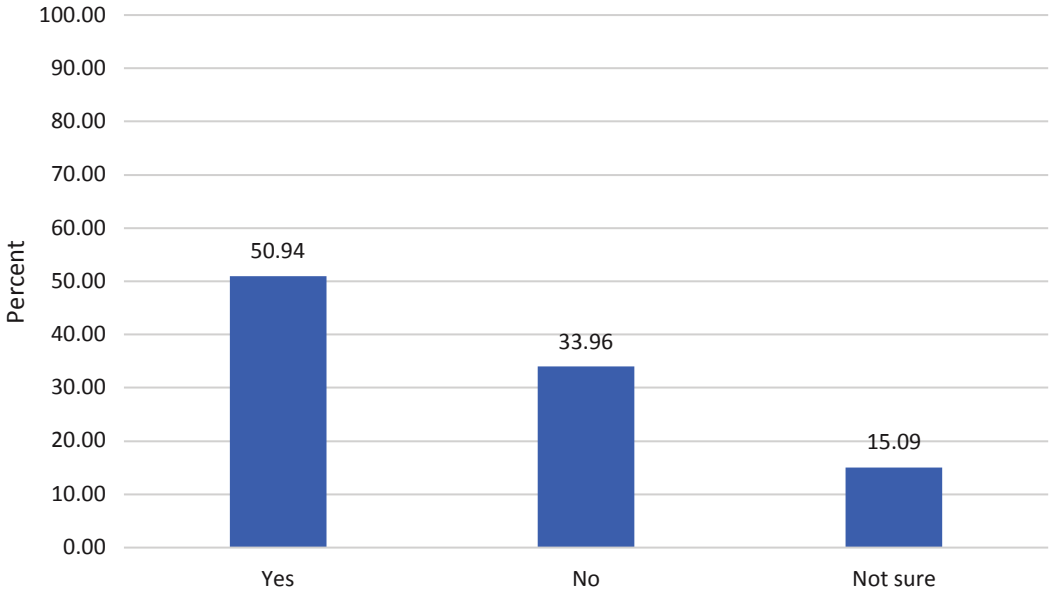


Figure 18. Do you feel that innovation is more difficult while working remotely?

To determine if airport employees think their career progression might be negatively affected by remote work, participants were asked, "Do you feel your career might suffer while working remotely?" In general, participants do not believe their careers will suffer while working remotely (47.17%, n = 25). However, one-third of participants (33.96%, n = 18) worry that their career might suffer while working remotely. A few (18.87% n = 10) were unsure (see Figure 19).

To determine how remote work may have affected esprit de corps, participants were asked, "Do you feel it is difficult to add new members to your team/department while working remotely?" The vast majority (71.70%, n = 38) do feel that new employee integration is difficult

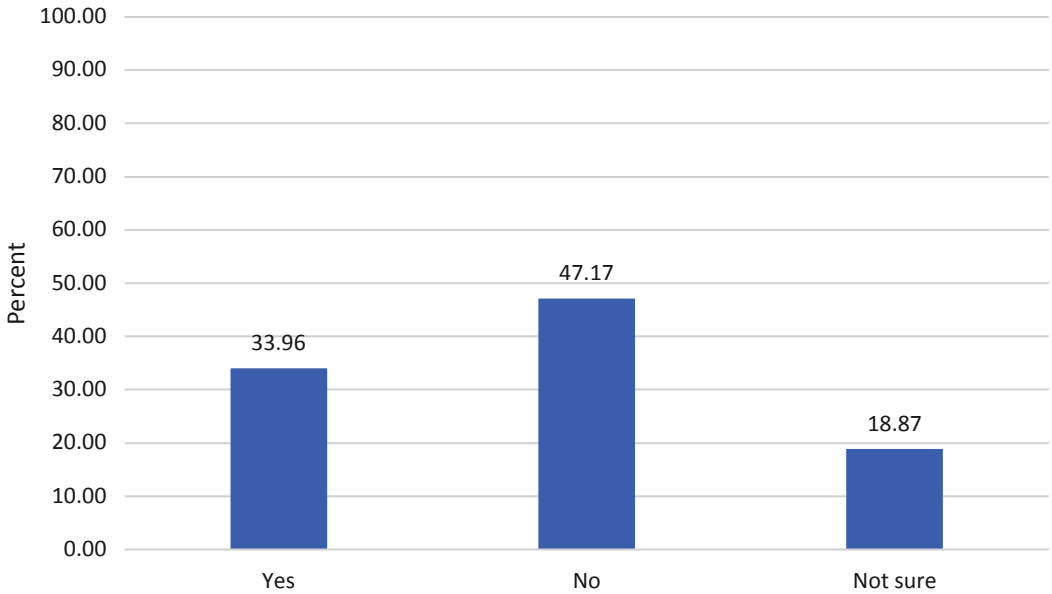


Figure 19. Do you feel your career might suffer while working remotely?

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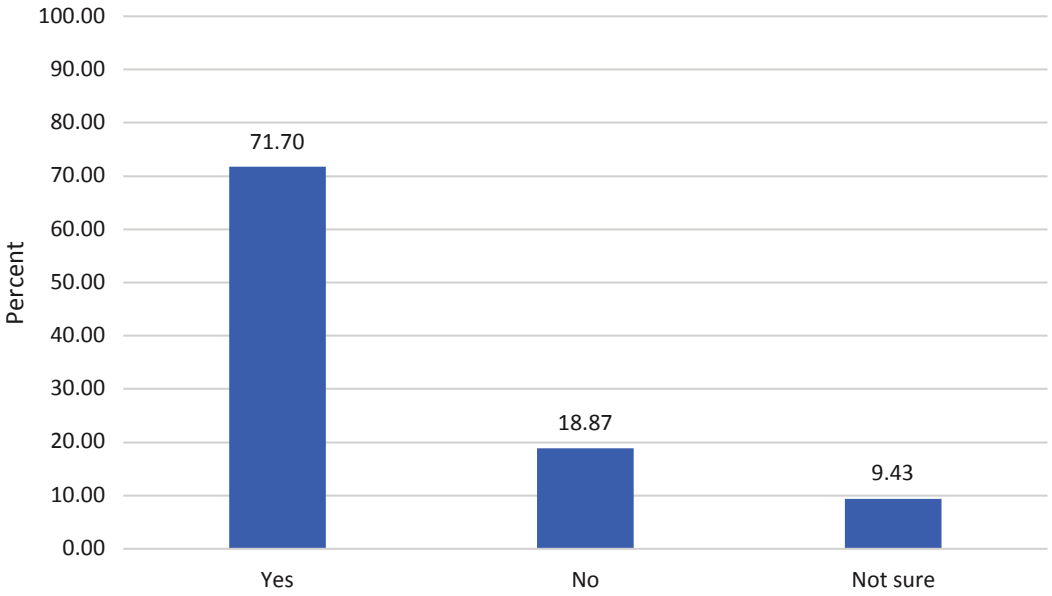


Figure 20. Do you feel it is difficult to add new members to your team/department while working remotely?

while working remotely. A few participants (18.87%, n = 10) do not feel that it is difficult to add new employees to the team/department while working remotely (see Figure 20).

Employees expect that managers will be available for them when needed. However, employees working remotely are physically distanced from management. The survey asked participants, "Do you feel that your manager is less available while working remotely?" Although 56.60% (n = 30) of survey participants do not feel that their manager is less available while working remotely, 39.62% (n = 21) do feel their manager is less available (see Figure 21).

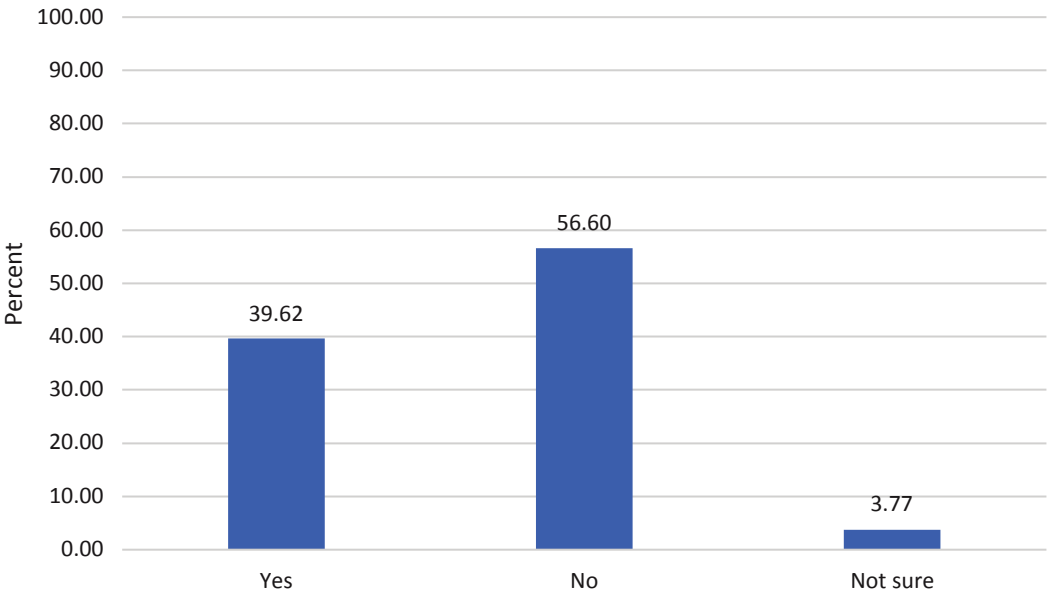


Figure 21. Do you feel that your manager is less available while working remotely?

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To gauge whether airports were intentional about maintaining workplace culture with remote workers during the pandemic, participants were asked, "Do you feel your airport has engaged in team and culture-building exercises while workers are remote?" Unfortunately, 47.17% (n = 25) of participants indicate their airport did not engage in team and culture-building exercises while workers were remote. That being said, 37.74% (n = 20) of participants indicate their airport did engage in team and culture-building exercises while workers were remote (see Figure 22).

To confirm whether one of the well-recognized disadvantages of remote work, diminished social interaction with co-workers, really affected airport employees, participants were asked, "Do you feel you are socializing less with co-workers while working remotely?" Unfortunately, 84.91% (n = 45) of participants felt they were socializing less with co-workers while working remotely. Only 13.21% (n = 7) felt they were not socializing less with co-workers while working remotely (see Figure 23).

Similarly, participants were asked, "Do you feel that morale among your team has suffered while working remotely?" Although 45.28% (n = 24) of participants indicated that morale did not suffer, 37.74% (n = 20) of participants indicated that morale among team members did suffer while working remotely (see Figure 24).

Participants were also asked, "Do you feel your team's culture has suffered while working remotely?" Although 48.08% (n = 25) felt their team's culture did not suffer, 42.31% (n = 22) did feel their team's culture suffered while working remotely (see Figure 25).

To provide survey participants the opportunity to share insight into how they may have done things differently during the pandemic, participants were asked, "If you were in charge of the organization/department during the pandemic, what would you have done differently, if anything?" This was an interesting question, as it provided participating employees the opportunity to "assume" a managerial role and suggest how things could have been done differently. In total, 43 responses were received, addressing numerous improvements. Some feel that remote work should continue. Some would have better supported remote employees. Others would have provided better communication about health and wellness opportunities. Acting upon employee

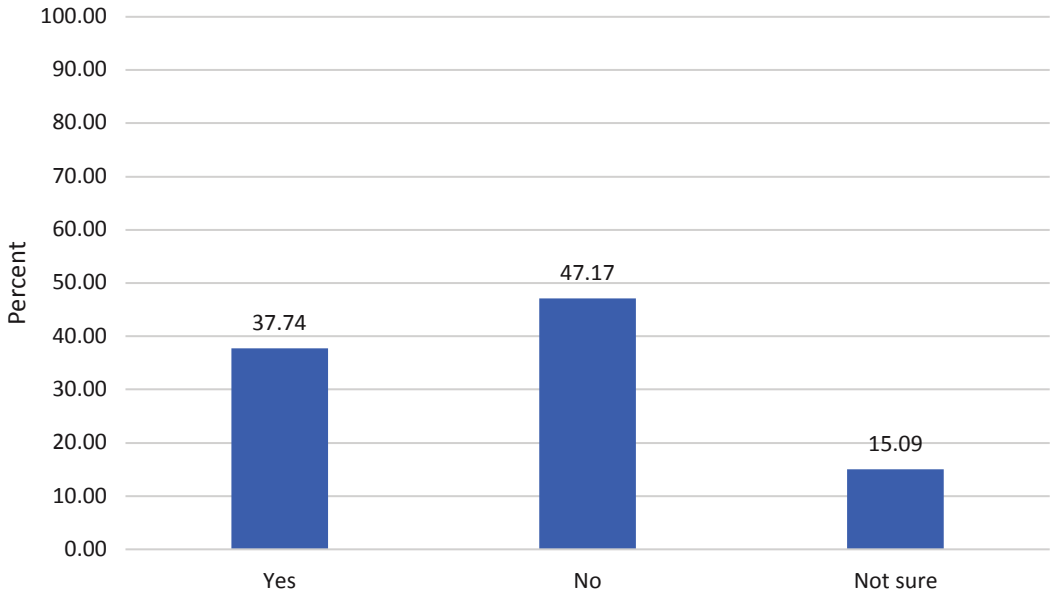


Figure 22. Do you feel your airport has engaged in team and culture-building exercises while workers are remote?

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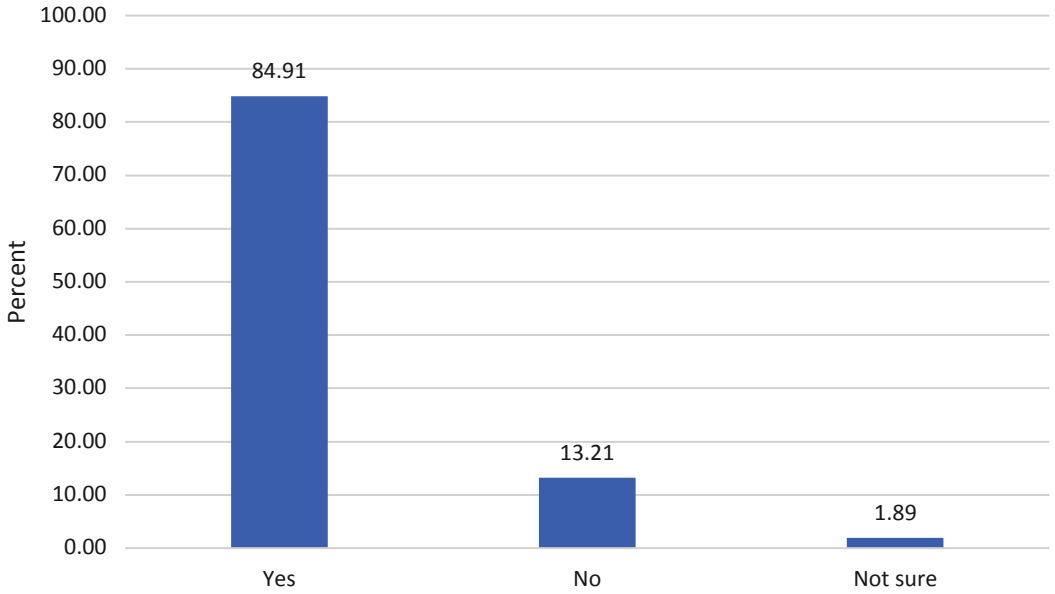


Figure 23. Do you feel you are socializing less with co-workers while working remotely?

feedback was also referenced as important. Some of the specific suggestions for handling their workplace's response to the pandemic offered by employees include:

- "I would have continued to allow a hybrid schedule so that people could continue to have some limited work from home options if they desired."
- "I would have increased interaction and made sure that all team members had the tools and equipment available at home for remote work, including ergonomic chairs, desks, etc."

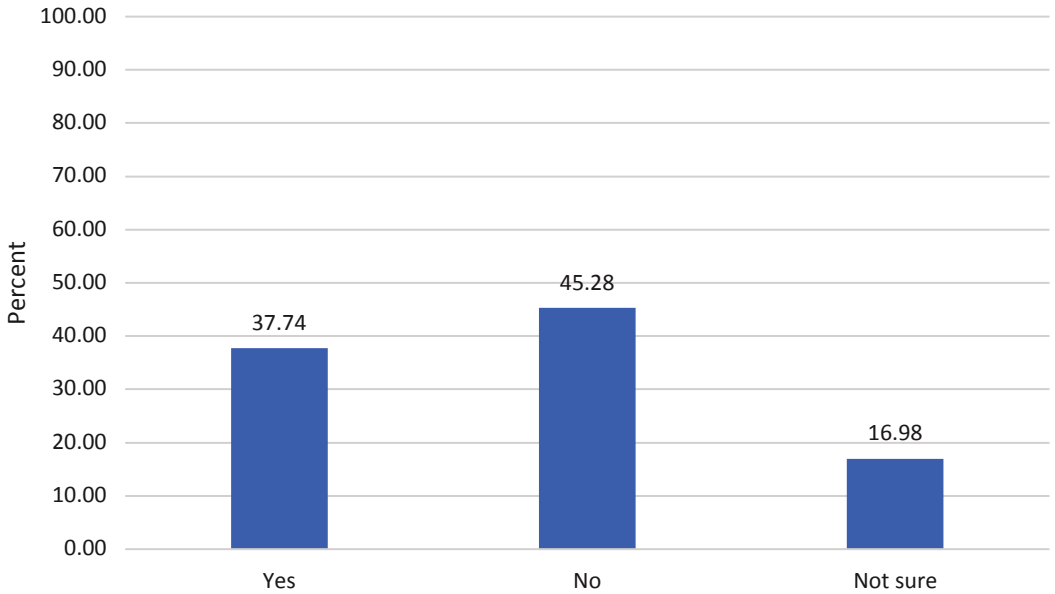


Figure 24. Do you feel that morale among your team has suffered while working remotely?

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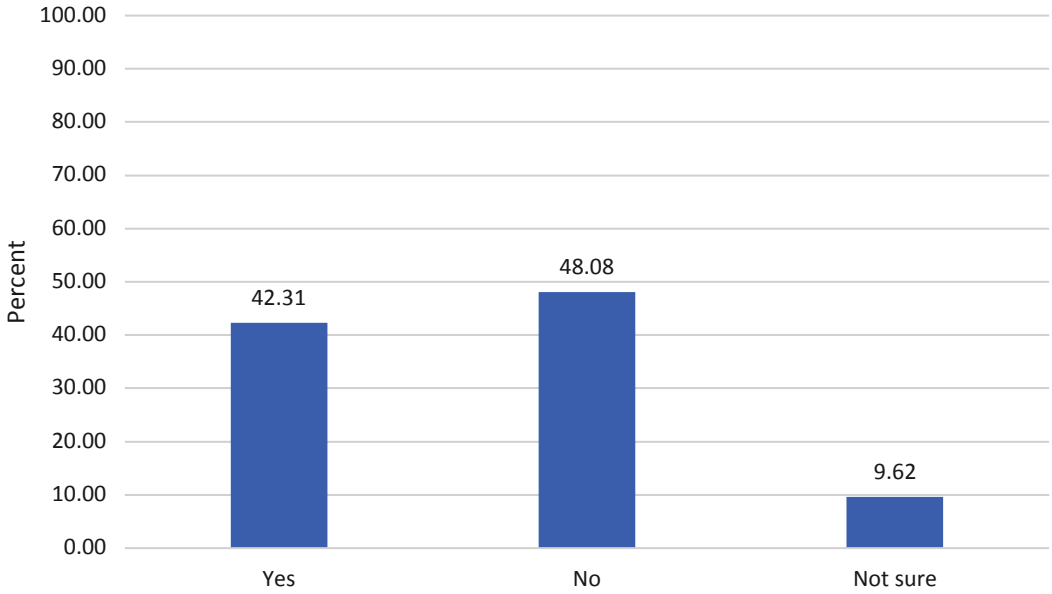


Figure 25. Do you feel your team's culture has suffered while working remotely?

- "Provided more and consistent communication about health and wellness opportunities. Though I did not experience this need, many did, and I feel were unaware of others feeling the same or of services offered."
- "Accepted the feedback received from surveys and had an informal dialogue to find a common ground to encourage and upbuild the morale first and then promote new and innovative opportunities for the organization in the future."

Participants were provided an opportunity to share any additional thoughts. Noteworthy comments include:

- "I believe the modern-day culture is anything but modern. From schedules, to work/life balance, to how people are compensated, including employer's flexibility in exchanging insurance and vacation benefits for wages at the individual level. We have a long way to go on modernizing our business staffing models!"
- "Airports should continue recognizing the many benefits to remote work and continue supporting their employees working remotely as much as possible, as it will INCREASE morale since it allows a better work/life balance."
- "Remote work is a valuable option necessary to attract talent. Employers should embrace this option or create hybrid work options when necessary or dictated by the job. Airports should not deny remote work for a department because another dept. cannot work remote (i.e., maintenance, ops, public safety). Generally, remote work would also make additional terminal space available to rent to tenants."

Overall, these comments were supportive of continued remote work and alternate work arrangements.

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CHAPTER 5

Case Examples

To better understand unique working arrangements at airports, specific airports were selected for additional follow-up and are showcased here as case examples.

Case Example 1—Atlanta Regional Airport, Falcon Field

Location: Atlanta, Georgia
Airport Identifier: FFC
FAA Region: Southern
Airport Hub Size: GA
Acreage: 350
Based Aircraft: 198
2021 Annual Aircraft Operations: 76,000

"At our GA airport, everything is very hands on, making remote work only appropriate for very few employees." – Airport Manager, Atlanta Regional Airport, Falcon Field

Owned and operated by the Peachtree City Airport Authority, the Atlanta Regional Airport is staffed by 18 employees, including four employees in customer service, two in airfield maintenance, four in administration, and eight in line service. This general aviation airport also owns and operates the fixed-base operator (FBO). During the COVID-19 pandemic, the airport never terminated, furloughed, or decreased the work hours of any employee. Additionally, the airport did not have any turnover during the pandemic. Even though the aviation industry slowed during the pandemic, "people were still flying, we were receiving fuel calls, and other phone calls," according to the Airport Manager. Although the terminal building was closed for three weeks at the height of the pandemic, airport staff continued working on other projects during this time. Once the terminal building reopened, a plexiglass screen was put in place for customer service personnel. Employees and customers were encouraged to maintain social distance. Only two employees worked remotely at any time during the pandemic. The bookkeeper worked remotely for several months, bringing in checks to sign on a regular basis. The Assistant Airport Manager worked remotely for several weeks to care for a newborn (personal communication, August 24, 2022).

Case Example 2—Albert Ellis Airport

Location: Jacksonville, North Carolina
Airport Identifier: OAJ
FAA Region: Southern
Airport Hub Size: Non-hub
Acreage: 675
Based Aircraft: 26
2021 Annual Aircraft Operations: 19,425

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"We continue to hold some virtual meetings and focus on quality of work life for employees to enhance airport employee retention." – Airport Director, Albert Ellis Airport

Owned and operated by Onslow County, the non-hub Albert Ellis Airport has 32 full-time equivalent employees. During the COVID-19 pandemic, the Airport Director offered the ability to work remotely for certain employees, such as administrative staff (e.g., Finance Manager and Accounting Clerk). However, staff were reluctant to work remotely and chose instead to remain on-site and in-person. The employees eligible for remote work expressed concern with disconnecting from co-workers, as well as distractions at home (i.e., children). Employees in certain departments, including Police/Fire and Maintenance, were not eligible for remote work due to the hands-on nature of these positions. At the height of the pandemic, the airport permanently raised the gates to the airport parking lot. Parking cashiers were then reassigned to custodial duties in the terminal. According to the Airport Director, Onslow County has encountered significant employee retention challenges and was not immune to the "Great Resignation." In 2021, the County had 1,300 employees, with 560 of these employees resigning over a two-year period. Even with 10% Cost of Living Adjustment pay raises for county employees (on top of 5% last year), the County, especially for skilled trades, is competing with other employers, such as Marine Corps Camp Lejeune and McDonald's (personal communication, August 24, 2022).

Case Example 3—Denver International Airport (non-management airport employee)

Location: Denver, Colorado
Airport Identifier: DEN
FAA Region: Northwest Mountain
Airport Hub Size: Large
Acreage: 33,531
Based Aircraft: 1
2021 Annual Aircraft Operations: 588,855

"We are designing a new office facility that will support employees working in a hybrid arrangement. This facility includes the ability for on-site employees to connect with remote employees, such as a variety of environments for shared spaces. By including a variety of small areas to either have private conversations or group conversations with a coffee lounge feel, we're hopeful that all employees will feel a sense of belonging." – Airport Employee, Denver International Airport

The interviewee, who is an Airside Engineering Supervisor at Denver International Airport, worked remotely (either fully or in a hybrid arrangement) from March 2020 to October 2021. According to this person, "Engineers did not have to be on-site all the time. Even when a site visit was necessary, the Engineer worked in the field, easily maintaining social distance." Even though the pandemic is now over, the interviewee continues working in a hybrid arrangement. She works two days in the office, and three days from home. At Denver International, which is a city-owned and -operated airport, supervisors have discretion as to which employees are considered essential and must remain on-site and in-person and which employees are considered non-essential and may work in a fully remote or hybrid work arrangement. She feels that one of the most significant benefits of remote or hybrid work is enhanced sustainability due to lower carbon emissions with fewer commuters on the road. Even so, based on her experience, the interviewee feels that fully online is not the most effective arrangement for employees in design and construction. "I believe a hybrid model is more appropriate." The airport hopes innovative design makes the work environment more comfortable and appealing. "Alternating when employees are in the office allows for more flexible space utilization" (personal communication, August 24, 2022).

Case Example 4—Texas Gulf Coast Regional Airport

Location: Angleton, Texas
Airport Identifier: LBX
FAA Region: Southwest
Airport Hub Size: Reliever
Acreage: 674
Based Aircraft: 95
2021 Annual Aircraft Operations: 77,981

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With 13 employees, Texas Gulf Coast Regional Airport, located south of Houston, Texas, maintained essential operations during the COVID-19 pandemic. Even at the height of the pandemic, there were always a customer service representative, a line technician, and several supervisors on duty at the airport. The airport rotated employees to minimize the number of employees present at any one time. The airport adopted standard COVID-19 protocols to protect both customers and employees, including requiring face coverings, and encouraging social distancing and use of sanitization stations. The Airport Director was focused on ensuring fairness among all employees during the pandemic, making sure not to allow only some employees to work remotely, and not others. The Airport Director suggests that all airport employees are needed on-site. The Airport Director also desired to maintain pre-pandemic levels of productivity and was concerned that remote employees would be challenged to do that. As the Airport Director shared, "There is a home environment and a work environment. They are not the same." (Personal communication, August 25, 2022).

"If certain employees need not be at the airport and can work from home, we must ask the question, 'Are we staffed correctly'?" – Airport Director, Texas Gulf Coast Regional Airport

Case Example 5—Orange County (John Wayne) Airport

Location: Santa Ana, California
Airport Identifier: SNA
FAA Region: Western Pacific
Airport Hub Size: Medium
Acreage: 504
Based Aircraft: 367
2021 Annual Aircraft Operations: 315,838

At the start of the pandemic, the leadership of Orange County Airport expressed concern about COVID-related illness among employees and passengers. Although some employees stayed home for a period of time, such as those with compromised immune systems or those with child-care issues related to closed schools, the Airport Director's main priority was to get everyone safely back to work. First, all airport employees were classified as "essential." This decision was made to ensure that no airport employees would be subject to county layoffs of "non-essential" personnel during the pandemic, to prevent employees from feeling unimportant, and to support the notion that all employees could be on-site. For employees in departments such as Police, Operations, and Maintenance, working from home was not an option, as their jobs could only be performed at the airport. Another concern was the level of productivity with at-home employees. Consideration of this dilemma motivated the Airport Director to get all employees back to work at the airport. "We all need mutual accountability, which is difficult in a remote work environment." Although COVID-19 surprised all airports, creating far-reaching and cascading secondary impacts, it is difficult to create exact plans for a future crisis. In general, the Airport Director explained, airports should be prepared for business interruptions, but this will likely require on-the-spot decisions and the ability to pivot based on the specific circumstances (personal communication, August 25, 2022).

"How can an airport offer one group of employees a benefit [i.e., working from home] that other employees don't get?" – Airport Director, Orange County (John Wayne) Airport

Case Example 6—Tampa International Airport

Location: Tampa, Florida
Airport Identifier: TPA
FAA Region: Southern
Airport Hub Size: Large
Acreage: 3,300
Based Aircraft: 79
2021 Annual Aircraft Operations: 212,973

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"By establishing a critical emergency management foundation well in advance of this unexpected threat, Tampa International has remained responsive despite these unprecedented circumstances." – Executive Vice President of Airport Operations and Customer Service, Tampa International Airport

The Executive Vice President of Airport Operations and Customer Service at Tampa International Airport (TPA) explains how the airport developed a "playbook" to ensure the airport remained open and operational for essential travel, while ensuring the health and resilience for each employee. This "playbook," he explained, included an A and B team of each workgroup, "identifying and placing key employees in a reserve status working from home as a back-up capable of stepping into critical functions, such as the airport operations center, if needed due to an outbreak occurring at the airport." To minimize overall risk to COVID-19, by March 16, 2020, TPA had moved all non-essential employees to remote work, including departments such as marketing, procurement, and finance. This "was made possible by the rapid response of the airport's IT department along with an in-place, but not broadly adopted, work-from-home policy." The use of Microsoft Teams and Box was crucial for enabling the productivity of remote employees. Essential employees (those required to work on-site due to their job function) included employees in airfield and terminal operations, airport operations center, maintenance, guest services, and concessions. TPA implemented a number of measures to ensure the health and safety of these essential employees, including having cleaning supplies readily available, following social distancing guidelines, separating team members where possible, and identifying alternate staff to cover for anyone unable to work. (Personal communication, August 26, 2022; Tiliacos 2020).

Case Example 7—Punta Gorda Airport

Location: Punta Gorda, Florida
Airport Identifier: PGD
FAA Region: Southern
Airport Hub Size: Small
Acreage: 1,934
Based Aircraft: 369
2021 Annual Aircraft Operations: 87,599

"Collaborative work best gets done face-to-face in conference rooms and office meetings. MS Teams was not as effective for us as face-to-face interaction." – CEO, Punta Gorda Airport

Located on the west coast of central Florida, the Punta Gorda Airport has 369 based aircraft and serves as a small-hub airport. When the COVID-19 pandemic first began, all airport employees that could work off-site (such as Accounting and Purchasing) were sent home. Personnel in Operations, Maintenance, Security, etc., as well as the Airport Director, continued working on-site at the airport due to the on-site nature of their work. As the pandemic continued, the airport examined employee requests to work from home on a case-by-case basis. For instance, as schools were closed, those employees with school-aged children either had to arrange alternate care for their children or remain at home to care for them and oversee their remote learning. The airport learned, however, that it was difficult to make accommodations for some employees and not others. This benefit could not be offered to everyone, as certain airport functions can only be performed on-site. Further, the airport discovered drawbacks to allowing some employees to work from home. The airport used the COVID-19 pandemic as an opportunity to update their work-at-home policy. Currently, the airport does not allow work-at-home unless the employee has a specific situation that requires unique accommodations. It is the airport's belief that the airport and employees most benefit when all employees are on-site and working collaboratively together (personal communication, August 30, 2022).

Case Example 8—Philadelphia International Airport

Location: Philadelphia, Pennsylvania
Airport Identifier: PHL
FAA Region: Eastern
Airport Hub Size: Large

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Acreage: 2,302
Based Aircraft: 25
2021 Annual Operations: 268,884

"Communication is key. Even though it evolved during the pandemic, we intentionally maintained high levels of communication with all employees–whether on-site or remote" – HR Manager, Philadelphia International Airport

Owned and operated by the City of Philadelphia, the Philadelphia International Airport is a large-hub airport with more than 15 million passenger enplanements annually. As the COVID-19 pandemic began, the airport considered which employees could work from home. It quickly determined that the vast majority of employees (approximately 80%) would be required to continue working on-site at the airport. Personnel in Operations, Facilities, and various trades could only perform their job functions on-site. According to the HR manager, determining which employees were eligible "depended on the type of work they performed. Can their work be performed remotely?" Those employees that could work off-site were initially eligible for 100% remote work. These employees received training in use of technology, logging work time (clocking in), etc., to support continued productivity—even in a remote setting. As the pandemic evolved, so did employee work arrangements. Shifting to a hybrid model allowed those employees who had been working 100% remote the ability to work on-site several days per week, which increased employee interaction and productivity. The pandemic required the airport to refine and improve their existing work-from-home policy. Moving forward, the airport will continue to offer remote work options for certain groups of employees, which has also aided with recruiting new employees who are more interested in flexible work arrangements. The airport's remote work experiment during the pandemic did have some challenges. Most significantly, the airport learned that remote employees often feel lonely and disconnected form colleagues. In an effort to be responsive to employee needs, the airport held open houses, providing opportunities for remote employees to connect with supervisors and colleagues. Weekly virtual sessions were held to keep staff updated on airport happenings. Paper letters were mailed periodically to employees to keep everyone up-to-date on airport/HR events. Virtual grief counseling sessions were arranged with employees who needed additional support for their emotional and mental well-being (personal communication, August 30, 2022).

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CHAPTER 6

Conclusions and Future Research

This report is intended to present a summary of current practices regarding remote work and the evolution of work strategies at airports. Conclusions can be drawn from the responses gathered to arrive at a state of current practice and provide insight to airports moving forward. These conclusions are the topics of this chapter.

1. **Eligibility for remote work.** Airports commonly determine employee eligibility for remote work based on the type of work employees perform. Some employees remain on-site and in-person due to the nature of their work and are classified as essential, such as those conducting airfield inspections, ensuring security, and responding to emergencies). Other employees can perform tasks remotely (at home or other off-site locations) and are classified as non-essential, such as those processing accounts payable, managing contracts, and maintaining information technology resources.
2. **Technological tools.** Rapid advances in cloud computing and technological tools such as webcams enabled significant numbers of airport employees to work remotely during the pandemic.
3. **Business continuity.** Whether or not an airport had an existing business continuity or crisis plan, airports were confronted with a crisis caused by the COVID-19 pandemic. The pandemic created a sense of urgency at airports nationwide to consider how to continue operating amid these challenges. Some airports have created a business continuity or crisis plan and have an emergency team structure in place to ensure they are equipped to handle future crises that may cause significant disruption of the business.
4. **Essential/Operational employees.** Airports benefit from determining a team of essential employees that need to remain at the airport during significant events, including a pandemic. Deciding who is on this team ahead of time minimizes the need to make these staffing decisions "in the moment."
5. **Back to the office.** Now that the pandemic is officially over, numerous airports no longer allow employees to work remotely and are requiring all employees to be back in the office. Leadership at these airports generally see greater benefits in face-to-face interaction.
6. **Remaining remote.** Some airports, recognizing the benefits of remote work to both airports and employees, are continuing to allow "non-essential" employees to work remotely. These airports often have unique considerations for new employees, possibly requiring the new employees work on-site and in-person for the first 9 months to one year before being eligible for remote work, for example.
7. **Long-term accommodation of remote employees.** As half of employees that have worked remotely at airports may like to continue working remotely, airports may consider how to accommodate these employees in a remote or hybrid fashion for the long term; otherwise, employee retention could be negatively impacted.
8. **Enhanced hiring practices.** Numerous airports have altered hiring practices for the foreseeable future to allow flexible options, such as conducting virtual interviews.

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9. **Generational similarities.** Generally, there is no difference among various generations regarding their level of success with remote work.
10. **Sustainability of remote work.** Almost half of those airports experienced with remote work feel that remote work is sustainable for their airport years into the future. Almost half of employees with remote work experience feel that remote work is sustainable for the long term. However, fewer than 20% of airports experienced with remote work actually have plans to support long-term remote work. Thus, the feasibility of long-term remote work does not correspond to the plans to maintain long-term remote work for employees.
11. **Employee training expectations.** Employees have high expectations for training delivered at a distance. Just over 40% of remote employees feel that training is more difficult while working remotely. Oftentimes, airports fell short in this regard, leaving employees to rate in-person training as more effective. Those airports receiving high marks from employees with distance training were intentional about delivering high-quality virtual training using motivated presenters, "live" webinars, and opportunities for employees to engage in the virtual environment.
12. **Benefits.** Generally, burnout was not experienced by airport employees working remotely. Employees with experience working remotely speak highly of numerous benefits, including less time commuting, more time with family, more time for hobbies, less stress, and, in general, better work-life balance.
13. **Challenges.** Employees with experience working remotely also speak of challenges, including less face-to-face interaction with co-workers and supervisors, distractions at home, and difficulty in accessing "paper" files at work. More than eight out of ten airport employees with some remote work experience feel they socialize less with co-workers while working remotely. Just over one-third of airport employees with remote work experience feel that morale among their team has suffered while working remotely. Four out of ten feel their team's culture has suffered while working remotely.
14. **Innovation.** In the dynamic airport industry, innovation can be a source of competitive advantage. However, more than half of remote employees report that innovation is more difficult while working remotely. Although a remote employee may have fewer distractions and spend less time commuting, remote employees also report having less interaction with colleagues, making in-person collaboration all but impossible. Perhaps innovation is more likely when in discussion and collaboration with colleagues.
15. **Career progression.** Generally, employees working remotely are not concerned that remote work will hamper their future career progression.
16. **New employee integration.** Although remote work has numerous benefits, employees report that integrating new employees onto their team while remote is difficult. Indeed, three-quarters felt that it is difficult to add new members to their team while working remotely.

Further Research

As the nature of work and ideas about what defines a workplace continue to evolve, the airport industry would benefit from continued research on the topic of work arrangements that include remote work, hybrid work, and in-person work. In the future, research will be less focused on pandemic-related remote work and more focused on innovative work arrangements and recruiting practices to support airport staffing needs and airport succession planning.

1. Future research on the use of remote/hybrid work arrangements will be helpful, especially considering that many participating airports believe that remote work is sustainable for the long term.
2. Airport staffing needs, including recruiting, selection, and retention, would be a worthy topic of future research.

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3. Research into airport work policies, including flexible schedules and new employee work incentives, would be appropriate.
4. Future research into how airports compete for the workforce with other businesses that allow employees to work remotely would be useful.
5. Research into airport irregular operations plans and business continuity plans and how these types of plans were used during the pandemic and can be used to prepare for future business disruptions would be insightful.

Research into each of these topics will provide insight into current airport practices to address staffing shortages, retention challenges, and flexible work arrangements. Once the pandemic is several years behind us, it would be helpful to know the degree to which flexible arrangements that began during the pandemic continue to exist, even in alternate models.

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APPENDIX A

Airport Manager/HR Manager Survey

Airport Work Models - Airport Manager/HR Manager Survey - ACRP S06-07

Intro & Consent

This survey is being conducted to collect data in association with the Airport Cooperative Research Program (ACRP) Synthesis project – Evolution of Airport Work Models and Strategies (S06-07). This project, which is funded by the National Academies as part of the Transportation Research Board, is designed to collect data regarding the remote work and remote learning/training practices in use by airports in response to the COVID-19 pandemic. The final report will highlight remote work and training practices for airport employees, with innovative practices presented as case examples.

Your responses will remain confidential. Final data will be reported in aggregate and your airport will only be identified as a possible case example of best practices or in a list of responding airports. If you choose to participate, you are free to withdraw from the study at any time without consequence or penalty.

The survey should take 10-15 minutes of your time. By clicking OK, you agree to participate. Any follow-up emails or calls will be to help clarify a response, respond to any questions, and to help us achieve a satisfactory response rate. You may also be contacted for more information if we choose to highlight your airport in a case example.

Thank you-
C. Daniel Prather, Ph.D., D.B.A., A.A.E., CAM
Principal Investigator
DPrather Aviation Solutions, LLC

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Airport Work Models - Airport Manager/HR Manager Survey - ACRP S06-07

Demographics

1. Is your airport FAA Part 139 certificated?

- ◯ Yes
- ◯ No

2. What category is your airport?

- ◯ Large hub
- ◯ Medium hub
- ◯ Small hub
- ◯ Non hub
- ◯ Nonprimary commercial service
- ◯ GA
- ◯ Reliever

3. In which FAA region is your airport located?

- ◯ Alaskan
- ◯ Central
- ◯ Eastern
- ◯ Great Lakes
- ◯ New England
- ◯ Northwest Mountain
- ◯ Southern
- ◯ Southwest
- ◯ Western Pacific

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4. At any time during the COVID-19 pandemic (March 2020-March 2022) did your airport allow (or require) any airport employees to work remotely for any period of time?

- ◯ Yes
- ◯ No

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5. If not, please describe the managerial decision-making process that determined remote work would not be necessary.

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Airport Work Models - Airport Manager/HR Manager Survey - ACRP S06-07

6. Which employee groups were required to work remotely at any time during the pandemic?

- [ ] Airfield Operations
- [ ] ARFF
- [ ] Concessions
- [ ] Environmental/Noise
- [ ] Finance
- [ ] Ground Transportation
- [ ] Guest/Customer Service
- [ ] Governmental/Legal
- [ ] Information Technology
- [ ] Maintenance
- [ ] Marketing
- [ ] Planning and Design
- [ ] Public Relations
- [ ] Police/Public Safety
- [ ] Procurement
- [ ] Properties/Contracts/Commercial Development
- [ ] Terminal Operations
- [ ] Other (please specify)

7. How did you choose the employee groups eligible for remote work?

8. Which employee groups did your airport consider essential, and thus in-eligible for remote work?

- [ ] Airfield Operations
- [ ] ARFF
- [ ] Concessions
- [ ] Environmental/Noise
- [ ] Finance
- [ ] Ground Transportation
- [ ] Guest/Customer Service
- [ ] Governmental/Legal
- [ ] Information Technology
- [ ] Maintenance
- [ ] Marketing
- [ ] Planning and Design
- [ ] Public Relations
- [ ] Police/Public Safety
- [ ] Procurement
- [ ] Properties/Contracts/Commercial Development
- [ ] Terminal Operations
- [ ] Other (please specify)

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9. What technology tools enabled employees to perform their jobs in a remote manner?

10. What considerations are there for new employees regarding possible remote work arrangements?

11. What are some methods your airport used to manage employees working remotely?

12. What percentage of your airport employees would like to continue working remotely?

0 Percentage 100

13. How have your hiring practices changed since, and as a result of, the COVID-19 pandemic?

14. How has your overall HR function changed since, and as a result of, the COVID-19 pandemic?

15. During your experience with remote work, did you notice one generation of employees were more efficient/productive when working remotely? (If so, which?)

- [ ] No/None
- [ ] Baby Boomer (57-75 years of age)
- [ ] Generation X (41-56 years of age)
- [ ] Millennial (26-40 years of age)
- [ ] Generation Z (25 years of age and younger)

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16. Do you feel a remote work model is sustainable for your airport, even years into the future?

- ◯ Yes
- ◯ No

Why/Why Not?

17. Does your airport have a long-term plan for remote work?

- ◯ Yes
- ◯ No

Please explain:

18. Morale took a hit during the pandemic, what "fun" thing did your airport do that you can share?

19. Please share lessons learned from your airport's remote work experience.

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20. Were employees trained (either initially or recurrently) via either fully online asynchronous or hybrid/blended approach at any time during the March 2020-March 2022 time period?

- ◯ Yes
- ◯ No

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21. Please describe how employees were trained during the pandemic.

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22. Do you have any other thoughts to share on this topic?

Airport Work Models - Airport Manager/HR Manager Survey - ACRP S06-07

Contact

23. We would like to highlight unique case examples in the report. Would you be willing to discuss your airport's remote work and training arrangements via the phone for this project? If so, please indicate your name, airport, email address, and phone number. Thank you!

Name

Airport

Email Address

Phone Number

If you would be willing to share your remote work policy, please email it. If you would like your submission considered anonymously, please indicate. Thank you for your time in completing this survey!

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APPENDIX B

Airport Employee Survey

Airport Work Models - Airport Employee Survey - ACRP S06-07

Intro & Consent

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Your responses will remain confidential. Final data will be reported in aggregate and your airport will only be identified as a possible case example of best practices or in a list of responding airports. If you choose to participate, you are free to withdraw from the study at any time without consequence or penalty.

The survey should take no more than 10 minutes of your time. By clicking OK, you agree to participate. Any follow-up emails or calls will be to help clarify a response, respond to any questions, and to help us achieve a satisfactory response rate. You may also be contacted for more information if we choose to learn more about your personal experience with remote work and remote learning/training.

Thank you-
C. Daniel Prather, Ph.D., D.B.A., A.A.E., CAM
Principal Investigator
DPrather Aviation Solutions, LLC

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Airport Work Models - Airport Employee Survey - ACRP S06-07

Demographics

1. Is your airport FAA Part 139 certificated?

- ◯ Yes
- ◯ No

2. What NPIAS category is your airport?

- ◯ Large hub
- ◯ Medium hub
- ◯ Small hub
- ◯ Non hub
- ◯ Nonprimary commercial service
- ◯ GA
- ◯ Reliever

3. In which FAA region is your airport located?

- ◯ Alaskan
- ◯ Central
- ◯ Eastern
- ◯ Great Lakes
- ◯ New England
- ◯ Northwest Mountain
- ◯ Southern
- ◯ Southwest
- ◯ Western Pacific

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4. What is your age group?

- ◯ Under 18
- ◯ 18-24
- ◯ 25-34
- ◯ 35-44
- ◯ 45-54
- ◯ 55-64
- ◯ 65+

5. Which airport department do work in?

- ☐ Airfield Operations
- ☐ ARFF
- ☐ Concessions
- ☐ Environmental/Noise
- ☐ Finance
- ☐ Ground Transportation
- ☐ Guest/Customer Service
- ☐ Governmental/Legal
- ☐ Information Technology
- ☐ Maintenance
- ☐ Marketing
- ☐ Planning and Design
- ☐ Public Relations
- ☐ Police/Public Safety
- ☐ Procurement
- ☐ Properties/Contracts/Commercial Development
- ☐ Terminal Operations
- ☐ Other (please specify)

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6. Were you trained via distance learning at any time during the pandemic (March 2020-March 2022)?

- ◯ Yes
- ◯ No

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7. If so, what is your perception of the quality of distance learning training versus in-person training?

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8. Did you work remotely at any point during the COVID-19 pandemic (March 2020-March 2022)?

- ◯ Yes
- ◯ No

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9. If not, would you have liked to?

- ◯ Yes
- ◯ No

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10. Which would you prefer?

- ◯ Work remotely
- ◯ Return to the office

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11. To what degree have you experienced "burnout" (physical or mental breakdown) while working remotely?

None	Minor	Somewhat	Significant	Extreme
★	★	★	★	★

12. What are the benefits to working remotely?

13. What are the challenges to working remotely?

14. How were you able to maintain work-life balance while working remotely?

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15. Do you feel working remotely is sustainable for the long-term?

- ◯ Yes
- ◯ No
- ◯ Not sure

16. Do you feel that training is more difficult while working remotely?

- ◯ Yes
- ◯ No
- ◯ Not sure

17. Do you feel that innovation is more difficult while working remotely?

- ◯ Yes
- ◯ No
- ◯ Not sure

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18. Do you feel your career might suffer while working remotely?

- ◯ Yes
- ◯ No
- ◯ Not sure

19. Do you feel it is difficult to add new members to your team/department while working remotely?

- ◯ Yes
- ◯ No
- ◯ Not sure

20. Do you feel that your manager is less available while working remotely?

- ◯ Yes
- ◯ No
- ◯ Not sure

21. Do you feel your airport has engaged in team and culture building exercises while workers are remote?

- ◯ Yes
- ◯ No
- ◯ Not sure

22. Do you feel you are socializing less with co-workers while working remotely?

- ◯ Yes
- ◯ No
- ◯ Not sure

23. Do you feel that morale among your team has suffered while working remotely?

- ◯ Yes
- ◯ No
- ◯ Not sure

24. Do you feel your team's culture has suffered while working remotely?

- ◯ Yes
- ◯ No
- ◯ Not sure

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25. If you were in charge of the organization/department during the pandemic, what would you have done differently, if anything?

26. Morale took a hit during the pandemic, what "fun" thing did your airport do that you can share?

Airport Work Models - Airport Employee Survey - ACRP S06-07

27. Do you have any other thoughts to share on this topic?

Airport Work Models - Airport Employee Survey - ACRP S06-07

Contact

28. Would you be willing to discuss your personal experience with remote work and training arrangements via the phone for this project? If so, please indicate your name, airport, email address, and phone number. Thank you!

Name	
Airport	
Email Address	
Phone Number	

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APPENDIX C

Additional Literature Review

History of Remote Work

Remote work, although a somewhat novel consideration for airports, is not a new concept. Although many of the technologies that support it were only recently developed, as Sako (2021) explains, remote work arrangements date back several decades:

> By the 1980s, remote office work—work-from-home—was considered an extension of flexible work arrangements alongside part-time work, which enabled workers, especially female workers, to balance work and childcare. With the Internet enabling connectivity since the 1990s, remote work morphed into offshoring to low-cost global locations. Offshored work included office work at call centers and software engineering centers, but also freelancing in design data entry programming, and translation using platforms such as Upwork, LinkedIn Profinder, and Fiverr." (p. 20)

Consider how various major events have seemingly pushed the workforce closer to remote work, only to return to the office. For example, a headline in the Los Angeles Times after the city's 1994 earthquake, read "TELECOMMUTING TAKES OFF AFTER QUAKE . . . MOST EMPLOYEES WELCOME THE CHANGE." Yet, in-person work soon returned.

COVID-19 was a game changer, however. Due to numerous restrictions in response to the COVID-19 pandemic, many employees who were previously unwilling, or likely never offered the opportunity, to work remotely were not only offered, but in many cases, required to work remotely. Indeed, "the COVID-19 pandemic triggered a rush towards remote work, making it crucial for business continuity" (Iwashita, 2021, p. 9).

"The pattern of modern-day work . . . in which employees commute to an office, has proved resistant to technological and social change. It outlasted the invention of the fax machine, the internet, and video-phones; the decline of former industrial regions, stratospheric real estate prices in coastal megacities; globalization; climate change; air pollution; record commuter congestion; and failing subway systems. Each of these transformations should have nudged the business world toward remote work; none did" (Cooke, 2021, p. 40).

Prior to the emergence of COVID-19, "at most 5% of Americans worked from home for more than three days per week" (Yang et al., 2022, p. 43). With stay-at-home orders, non-essential personnel instructed to stay home, and employee hesitance to return to the office for fear of COVID-19 exposure, the pandemic accelerated changes already underway, and pushed things over a 'tipping point'" (Sako, 2021, p. 20). "Hardly anyone made it through the pandemic with their work life unchanged" (Maslach and Leiter, 2022, p. 64).

"The unprecedented outbreak of COVID-19 pandemic in 2020 has required millions of people across the world into being remote workers, inadvertently leading to a de facto global experiment of remote working" (Wang et al., 2021, p. 17). Prior to COVID-19 and the need for workers to vacate the office due to health concerns, "most workers had little remote working experience; nor were they or their organizations prepared for supporting this practice" (Wang et al., 2021, p. 17).

According to Yang et al. (2022), by April 2020, as many as 37% of Americans were working from home full-time. Sako (2021) reports that 35–50% of all U.S. employees were working entirely or partly from home by early May 2020. By late May 2020, only several months into the COVID-19 pandemic, the Gallup organization reported that 65% of the U.S. workforce was working remotely at least some of the time. By October 2020, 71% of U.S. workers who could perform their jobs remotely were, in fact, doing so (Gallup, 2020).

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Remote Work Defined

According to the U.S. Office of Personnel Management, remote work is defined as "a work arrangement in which the employee resides and works at a location beyond the local commuting area of the employing organization's worksite. The arrangement generally includes full-time telework and may result in a change in duty location to the alternative worksite" (U.S. Office of Personnel Management, 2013, p. 259).

Jamsen et al. (2022) define remote work as "a form of work that employees perform outside the physical premises of the organization by utilizing communication technology" (p. 2). Wang et al. (2020) share, "Remote working is defined as a 'flexible work arrangement whereby workers work in locations, remote from their central offices or production facilities, the worker has no personal contact with co-workers there, but is able to communicate with them using technology' " (p. 17).

Although definitions vary somewhat, each is united in that remote work allows employees to perform their work at an off-site location. This contrasts with the traditional work model in which employees work on-site at their employer's location.

Remote Work Models

Traditional work-from-home models offer the employee temporal flexibility but expect them to continue working in the local area. Even so, work-from-home arrangements allow workers greater control over ambient workspace elements, such as clothing, layout, music, ventilation, etc.

An emerging form of remote work, known as "work-from-anywhere," allows the worker to choose to live in a preferred geographic location, even if it is outside the local geographic area of the home office. By allowing workers this greater flexibility in choosing where to live, housing costs can be reduced, family time can be increased, and workers can have greater control over their lives (Choudhury et al., 2020).

Enforced remote work, an arrangement in which employees work from home or effectively resign, is much different than traditional remote work in which employees are given the option to work remotely. First, the traditional remote working arrangement has commonly been a blended or hybrid approach, whereby the employee works from home several days and in the office several days each week. Splitting time between remote and office work offers great flexibility and allows employees to enjoy the benefits of both working locations. Second, the traditional remote employee has considered the implications of remote work on his or her personal situation and preferences. In essence, the employee choosing remote work likely does so for the benefits it provides, whereas the employee required to work remotely may experience unique challenges, stemming from, among other things, the person's lack of control over their work situation (Zhang et al., 2021).

Rather than an all-or-nothing approach, a hybrid approach may be more reasonable for both employees and the employer. "Hybrid work involves providing the employees the freedom to decide where and when to work" (Finnegan, 2021). According to a survey conducted by Morning Consult in 2021, "87% of respondents want the flexibility to continue some form of remote work" (Peppercorn, 2021, p. 2). Gallup data shows that hybrid work arrangements are by far the preferred work arrangement for remote-capable employees—those whose current jobs can be done off-site at least part of the time. Nearly 60% of remote-capable employees prefer a hybrid model, nearly twice the 32% who would like to work remotely all the time (McGregor, 2022). Certainly, employers can require specific days in the office, but by allowing greater flexibility, "employees have the ability to fit their work around the rest of the things in their life, instead of structuring their day around the time they spend in the office" (Finnegan, 2021, p. 176).

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Many organizations, airports included, find it possible for some employee groups to work remotely, but require other employee groups to work on-site. This "mixed-mode" model consists of a mixture of full-time remote employees and full-time office employees (Yang, et al., 2022). Admittedly, this arrangement may extend this benefit to some employees, but not others, creating either perceived or real inequity in the workplace.

Training

The aviation industry is one characterized by extensive training. The literature presents three generations of aviation training since the industry began. First, apprenticeship was the primary means of training from 1903 to 1929. Second, simulation, initially in very crude forms, was the primary means of training from 1929–1979. Third, since 1979, safety has been the primary focus of aviation training, although some degree of simulation and apprenticeship may still be used (Sun et al., 2021).

In the modern era, computer-based training and distance learning have enabled all industries, including aviation, to train employees more efficiently and in some instances, without regard to time and space requirements. As Sun et al. (2021) share, "Distance learning has a huge potential to herald the start of a fourth generation of education training in the aviation sector" (p. 111).

Even so, distance learning has presented unique challenges. First, distance learning courses vary dramatically in quality. Second, the preparation of quality online learning materials is a significant investment of time and effort. Third, the availability of technology drives the adoption of distance learning (Sun et al., 2021).

Industry experience has indicated that, due to COVID-19 stay-at-home restrictions and degrees of remote work arrangements, employee training at airports was significantly impacted. For example, employees needed technological assets such as computers, mobile devices, high-bandwidth internet connections, etc. to access training offered at a distance. Would the airport provide these tools for employees or expect employees to arrange such access with personal resources? Additionally, unless the training was provided in a synchronous ("live") format, there would be a lack of interaction between employees and the trainer. Further, how would airports ensure that assessments provided to employees would be fair and prevent cheating (Sun et al., 2021)?

Airports that discovered the benefits of a "blended learning strategy" were likely less impacted by the various COVID-19 restrictions and the impacts on employee training and development. A blended approach incorporates in-person and computer-based training. Even when pandemic restrictions were in effect, making in-person training difficult, cloud-based platforms such as Zoom, Webex, and Microsoft Teams could be used to provide a synchronous, albeit virtual, "in-person" component. A live training session could be conducted one-on-one or with multiple employees from any location desired. In fact, both the trainer and employees being trained could virtually meet from the comfort of their own residences (Sun et al., 2021).

Work Philosophy

The COVID-19 pandemic solidified existing views of work for some, while completely upending it for others. The literature seems to indicate that remote work, or some form of it, is here for the long term. Granted, the literature does not only consider airports, but the economy as a whole. This view in support of remote work may be due to a changing philosophy of work.

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In the West, "going to work has defined modern life and modern lives, along with the labor conditions, cities, relationships, and patterns of movement that make them up" (Cooke, 2021, p. 42). In essence, the view that one must "go to work" has been very real. However, Cooke (2021) proposes that, "Doing work and going to work are very different things" (p. 42). He suggests that some employees need not be at a specific geographic location to perform "work."

Flexibility is now more important to employees in our post-pandemic era. As Evans et al. (in Choudhury et al., 2020) state, "Flexibility in the employment relationship is defined as 'ceding control to workers over the circumstances of their work by enabling them to vary those circumstances to address personal and family needs and uncertainties'" (p. 659).

"We should not care about how many days or hours anyone works. Every job and task should have objective metrics, which are output-based, and if an employee can perform those metrics in two days, so be it. . . . We should give people the flexibility to work when they want to, whichever hours they want to, whichever days they want to, and care only about their work." (Choudhury in Ren, 2022 paragraph 23)

Ren (2022) makes a bold prediction, "A decade from now, offices shall be used for one thing only: quality time with colleagues" (paragraph 1). Choudhury (in Ren, 2022) states, "We will probably in 10 years stop calling this 'remote work.' We'll just call it work and work is something you do, not where you go or where you live" (paragraph 12).

Although some may disagree with these statements, it is clear throughout the American economy that businesses of all sizes and in all segments have adopted a long-term view that "work" can be performed anywhere. Yet, as we know, certain tasks (such as airfield inspections and responding to aircraft emergencies) cannot be performed from anywhere. Thus, like anything else, each airport will need to evaluate their unique human resource needs and develop policies to support employees, as well as the employer, which may or may not be in line with leading thinkers of the day.

Great Resignation

During the pandemic, as businesses significantly altered their employee work model and workers confronted the stress of stay-at-home orders and social isolation, many employees, resigned from businesses in droves. For example, in November 2020, 3.2 million Americans quit their jobs. In November 2021, 4.5 million Americans quit their jobs. A 2021 Microsoft study—the 2021 Work Trend Index—found that 41% of the labor force was considering leaving their employers at the end of the year (Esterhuizen, 2021). In March 2022, 4.5 million Americans quit their jobs. This "Great Resignation" was underway in full force and was long-lived.

By May 2022, after more than two years of the pandemic, "61% of young professionals between the ages of 25-35 changed jobs in the last two years or plan to do so within the following two years" (Genovese, 2022, paragraph 7). Many of these employees had been lured away to organizations offering higher pay, better titles, and better perks. Interestingly, as employees resigned, businesses had to sweeten their offerings to attract replacement workers.

The Great Resignation also resulted from individuals seeking more flexibility and control over their lives. Zubernis (2022) proposes that this record rate of employees walking away from their jobs is due to burnout. "Recent surveys of workers find that half feel burned out and two-thirds believe their feelings of burnout have worsened" (p. 56).

The grass is not always greener, as experienced employees know. As Genovese (2022) shares, according to a recent survey by Joblist "About 26% of people who quit their job during what was dubbed the Great Resignation already regret it" (paragraph 3). These workers likely quit because of unfulfilled needs at work and realized after the fact that the other option—unemployment—may not be any better. For those who did secure another job, 42% said "the new gig didn't live up to their expectations" (Genovese, 2022, paragraph 3). As a result, the "Great Resignation" appears to be turning into the "Great Regret."

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Employee Desires

Surveys consistently show that "remote workers overwhelmingly want to stay remote" (Colvin, 2022). Some companies, especially tech firms such as Twitter and Facebook, have said if an employee's work can be done from home, the employee can work from home forever.

Close to one-third of U.S. workers, a substantial amount, are projected to continue working remotely even beyond the need to do so due to COVID-19 (Zhang et al., 2021). According to research by Routley (2020), 98% of surveyed employees would like the option to continue working remotely for the rest of their careers.

Several surveys show that 91% of remote employees would like to continue their hybrid or remote working, and 76% say their employer will allow them to work remotely going forward (Shreedhar et al., 2022).

A number of companies have learned the hard way that today's employees desire flexibility in their work arrangements. Employee recruiting and hiring have had to change to meet these new desires. Colvin (2022) explains, "If you don't offer job candidates the freedom they demand, the best ones won't even consider you" (p. 43).

Advantages and Benefits

All types of working arrangements have advantages and disadvantages. Oftentimes, two employees in the same organization and in the same position have different views of the workplace, depending on their perspective. As such, when considering the benefits and challenges of remote work, it is important to realize that employee attitudes vary, and it is difficult to apply specific advantages and disadvantages to all employees in all organizations.

Advantages of remote work, according to Gandhi et al. (2021, p. 3312), include:

- Employer
  - Astonishing productivity
  - Increased employee retention
  - Decrease in sick days and employee time off
  - Reduction in cost for employers
- Employee
  - Reduces stress
  - Prolong the careers of older workers
  - Sense of freedom
  - Eliminate travelling expenses
- External stakeholders
  - Reduce traffic on roads
  - Reduce migration leading to a reduction in overpopulation of metropolitan cities
  - Reduce the emission of greenhouse gases

According to Choudhury (in Ren, 2022):

> Remote work is often pitched as something employees want and employers don't. My research has showed that this is a win-win. For employees, it's great to work from anywhere because you can move to a cheaper location. You can live where you want to. For employers, it's a win as well because you are not constrained to hiring from the local labor market – where you have an office. The other big benefit is productivity. In the U.S. Patent Office, we documented a 4.4% productivity gain back in 2012, when they allowed patent examiners to work from anywhere. The final win for employers is that work from everywhere leads to a fairer workforce, especially on the dimensions of gender and race. There's at least two decades of research showing women have lost out on career opportunities because of geography. But if the company lets you work from anywhere, you don't have to relocate (paragraph 7–9).

Gajendran and Harrison (2007) and Zakaria et al. (2021) discovered that, in general, remote workers experienced higher levels of job satisfaction and performance. These employees also experienced less stress and had fewer turnover intentions. Zakaria et al. (2021) point out that "remote work allows the employee more flexibility and autonomy in doing their work, meeting their job demands, and personal needs" (p. 93). In a study of remote work-related tweets on the Twitter social media platform, Zhang et al. (2021) discovered that "overall, attitudes conveyed by remote work-related tweets were mildly positive" (p. 802). In this same study, while some tweets viewed working remotely as beneficial, other tweets conveyed higher stress when working from a home office. "Some are loving it, with flexible schedules, no long commute, and more time with family. But others are unhappy with loneliness and the blurred boundary between work and leisure" (Sako, 2021, p. 20).

According to Vaillancourt (2021), "During our global work-from-home experiment, many people enjoyed improved focus, less time commuting, reduced office drama, more flexible schedules for work and life responsibilities, and lower out-of-pocket costs for things like

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transportation, clothing, coffee, and lunch. For some, though certainly not all, work-from-home options increased morale and productivity" (p. 48).

In study exploring perceptions of remote work by HR managers, Perez et al. (2002) discovered that benefits of remote work include labor time flexibility, decreased commuting, and autonomy. Iwashita (2021) proposes that remote work stimulates productivity and well-being. "Companies that previously forbade remote work can no longer deny its benefits" (Limoncelli, 2020, p. 110).

Remote work has also benefitted the HR function with reduced costs and greater efficiencies. "We believe that remote work is hastening a shift that reduces hiring, screening and monitoring costs for managers and lessens burdens on workers to understand firm-specific context" (Ozimek and Stanton, 2022, paragraph 21). Additional benefits for the employer include "decreased employee turnover, increase in the ability to attract employees, reduced sick time, reduced overhead, and reduced carbon footprint" (Finnegan, 2021, p. 175).

Specifically with regard to environmental benefits, with many employees working remotely during the pandemic, the daily commute all but disappeared. Although global CO_2 emissions dropped by 17% in April 2020 compared to peak 2019 levels, by March 2020, emissions were almost back at pre-pandemic levels, even though employees weren't (Shreedhar et al., 2022). As environmental footprints vary across industries and companies, "it may be more appropriate to focus on emissions reductions from cooling versus heating, or both" (Shreedhar et al., 2022, p. 6). In other words, a one-size-fits-all approach to environmental sustainability will not work.

Disadvantages and Barriers

Disadvantages of remote work, according to Gandhi et al. (2021, p. 3312), include:

- Employer
  - Demotivation among employees
  - Hard to communicate the work
  - Low reliability
  - Difficult to manage and maintain accountability
- Employee
  - Bad exercise regime
  - Higher bills of electricity, internet, etc.
  - Lack of proper equipment
  - Ergonomic issues
- External stakeholders
  - Adverse effects on auxiliaries to trade
  - Lack of personalization for the customer

The literature speaks widely of barriers or challenges experienced by remote workers. This alternative working arrangement affects employees differently, but several barriers are commonly noted in the literature.

First, a blurring of the work-life boundary is a common experience for remote workers. This is less pronounced with on-site employees, who can leave work and then return home. Routley (2020) explains that remote employees often experience difficulty in "unplugging" from work. The workday hours blur as the work piles up, and before we know it, "we're tethered to our desks and working around the clock" (Gaines, 2021, p. 8). As one employee shared with Yarberry and Sims (2021), "My home life and work-life are the same" (p. 247). Employees with home offices may feel as if they can never truly step away from their jobs. When they are home, they are at "work." Employees with home offices in their bedroom are highly susceptible to feelings of being at work 24/7. Indeed, remote employees are challenged in dividing personal and professional time, thus creating an upset in work-life balance. These "blurred lines between work and home can lead to burnout, stress, and other career inhibitors" (Yarberry and Sims, 2021, p. 246). The U.S. Bureau for Economic Research found that the average workday increased by 48.5 minutes during the COVID-19 lockdown, which equates to two extra workdays per month (Esterhuizen, 2021).

Employers can address this "blurred" work-life boundary experienced by remote employees in several ways. Employers can develop work-from-home guidelines, to include placing the home office desk outside the bedroom. Employers can also respect the employee's time, avoiding phone contact or expecting email responses after normal working hours. It is also beneficial to encourage remote employees to make time for self-care, to include taking breaks throughout the day, making time for hobbies, and avoiding working after normal business hours.

Second, remote workers may experience distractions throughout the day. This was certainly more pronounced when schools were closed, and school-age children were at home attempting

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to learn remotely, all the while seeking the attention of caregivers. Even without children at home, distractions can be more prevalent at home. "There are countless distractions, from kids, to pets, to partners trying to get their own work done" (Gaines, 2021, p. 8).

Employers can address these distractions to some degree by clearly setting expectations and developing guidelines for employees to consider in locating their home office. Much of this depends on employees to manage home life and minimize distractions that can reduce productivity.

Third, remote workers may experience social isolation while working at home alone, away from colleagues. As explained by Wanberg et al. (2020), as employees begin remote work an abrupt transition is often required, where employees may lose resources at work, such as co-worker interaction and support, but also in their personal life, such as social isolation. Yarberry and Sims (2021) discovered that employees working remotely are less likely to engage with co-workers, leading to feelings of isolation. As Jamsen et al. (2022) discovered, "The absence of social lunch and coffee breaks, as well as spontaneous encounters in the hallways, led to a drop in the amount of relational communication, decreased the amount of humor and support, and raised the threshold to contact others" (p. 4). These feelings of professional isolation also resulted in less organizational learning (Jamsen et al., 2022). As Manko and Rosinski (2021) share, "The most salient feature of remote work is physical isolation" (p. 42). While introverted employees may actually appreciate such isolation and achieve at higher levels, extroverted employees that thrive on social interaction may dread remote work. "There's something unique that humans get from interacting with one another that doesn't come across as well through technology" (Semuels, 2020, p. 44).

Organizations that intentionally create opportunities for employees to connect, even if remotely or virtually, will generally see fewer instances of employees feeling isolated. Even during COVID-19, some organizations held outdoor gatherings, such as barbecues, which allowed employees to connect face-to-face with their colleagues in an outdoor setting in which social distancing was possible.

Fourth, the technology that has enabled remote work can also be burdensome. Employees commonly voice concern with online meetings throughout the day. Experiencing "Zoom fatigue" was very real for many employees. This results in distraction and loss of effectiveness. Holding shorter online sessions, even if they may be more frequent, can be effective. In other words, by allowing plenty of breaks throughout the day, even designating portions of the day as no-Zoom zones, employees won't burn out as quickly. "Successful online meetings move briskly and stay on purpose" (Manko and Rosinski, 2021, p. 49).

Employers can address "Zoom fatigue" in a number of innovative ways. Zoom "watercooler meetings" can use the same technology that causes employee fatigue to create a more supportive environment. It's not so much the technology, but how the technology is used that matters. As Manko and Rosinski (2021) share, "There appear to be cases in which online interactions can actually work better than face-to-face meetings (p. 43). Having weekly online check-ins or incorporating games or family time during virtual meetings may also prove effective.

In addition to these four primary challenges experienced by remote workers, these employees may also face difficulty navigating different time zones, difficulty in collaborating and communicating, unreliable wi-fi, and lack of motivation. The unique challenges experienced by remote workers are as varied as the types of work arrangements. The key is to listen to employees, regularly seek a sense of each employee's well-being. What can the organization do to better support the remote employee? Remote work may not "work" for everyone equally. Some employees will thrive, while others will suffer. It is up to the organization to gauge which employees need additional support to ensure success with remote work.

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APPENDIX D

Remote Work Policy and Agreement Samples

Remote Work Policy Sample 1

Source: City of Philadelphia, Tamara Whitfield

Virtual Work Policy for City Employees

Effective Date: June 11, 2021

Policy Version No.: 5

Point of Contact:

Statement of Policy

The City of XXXX workforce provides essential services to over 1.5 million City residents, including protecting residents' lives, providing residents with educational opportunities, and improving residents' health. Although providing those services often requires onsite presence, the City remains committed to enhancing the work-life balance of its employees and offering flexible working environments when possible. To that end, this policy applies to all City employees who have been directed or approved to work remotely to support City operations and sets forth the circumstances under which City employees may work remotely, and the responsibilities of all parties to any remote work arrangements.

Virtual Work Definition and Eligibility

"Virtual work" is defined as conducting work remotely, whether from home or an alternate location away from the office. All employees may be eligible for virtual work, either intermittent or regularly scheduled, depending on their job duties and the operational needs of the Appointing Authority. When necessary, Appointing Authorities may require employees work virtually. All virtual work requires prior approval by an Appointing Authority.

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Permissible Virtual Work Arrangements

The primary goal of this policy is to enhance workplace flexibility while ensuring that City employees continue to deliver high-quality work product that meets the needs of City government and City residents.

Appointing Authorities may decide how many days per week an employee may work virtually, up to a maximum of three days (or 22.5 hours), unless the Appointing Authority has received authorization to exceed this maximum from the appropriate cabinet member. In exigent circumstances, for example, in the case of an emergency declaration, employees may be authorized to work virtually for an extended period of time beyond this maximum. If employees are allowed to decide which weekdays they work virtually, Appointing Authorities may require that they designate virtual workdays with sufficient advance notice to allow the Appointing Authority to accommodate operational needs.

After an Appointing Authority has approved virtual work for an employee, the Appointing Authority and employee should select an arrangement from those detailed below that best fits the work pattern of that department and the work which the employee in question will be expected to complete. An Appointing Authority may authorize a temporary deviation from a selected arrangement to accommodate unforeseen circumstances on the part of the employee.

A. *Set Schedule*. Appointing Authorities may use a set schedule for approved virtual work. Set schedule virtual work requires the Appointing Authority develop a schedule for the virtual work to which the employee will be expected to adhere (e.g., 8 a.m. to 4 p.m., 9 a.m. to 5 p.m., etc.) Set schedule virtual work should be used when an employee is expected to provide services, either internally or externally, during a set period, and not expected to work outside that period.

B. *Core Hours*. Appointing Authorities may designate certain "core hours" during which a virtual work employee is expected to be working (e.g. 10 a.m. to 2 p.m., 10 a.m. to 12 p.m. and 2 p.m. to 4 p.m., etc.) Employees are then permitted to flex the remainder of the workday. Core hours virtual work should be used when employees are expected to be available to work collaboratively or answer questions, but also have self-directed tasks that may be completed without direct interaction or supervision. Employees are required to complete the minimum work hours according to their work schedule (e.g., 7.5 or 8.0 daily, 75 hours biweekly, etc.)

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General Guidelines

The City recognizes that its departments, offices, and agencies provide very different services with very different attendant operational needs. The following guidelines are intended to provide a baseline for all virtual work engaged in by City employees and are not intended to prevent Appointing Authorities from issuing additional specific and reasonable rules, restrictions, or policies.

C. *Work Location and Assessment*. Employees who work virtually on a regular basis are expected to do so from a location that allows the employee to work in an ergonomically appropriate workspace free from distractions. It is recommended that all regular virtual work arrangements be approved by a departmental safety officer after an assessment of the employee's home workspace conducted in concert with ergonomic guidelines provided by Risk Management. Appointing Authorities may approve sporadic virtual work without such an assessment.

D. *Virtual Work Technology*. Employees must have access to appropriate technology to facilitate virtual work.

1. City-owned Devices. Virtual work must be conducted using City-owned devices. Appointing Authorities should work with the Office of Innovation and Technology (OIT) and their office's IT manager or consultant to ensure that City-owned machines are provided to employees who are expected to engage in virtual work.

   i. **Computers.** It is the City's policy to provide laptop computers to employees working virtually. In exigent circumstances, Appointing Authorities may authorize employees to take home desktop computers to facilitate virtual work.

   ii. **Phones.** Appointing Authorities should provide cellular telephones for employees expected to perform customer-facing functions that involve telephone contact virtually. These cellular telephones may be voice-only if the virtual work does not require data usage. Appointing Authorities should consider their department's ability to support the cost of monthly data plans when requiring cellular phone use or the type of phone (voice-only vs. smart phones with data plans and apps).

   iii. **Peripherals.** When necessary, the City will provide each employee one set of peripherals (keyboard, mouse, monitor). That set of peripherals should be used by the employee at his or her principal work location (e.g., home or office).

   iv. **Multi-Factor Authentication and Virtual Private Network.** Employees are required to adhere to security measures and must use both multi-factor authentication (MFA) and virtual private network (VPN) client, both provisioned by OIT, to strictly manage access to their City email and network credentials as to preserve and protect City data and technology operations vulnerable to compromise.

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2. Personal Electronic Devices. Employees are not allowed to use personal devices for routine or daily work assignments. Employees must use City-owned devices for communications and work assignments except to check emails or Office 365 communications on a limited or as-needed basis. Appointing Authorities may not approve use of personal devices under any circumstances for employees whose work uses programs and databases that do not have web-based interfaces (e.g., FAMIS, etc.), or employees whose work involves sensitive and/or confidential information protected by federal, state, or local statutes (e.g., HIPAA or CHRIA-protected information, taxpayer information, etc.). Employees using a personal device, as stated above, must follow all guidelines set forth in the Office of Innovation and Technology's Access Control policy found on the City's intranet site, including the use of VPN and MFA and the Personal Device Use Policy.
3. As of July 6, 2021, any employee working from home will be required to participate in OIT's mandatory Cyber Security training through the City's Learning Management System (LMS) on an annual basis. Any employee failing to complete such training on an annual basis may have their work from home privileges revoked.

E. *Reimbursable Expenses*. Appointing Authorities may not choose to reimburse remote employees for consumables ordinarily provided to employees while working onsite, such as paper, writing utensils, or other office supplies. Employees should be directed to obtain office supplies onsite, via existing City purchasing processes. The City will also not reimburse employees working virtually for the following expenses:

- Expenses related to home internet services, including installation costs, ongoing service costs, or data overages;
- Expenses incurred in purchasing, using, repairing, or replacing personal devices, including computers, cellular telephones, and peripherals such as printers, keyboards, and mice;
- All other expenses not specifically enumerated as reimbursable in this policy.

F. *Dependent Care*. Employees are expected to obtain care for dependents as necessary to ensure that employees may effectively work virtually. Appointing Authorities may relax this care expectation during exigent circumstances (e.g., sporadic school closures, illness, or other emergencies that prevent employees from obtaining dependent care).

G. *Timekeeping*. Employees may combine virtual work with onsite work and/or leave time when appropriate and approved. All virtual work should be tracked on the employee's timesheet using the appropriate time code provided by payroll.

H. *Discipline*. Employees working virtually must abide by the same rules, regulations, and policies regarding conduct and work performance that they would be expected to adhere to while working onsite. Employees who fail to follow such rules, regulations, or policies, or who abuse virtual work may have their virtual work privileges be revoked and/or be disciplined, with sanctions up to and including termination.

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I. *Signed Acknowledgement*. As of July 6, 2021, all employees working virtually on a regular basis shall sign the acknowledgement form found at the end of this policy. Additionally, Appointing Authorities may also request an employee complete the virtual work agreement form found in Appendix A of this policy that delineates any restrictions on that employee's virtual work and memorializes that employee's agreement to abide by this policy and any other applicable department-specific policies. Appointing Authorities shall maintain signed copies of all certifications and agreements for any employees permitted to work virtually on a regular basis.

Role Responsibilities

Employees, supervisors, and Appointing Authorities must work cooperatively to ensure that employees' virtual work meets the highest standards of quality and quantity. This section delineates the responsibilities of each role in the virtual work arrangement.

J. *Appointing Authority's Responsibilities*.

1. Appointing Authorities are expected to work diligently to arrange work so that it can be completed virtually when possible.
2. Appointing Authorities should provide virtual work schedules, and deviation from those schedules should be pre-approved.
3. Appointing Authorities must ensure that all overtime earned while working virtually is accurately recorded and timely paid, regardless of whether such overtime was approved. If an employee worked unapproved overtime while working virtually, the Appointing Authority should discipline the employee appropriately.
4. Appointing Authorities must ensure that all employees working virtually are provided with appropriate performance metrics and/or deliverable requirements to ensure that those employees produce the quality and quantity of work expected.
5. Appointing Authorities must ensure that all employees working virtually participate in City-offered trainings on virtual work.

K. *Supervisor's Responsibilities*.

1. Supervisors (or other individuals designated by the Appointing Authority) must approve all virtual work that would result in an employee earning overtime. Supervisors should approve such overtime work only if that employee is working on time-sensitive matters that cannot be completed during a standard work-day.
2. Supervisors will review all work product delivered and functions performed to ensure employees are completing the appropriate quantity and quality of work while working virtually.
3. Supervisors should ensure that employees working virtually may participate in the organization's work to the same extent they would be able to participate if working onsite. This may include ensuring that meetings may be joined virtually, providing call-in numbers for important conference calls, or ensuring that important communications are timely sent by electronic means.
4. Supervisors will immediately document and work to resolve any performance issues.

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L. *Employee's Responsibilities*.

1. Employees must ensure that they are prepared to work virtually prior to engaging in virtual work, including ensuring that they have access to necessary devices, documents, applications, information, and communication tools required to perform work effectively.
2. Employees must preserve work material created while working virtually as they were working onsite, including preserving work material created using personal devices.
3. Employees using personal devices for virtual work must surrender any such device at the City's request so that the City may obtain information for potential disclosure in responding to City obligations, such as Right-to-Know requests, internal or external investigations, or lawsuits.
4. Employees provisioned City-owned laptops or computing devices must regularly connect, no less than once per week, to the City's network so that devices receive software and antivirus updates to ensure operability, antivirus readiness, and installation of critical patches to resolve security vulnerabilities.
5. Employees working virtually must regularly submit such records of work deliverables as their Appointing Authorities deem appropriate.
6. Immediately report any and all technical issues related to virtual work to the OIT Support Center. Individuals may report technology emergencies by calling (xxx)xxx-xxxx. OIT cannot provide support for virtual work being conducted on personal devices. Departments not consolidated with OIT and who have their own desktop/laptop IT support units may contact that unit directly for assistance.
7. Employees should communicate as necessary with supervisors by email and telephone and plan to participate in scheduled business meetings as if working onsite.
8. Employees are expected to timely attend all trainings targeted towards virtual work.
9. Employees will provide supervisors and HR staff with a contact cellular telephone number if they have not been provided a City-issued mobile phone by their Appointing Authority. Supervisors will see to it that this information is recorded and available in City administrative systems utilized by human resources and OIT as to contact in case of emergency or cyber security incident.

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VIRTUAL WORK POLICY FOR CITY EMPLOYEES

EMPLOYEE ACKNOWLEDGEMENT FORM

I acknowledge that I have received and read a copy of the Virtual Work Policy for City Employees and understand that this document describes the policies that apply to virtual work as well as my responsibilities as an employee working virtually.

Employee **(print name)**	Appointing Authority Designee
Employee **(signature)**	Appointing Authority Designee
Date	Date
Employee Payroll Number	

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Remote Work Policy Sample 2

Source: Naples Airport Authority, Virginia Bendle

AIRPORT AUTHORITY EMPLOYEE PROCEDURES

PROCEDURE: TELECOMMUTING

DATE: MAY 13, 2022 PAGES: 3

SUPERSEDES: MARCH 19, 2020 **PREPARED BY: HR DEPARTMENT**

EXECUTIVE DIRECTOR SIGNATURE:

Declaration:

Due to the need to be responsive to changing circumstances, the XXX Airport Authority is implementing telecommuting arrangements on a limited, as-needed basis, for employees whose job duties are conducive to working from home, but who do not regularly telecommute. However, most positions require the employee to be physically present in the workplace.

The following classifications may qualify for telecommuting:

- Administrative staff may work remotely up to 2 days per week.
- Members of the Leadership Team may work remotely up to 1 day per week.

Employees should not assume any specified period of time for telework, and the Authority reserves the right to require employees to return to regular, in-office work at any time.

Procedures:

Preparations should be made by employees and managers well in advance to allow time to set up remote access. This may include appropriate equipment needs, such as hardware and software, and any necessary restructuring of job duties.

For voluntary telework arrangements, either the employee or department manager can initiate a telecommuting agreement. The employee and manager will discuss the job responsibilities and needs of the department, to determine if the job is appropriate for a telecommuting arrangement, including equipment needs, workspace design considerations and scheduling issues.

The manager will discuss with the department director and if approved, a telecommuting agreement (see attached) will be prepared by and signed by the employee and his or her department director.

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The employee will establish an appropriate work environment within his or her home for work purposes. The XXX Airport Authority will not be responsible for costs associated with the setup of the employee's home office, such as remodeling, furniture or lighting, phone or data lines, nor for repairs or modifications to the home office space.

The Authority will determine the equipment needs for each employee on a case-by-case basis. Equipment supplied by the organization is to be used for business purposes only. Consistent with the organization's expectations of information security for employees working at the office, telecommuting employees will be expected to ensure the protection of proprietary company and customer information accessible from their home office.

Conditions:

- The employee must not be on probation and must be in good standing.
- The employee will remain accessible and productive during scheduled work hours.
- Non-exempt employees will record all hours worked and meal periods taken in accordance with regular time-keeping practices.
- Non-exempt employees will obtain supervisor approval prior to working unscheduled overtime hours.
- The employee will report to the employer's work location as necessary upon directive from his or her supervisor.
- The employee will communicate with his or her supervisor as pre-determined before the commencement of telecommuting. The employee will communicate regularly with co-workers.
- The employee will comply with all Authority rules, policies, procedures, practices, and instructions that would apply if the employee were working at the Authority's work location.
- The employee will maintain satisfactory performance standards.
- The employee will make arrangements around caregiving responsibilities so that work may be performed.
- The employee will maintain a safe and secure work environment at all times.
- The employee will allow the employer to have access to the telecommuting location for purposes of assessing safety and security, upon reasonable notice by the Authority.
- The employee will report work-related injuries to his or her manager as soon as practicable.
- The Airport Authority equipment will not be used by anyone other than the employee and only for business-related work.
- The employee will not make any changes to security or administrative settings on the Authority equipment.

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- All tools and resources provided by the Authority shall remain the property of the Authority at all times. The employee will protect Authority tools and resources from theft or damage and to report theft or damage to his or her manager immediately.

- The employee must comply with the Authority's policies and expectations regarding information security.

- The employee will be expected to ensure the protection of proprietary Authority and customer information accessible from their home offices.

The employee is required to return Authority equipment and documents within five days of termination of employment.

*****

Please note: As with all other policies, programs and benefits, the Authority retains the right to change, augment, or discontinue this program at any time without notice.

This policy does not form or imply a contract of any kind. The foregoing procedure is intended to be comprehensive. The Authority recognizes, however, that no plan can cover every conceivable situation or problem and that there may be special circumstances which warrant consideration. Employees having an individual problem not addressed by the procedure should bring it to the attention of the Deputy Executive Director for review.

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Remote Work Agreement Sample 1

Source: Anonymous

Sample Virtual Work Agreement

INSTRUCTION

This form is to be completed by an employee who has been approved for a regular, recurring virtual work arrangement. The agreement needs to be reviewed and signed by the employee's supervisor, then emailed to_____

AGREEMENT

The following outlines the specifics and expectations of the Virtual Work arrangement which you will assume as of _____. Please note that any Virtual Work Arrangement which exceeds a maximum of three days (or 22.5 hours) must be approved by the

department's appropriate Cabinet official.

1. The details of your arrangement are as follows:
2. The following changes in my approach to communicating work expectations and/or office policy and practices will be implemented to accommodate this change:
3. The following changes will be made in the way you communicate your questions about work, provide finalized work and maintain connections with this office to accommodate this change:
4. We will meet every _____ to discuss how this arrangement is going and to adjust as needed.
5. (Remove if arrangement has already been piloted for 6 months). This arrangement will be piloted for _____ amount of time, at the end of which time, we will review the arrangement and determine whether it will continue.
6. If at any time this arrangement no longer serves your purposes or the needs of the organization, the arrangement may be discontinued.
7. This work arrangement will have no effect on your salary or job responsibilities.
8. It is expected that this work arrangement will not reduce your productivity.
9. If organizational needs require, there may be times when it will be necessary for you to forgo your virtual work arrangement to support unusual projects or conditions in the office. It is expected that you will make every attempt to adjust your schedule accordingly. Similarly, I will take into consideration your virtual

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work arrangement when scheduling meetings or gatherings that require all staff to be in attendance or when distributing workload.

10. The following access will be necessary for this virtual work arrangement, and will be provided by the City for your use:

11. The following equipment will be necessary for this virtual work arrangement; if you do not have access to any of these items to complete your virtual work, please inform your Supervisor:

    a. City-issued Computer

    b. Internet Connectivity

    c. Phone

12. Additional provisions for this arrangement include: (Please List)

I have read the above and discussed the terms and conditions with the undersigned supervisor and agree to all aspects of this agreement.

Requestor's Name	Requestor's Signature	Date
Supervisor's Name	Supervisor's Signature	Date
Appointing Authority Designee Name	Appointing Authority Designee Signature	Date

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Remote Work Agreement Sample 2

Source: Anonymous

Telecommuting Agreement

Employee Information

Name: _____ Hire date: _____

Job title: _____

Department: _____

FLSA status: ☐Exempt ☐Nonexempt

Temporary work address: _____

Phone number at work location: _____

Employee schedule: _____

The employee agrees to the following conditions:

The employee will remain accessible and productive during scheduled work hours.

Nonexempt employees will record all hours worked and meal periods taken in accordance with regular timekeeping practices.

Nonexempt employees will obtain supervisor approval prior to working unscheduled overtime hours.

The employee will report to the employer's work location as necessary upon directive from his or her supervisor.

The employee will communicate regularly with his or her supervisor and co-workers as predetermined by the supervisor.

The employee will comply with all Authority rules, policies, procedures, practices and instructions that would apply if the employee were working at the Authority's work location.

The employee will maintain satisfactory performance standards.

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The employee will make arrangements around caregiving responsibilities so that work may be performed.

The employee will maintain a safe and secure work environment at all times.

The employee will allow the employer to have access to the telecommuting location for purposes of assessing safety and security, upon reasonable notice by the Authority.

The employee will report work-related injuries to his or her manager as soon as practicable.

The Airport Authority will provide the following equipment: _____

_____

The employee will provide the following equipment: _____

_____

The employee agrees that Authority equipment will not be used by anyone other than the employee and only for business-related work. The employee will not make any changes to security or administrative settings on the Authority equipment. The employee understands that all tools and resources provided by the Authority shall remain the property of the company at all times.

The employee agrees to protect Authority tools and resources from theft or damage and to report theft or damage to his or her manager immediately.

The employee agrees to comply with the Authority's policies and expectations regarding information security. The employee will be expected to ensure the protection of proprietary company and customer information accessible from their home offices.

The employee understands that all terms and conditions of employment with the Authority remain unchanged, except those specifically addressed in this agreement.

The employee understands that management retains the right to modify this agreement on a temporary or permanent basis for any reason at any time.

The employee agrees to return Authority equipment and documents within five days of termination of employment.

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By signing this Agreement, I acknowledge I have reviewed Procedure #200-073, Telecommuting, and that I agree to the conditions in the Procedure and this Agreement.

Employee signature: _____ Date: _____

Manager signature: _____ Date: _____

Director signature: _____ Date: _____

Deputy Director of Aviation signature: _____

(FBO, Ops and Facilities) Date

Deputy Executive Director signature: _____

(Finance & Admin, HR, Communications, Leasing, IT) Date

Executive Director signature: _____

(Executive Department Only) Date

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APPENDIX E

Verbatim Comments, Airport/HR Manager Survey

[If your airport did not allow (or require) any airport employees to work remotely for any period of time during the COVID-19 pandemic (March 2020–March 2022)], please describe the managerial decision-making process that determined remote work would not be necessary.

- Single Employee.
- Only airport manager employed at airport.
- No employees. Only one volunteer.
- The airport was considered essential by the city.
- We only had 1 employee and or 1 additional part time employee.
- Work could not be done from home.
- Remote ops not conducive to maintaining continuity of operations.
- Airport operations are 24/7.
- Small GA Airport-The work is servicing airplanes.
- The employees working at the Airport require physical presence for operation of the Airport.
- All employees are considered essential for day to day operations.
- Only have 1 employee so distancing was not an issue.
- We do not have any Airport Employees.
- Only have one employee other than myself.
- All work is physical and presence is essential.
- We have part time staff and they rotated shifts to complete work to support airport operations.
- Short Staffed before Covid.
- Essential Workers.
- This is a one-man operation here at this small airport. Someone had to be physically here to ensure safety of the airport, fuel resources, and other necessary details.
- It was their decision to work.
- Able to self isolate at work office majority of the time. Primary duties require onsite ARFF and 139 requirements.
- Someone has to be on site for full service fueling.
- We followed City remote work guidelines and determined that working from home was not feasible.
- Single employee.
- Airport has one non paid employee.
- We kept employees separated with minimal interaction with each other and customers.
- Work was outdoor work, assist fueling aircraft, moving aircraft, maintaining grounds, mowing, weedtrimming, etc.
- No airport employees.
- We're a one man show.

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- Only one employee.
- Extremely low manned airport, kept our distance while doing the work needed.
- We have only one airport employee.
- Small staff- 1-5 employees. Instituted an every other day work schedule to split staff up. This ensured qualified people were available at all times for coverage. If one person got sick then the whole staff was not affected.
- Small rural airport with airport manager residing on site and primary work was conducted alone outdoors.
- Only two employees, whose duties included fueling of aircraft and airport maintenance.
- Staff of only two to operate and maintain airport grounds and facilities. Impossible to do that from a remote setting.
- Airport employees are essential workers.
- skepticism over necessity of doing so.
- Maintaining distance from transient customers would be possible. Physical presence at airport necessary to maintain fuel equipment and tow aircraft mandatory for business.
- Cannot fuel planes remotely.
- All personnel have an on-site role that cannot be fulfilled from home.
- Work could not be done remotely.
- Our FBOs that are located out at the airport are privately run.
- Only had two full time and one part time employee. All work was done on site.
- Airport was considered essential for employee staffing. Unable to accommodate airfield operations remotely.
- The books were the only work that could have been done remotely and I came in every day so no need for that.
- We work from the city offices. We have 3 employees in the office
- I was not the manager at KMWL at the time. I know as an employee at KRBD, we were deemed essential. Since some had to work on premises, we all did.
- Due to the level of staffing at the airport all personnel were needed on site.
- Small staff so we were able to plan out much of work to not work in close proximity of each other. Wore mask when appropriate. Emphasized hand washing and sanitizing practices.
- Our airport is staffed by 2 individuals. Our revenue depends on the airport being open daily to the public. Refuel of aircraft cannot be accomplished remotely. The airport followed all guidance set forth by the city and we remained open and COVID free the entire time.
- If you are sick, stay home. If you are afraid, stay home. Otherwise, it is business as usual.
- Only 2 employees at the airport so no need to work from home.
- Airport staff were needed on field to operate and manage the daily operations of the airport. Remote work for the airport was impossible due to the nature of the assignments of staff.
- No employees at this airport.
- Single employee. Airport Manager, only.
- Airport is not regularly staffed so the only time employees were there was for repairs and maintenance.
- That decision was easy. There are no job positions at this airport other than direct customer and/or aircraft transactions, like customer service and line service. Infrastructure related tasks like mowing airport grass and maintaining the airport terminal also require employees to be present to accomplish those tasks.
- The ability to have distance between member using the airport as our airport is unmanned and a GA only.
- Only one employee.
- Social distancing, making sure lobbies were clean and provided hand sanitizer. Making arrangements by phone was the preferred method of operation early on during the pandemic.
- Only 2 employees to manage and run the FBO. Wasn't an option to work from home.

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- It was decided by the Public works director that we would work here at the airport to better serve our tenants. We have an air med helicopter based here on the airport.
- With little to no traffic during COVID the airport decided to rotate office staff. Our front desk person worked 3 days per week and the office manager worked two days per week. We paid them 40 hour but it was important to us to minimize possible contamination of both office staff at the same time. This schedule allowed for the office staff to not be here at the same time during COVID
- The day to day operation of our GA airport does not allow for working remotely.
- No employees located at the airport.
- We depend upon frontline people that are unable to work from home. We set the example. We reduced contact but did not give the option of working from home.
- No employees.
- Secured building, no physical public access, all work done by phone and computer in locked office. Only two employees total.
- All workers were considered essential & were required to report to work.

How did you choose the employee groups eligible for remote work?

- The decision was based on the operational need to be physically located at the airport. Those employees who complete majority of work by computer or phone were eligible for remote work.
- Need.
- Employees were eligible for remote work based on their work assignment and in some cases according to their work locations in which it was impossible to social distance.
- Their close contact with others.
- Was only allowed for those that had enough to rotate the shifts from in person to remote dealing with work that could be done over the phone and internet (getting Purchase Orders, Lease updates, tenant notifications, etc.).
- I can answer phone and email from home, so while the initial shutdown was going on, I continued to run my airport from home.
- Ability to perform work remotely.
- The Administrative office is very small with 5 people. The lead positions were requested to work remotely when possible and to have the least amount of personnel on duty to reduce the staff.
- Based on each individual's capability of working remotely.
- Not safety or customer service related that was needed on site.
- First was to gauge interest (Very low). Next was could the majority of their work be completed remotely?
- Staff with jobs that didn't require them to be on-site were allowed to remote work on a rotation basis, some remote, some in, so that work could continue with the staff that had to be on-site.
- Employee has no reason to be on site and possibly exposed. All work could, and was completed off site.
- Based on the ability to work from home provided technology to log into their work computer.
- Employees were eligible for remote work based upon their technical capabilities to work remotely. The employees were largely administrative functions.
- Based on necessity for work on the physical plant.
- Any employees who were not required to be on site to perform work were eligible to work remotely.
- By determining if their job tasks could be completed by working remotely if provided the necessary equipment.
- Whichever could perform the essential duties of their job remotely, we allowed during the pandemic at some point. We also offered hybrid option.

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- Those that could.
- It was protocol from upper management
- If it was possible for them to do the work remotely, we were agreeable. Most jobs, custodians, parking lot attendants, maintenance personnel, and public safety could not, but most administration staff could.
- Front line (field) staff were not able to work remotely. Most other staff were able to be considered to telework.
- Groups that did not have day-to-day operational requirements. Individuals that had childcare problems relating to school closures and those living with people in high-risk health groups.
- Based on job function, i.e. fire, police work can't be done from home.
- Nature of work. Some staff were entirely remote while others worked a hybrid, and some did not work remotely at all.
- Ability to perform duties away from the office.
- We picked the employees that would have the most challenging time being able to social distance.
- Anyone who didn't require face to face customer or employee contact could work remotely.
- All qualified, reduced hours and closed offices, minimum authority presence.
- These groups while essential, were not considered to be essential to be on site and could use available technology to work remotely.
- Work did not require an on-site presence.
- Job description.
- Regulatory compliance, essential workers, safety and security.
- Presence at the airport was not necessary.
- Essential versus non-essential.
- Positive Covid Test, City Standards.
- Left up to each manager.
- It was an option
- It was simple. Can your work be done from a computer at home?
- Those with office duties.
- Could perform more than 80% of job description on line.
- We chose a skeleton crew to be at airport as needed and as required since we were an essential governmental function. The rest were allowed to stay home or come in and distance.
- Job duties allowed them.
- Hands on work like maintenance and inspectors had to stay at airport. Office workers remoted.
- Employees in jobs that cannot be performed remotely (maintenance, operations) were generally not eligible for remote work. All other jobs were allowed to perform work remotely at times.
- We have very few employees that don't have to be here to work. We have one accountant and she was able to access her computer remotely, so it was easy for her to work at home.
- Office workers and admin workers.
- Within close proximity of duties
- Chose employees who could work remote did not perform airfield, security, etc.
- Employees requested to work from home and it was reviewed by the senior management in their department to see if they would be able to manage their workload from home. Some employees who were eligible (an attorney and the CHRO + assistant) elected not to work from home.
- If the position could be conducted remotely (using laptops, technology, etc), employees were sent home to work. We ask what work can be done remotely and how often?

What technology tools enabled employees to perform their jobs in a remote manner?

- Computers, email, internet and cell phones.
- Internet, laptop, and cell phone

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- Employees who worked from home were provided with laptops, additional computer monitors, and in some instances cellular devices. In addition, employees were provided with access to VPN into the Airport's shared computer drives and other software.
- All safety equipment provided.
- Zoom, Microsoft Teams, and other web-based meeting platforms. The Microsoft suite of applications also helped.
- Computer and wifi.
- Telephone, Email.
- Linking to on-site computers remotely as well as cell phones.
- Laptops and internet
- We use the google platform (google meet + docs, slides, etc.)
- Laptops with webcams, high speed internet and a virtual environment (SharePoint/Office365).
- Laptop/tablet computers; Microsoft Teams for meetings.
- Laptop & Teams Meetings.
- We utilized a software provided by our outsourced MIS company.
- VPN; Zoom and other web meeting interface software.
- Remote internet access to airport secure servers.
- All staff were provided laptops with VPN connections to the airport network. The airport network servers had previously been upgraded to allow for remote connections while staff travels or is otherwise off site.
- Desktops or laptops, telephone or cell phone, internet.
- Laptops and cell phones.
- Good IT support, VPN access.
- Laptop.
- Computers that were tied into our computer network.
- Laptops and tablets, phones forward to employee cell phones (often their personal cell).
- Laptops, cellular phones and remote licenses for certain computer programs.
- Basic laptops, MS surface device, cell phones.
- Laptops, virtual meetings, mobile phones.
- Laptops and VPN's as well as Teams.
- Laptop computers.
- All had PCs or laptops equipped with remote video conferencing applications, such as Zoom or Teams. All had to have cell or landline phones during all working hours.
- Microsoft Teams combines with regularly used communication mediums
- Laptops, Google Meet, and a multitude of software specific to the job.
- Laptops, internet accommodations, ability to take office furniture and equipment home, virtual meeting software.
- Zoom, Teamviewer (remote access), laptop, Microsoft teams.
- MS Teams.
- VPN technology.
- The employee was allowed to take his computer and printer home. He could work through email and by scanning documents.
- Laptops, remote login, cameras.
- Laptops and mobile devices.
- Computers, Apps for team meetings.
- Zoom, Teams, cameras, wifi, laptops.
- Laptops, a VPN and cell phones.
- Microsoft teams, virtual conferencing software.
- Laptop provided and ability to log into email system and files remotely.
- Phones, computers, internet, VPN.
- Laptop and WiFi

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- MS Teams, VPN remoting in to office desktop, cell phones.
- Laptops using a VPN, telephones forwarded to cell phones.
- Laptop computers.
- Computers, phones and zoom.
- Email/Phone/Virtual.
- Zoom, computers, internet.
- Employees were issued laptops, iPads, chromebooks, computer screens, connection cables to their TV, wireless keyboards, scanners/printers.
- MS Teams, Cloud Systems, Wi-Fi, Laptops, Monitors, Multi-factor networks,

What considerations are there for new employees regarding possible remote work arrangements?

- We no longer are offering employees the option for remote work.
- None.
- For new employees, considerations regarding remote work arrangements are given in providing equipment while also ensuring they are in compliance and received onsite training and work requirements.
- None in place.
- No new employees - does not apply.
- We did not hire additional personnel during the 6–9 months from the original shut down in March 2020.
- We offer a flexible work arrangement to all staff. They can share how and when they'd like to work and if it can be done, we do it.
- It's considered in job design, IT equipment, and our business continuity planning (COG). It is also discussed in the hiring process. We are County operated and most employees are expected to work during natural disasters. Some have duties off airport such as shelters or PODs.
- No additional employees considered in the near future.
- None at this time.
- None.
- None.
- Most staff have transitioned back to on-site work. New employees are handled on a case by case basis.
- Flex time and remote work allowances for applicable positions.
- Currently, we do not have remote work options.
- We are trying to be more flexible.
- None for new employees.
- No different considerations for new vs. old employees in this regard.
- Usually have a mandatory on boarding period (e.g. 3 months) where must be in the office, then can be considered to telework.
- Very few and dependent on individual circumstances.
- Remote work will be considered on a case-by-case basis but prefer staff work from office
- Our airport authority ended all remote work Jun 2021; we are revisiting it to enhance employee attraction and retention and have drafted a remote work policy that we expect to implement in the next couple of months.
- None.
- None.
- None.
- All were eligible.
- Still considered if necessary but on site is preferred. Airports require a high level of in person interaction from all depts. most of the time.
- Employment within one of the groups eligible to work remotely.

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- Job description.
- Remote work is not standard and is no longer allowed. Employees can request ADA alternate work arrangements but that typically involves workplace modifications not locations.
- None.
- Depends on their roles and responsibilities.
- Must be in a position that would be able to work from home. No customer facing positions, or maintenance positions were eligible on a daily basis.
- Flexible time.
- Since COVID and prompted by the competitive employment environment and gas prices, we instituted a permanent limited remote work policy allowing certain regular employees to work up to 2 days per week remotely while managers and directors can do so no more than 1 day per week.
- None.
- On as-needed a basis only.
- None.
- We will not allow remote work again. It was a one-time thing.
- We have a small staff, this did not come up. However, some remote work would be considered for any employee able to accomplish their job without being on site.
- None, the City of Midland no longer supports remote work instead of working in office.
- Not at this time.
- Laptops/Cellphone or Allowance.
- Will consider job duties to determine if a flexible/remote work arrangement is possible.
- We currently have returned to a hybrid schedule. Those that can work from home are able to do so on M and F. They must be in the office T, W and Th. Some departments are already deviating from that schedule. Some people are back full time, others are staying at home up to four days. New hires are working up to five days a week depending on training needs.
- Managers designate prior to hiring whether the position can work remotely. We have policies and procedures in place for remote work arrangements, how to manage in a remote environment, and how often. We also have hybrid work arrangements - employees work part time in the office and part time remotely (primarily those in office environments).

What are some methods your airport used to manage employees working remotely?

- Use of virtual meetings and use of daily progress/job reports.
- Need.
- We utilize the City's Virtual Work Policy, as part of which is an agreement established in between supervisor and employee. In addition, we conducted weekly supervisor meeting and increased communication via email and letters mailed to all employees to provide updates.
- Kept them separated.
- Daily reporting of work accomplished and virtual meetings.
- I signed in from my phone.
- Telephone, Email.
- Virtual meetings weekly as well as cell phone access.
- Nothing other than normal KPIs for all staff.
- Goal setting and regular check ins.
- Work performed and cell phone contact.
- They were required to come in once a week on different to meet with their manager and pick up work.
- Some of the methods our airport used to manage employees working remotely included using daily "catch up" meetings, email, and conference calls.
- Staff worked in office two days a week. Supervision performed then as well as via telephone and email.

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- Office 365 allows for collaboration in a team environment. Daily meetings of the entire remote staff with those on site. Other meetings as needed to link staff on specific projects.
- Aviation Department's management meetings occur on Wednesdays, so we have all personnel working in person on that day. The other weekdays, remote work employees rotate. Typically, all remote work employees have 2 days in office and 3 days working remotely. This proves to be a big incentive—if the tasks gets done, remote work can continue. If things start to slip, in-person work might return.
- Management check ins, productivity and time-keeping.
- Regular check-in and touch-base.
- Laptop computers with access to work related facilities, city hall, etc.
- We trusted these employees to get their work done the same way they would if they were on site, and they did.
- Daily check in with supervisor, our work group developed a daily staffing report which identified staff attendance as well as who was on site and who was teleworking. This helped us track when call off rates were escalating and continuity of operations planning needed to be implemented to meet staffing requirements.
- Frequent check-in calls and expected work product.
- Required duties complete, reply to emails.
- Employees were required to document daily work (by hour) and submit weekly to supervisor. These were tabulated and held by Finance for payroll and audit purposes.
- Daily check in with managers.
- Teams, Skype and Zoom meetings.
- Employees were required to converse daily with their supervisors daily to describe tasks that were accomplished during each day.
- Regular check in and frequent communication.
- Trust.
- Virtual check-ins, Microsoft Teams chat/video/etc., frequent communication.
- Constant communication.
- Morning daily calls but other than that no real way to evaluate continued performance as this was all new to our airport.
- Expectations versus work product.
- Communication was essential.
- Integrity, honesty, daily contact.
- Email and Microsoft teams.
- Zoom and Teams check in meetings.
- More regular check-ins and communication.
- Timesheet tracker, regular phone calls.
- Daily calls.
- Voice check ins. Most were not able to do their normal jobs since their functions required them to be on site, but no work available since we had virtually no aircraft traffic.
- Teams
- Schedule and phone check in.
- Employees working remotely are expected to be available by telephone or email and to accomplish their work. Our staff did this with minimal oversight required.
- Phone calls and emails.
- Work log, reports and email check-in.
- Phone/Email/Virtual.
- Communications.
- We use Google Meets and Hangout. I am sure there are other tools departments are using. I keep a Hangout chat open with each of my staff and a group chat. This way I can see when they are away from their desk or I can ask them an individual question. I also liked to have

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a Google Chat to see each other face to face rather than just a phone call. I am not sure what worked for all others.
- Policies, handbooks, manager training, clear expectations.

How have your hiring practices changed since, and as a result of, the COVID-19 pandemic?

- Many of our hiring practices have gone virtual due to the pandemic. For example (interviewing).
- We've reevaluated a lot of the job descriptions and removed anything that wasn't truly required and had likely been rolled over year after year (Degrees, driver's license, certifications . . .) in hopes of having a bigger pool to choose from.
- Minimally, however we discuss the requirement to work during disasters.
- It is harder to attract candidates.
- They have not but it is harder to actually find candidates to fill positions.
- To date they have not. Still evaluating situation.
- Many of our tools for hiring are now digital including remote interviews. This allows us to recruit and hire more quickly if the hiring team is not all on site at the same time.
- If we are hiring for a position that can work remotely, we advertise it.
- We conduct first round via Teams and then in-person for final round.
- Our requirements for education and experience have been lowered, our wages are increasing and we are trying to be as flexible as possible about shifts and wfh.
- All done online.
- They are about to include remote work options in some jobs. We conducted virtual interviews to shortlist candidates and continue to offer that option for non-local applicants.
- Nothing comes to mind, we still have the same minimum requirements. We do a lot more virtual interviews.
- We have delayed in hiring and backfilling because of budgetary concerns.
- More online, but that was the initial stage for us anyway. It just went thru more online steps until a final face to face interview.
- To hire quality workers in such a competitive market, flexibility is necessary.
- Virtual interviews are common, but the final must be in person. Being an airport and FBO all job descriptions require employees to be physically present, so we will not hire employees who cannot commute into the airport.
- For a couple of months the interviews were held by phone. No more paper resumes, online only.
- We can't hire. [. . .] We are at 2/3rds manning.
- Not really, although it has been more difficult to fill vacancies (few as they are).
- We haven't changed our hiring yet, and it shows, we are getting 10% of the applications we used to get for Airport Operations positions. When I've contacted universities, I've been told that new graduates want jobs where they can work from home.
- We do a Google Meet before we bring them onsite and use the video chat rather than a phone call. Otherwise, everything pretty much stayed the same.
- Evaluation of jobs to determine if it can be performed remotely, interviewing remotely, and onboarding (severely impacted).

How has your overall HR function changed since, and as a result of, the COVID-19 pandemic?

- Vaccinations were required initially, however that is no longer a requirement.
- The overall HR function has changed by converting many of our processes and the processes utilized by our employees to digital.
- Time off management is the biggest change.
- The economic impacts of the pandemic on budgeting, recruitment and retention have impacted human capital significantly and will have a lasting effect. HR is recruiting for many

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more positions and the pool of applicants is shallow which creates greater demand on training resources, some have had limited availability until recently.
- We have not hired any additional employees in the last five years.
- The additional duties as they relate to COVID protocols and ensuring the safety of all of our employees and customers.
- Other than having to develop, and adjust/refine a COVID policy, it really has not.
- This is better answered by HR. From the HR customer perspective, they have not adapted fast enough nor innovatively enough to meet the changing conditions of job/hiring market, responding with policies and procedures to support the field in a timely manner, etc.
- Greater turnover of employees forcing the need to hire a HR generalist.
- Having to rethink employee attraction.
- We interview constantly.
- HR was great providing the information regarding public health orders and being creative with work schedules and identifying challenges of remote work at the onset.
- More remote involvement, less in-person engagement.
- We are very siloed so I am not sure what has changed within HR but our interactions with them have been significantly reduced because of their individual remote work schedules.
- We've hired a recruiter to remain competitive and due to the dramatic increase in activity in Florida.
- Mostly virtual interactions.
- We are increasing base pay to entice new hires.
- Interviews have been conducted remotely. Hiring decisions have been accelerated in an effort not to lose good candidates to competing opportunities.
- We are understaffed.
- We spend more time talking to employees on Google, on the phone, by email, and always following up verbal conversations with an email to be sure the employee understood the conversation. As the CHRO I became the "mask police" keeping employees on track with wearing masks.
- The great resignation has impacted our organization, hiring, attrition - all has increased. HR burnout is high. How we communicate with our employees has changed. How we provide training has changed (learning opportunities are hybrid or remote). This is in addition to most HR jobs being primarily remote now - almost 100% of the HR staff works remotely part or full time. Customer service to the organization is impacted. More reliance on data. At the same time, some learning opportunities and organization-wide meetings have more attendees by offering virtual options.

Morale took a hit during the pandemic, what "fun" thing did your airport do that you can share?

- For our onsite workers we had an appreciation day where we provided them with food and snacks. In addition, we offered a number of online offerings including virtual wellness, and Thursday Water Cooler Conversations.
- Nothing really but our morale was never that bad.
- Online team building with scavenger hunt app. We competed against other departments and all the items were found outdoors or something you could do at home.
- Not so much fun, but communications changed. The County Manager now sends a weekly update that not only provides news, new employees, exiting employees, but also shout outs from employees, the public etc. The employees appreciate the communication.
- I wish I could pick one, but the truth is it hasn't been much fun.
- With all the restrictions we really had no opportunity to do "fun".
- We celebrate birthdays/holidays and have team building events at least quarterly.
- Holiday luncheon by department, team building, catered meetings.

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- We cooked out a few times but social distancing was in force.
- We didn't do anything different (in terms of fun). . . . . most employees were happy they still had a job to go to every day and that their jobs were stable.
- I produced videos to take the place of regular "All Hands" meetings. When able, we met (socially distanced and with the doors open) in a hangar to create a sense of community and physical togetherness.
- Employee pig roast, September 2021.
- Employee engagement such as inviting staff to share internally what they did as a family while everyone was at home.
- When the pandemic was over we had more in person lunches and get togethers than we did before.
- All-hands BBQ was held once all employees returned to their normal workstations.
- Our operations requirements didn't change and I don't believe morale suffered. While our commercial service PAX #'s declined about 30% our FBO experienced a 30% increase and the ATCT operations count for all groups increased accordingly. We are a Non Hub Primary and serve mainly destination resort clientele. Overall we were busier in 2020/2021 than previous years.
- No change - we try to make every day, pandemic or not, fun and embrace a team atmosphere.
- Nothing. There was a very strict crack down on essential employee interactions meaning no more team meals, birthday celebrations or even in person shift briefings. It really isolated staff and made a sometimes stressful job harder.
- Being more disjointed, one of the best things to come from the pandemic was instituting a bi-weekly all employee huddle where each department gives an update on their activities/ projects. Out of 135 total employees, we regularly have 40-50 who participate.
- More flexibility to give away ""free"" free time.
- Fun was CNX.
- Nothing about the pandemic was 'fun'. During the pandemic (April/May 2021) we moved into a new facility with an open office plan. The building has a 2nd floor outdoor observation deck; we bought a BBQ grill and use it twice a month to cook lunch for the staff and we often invite other city departments to join us.
- For Tenants we had poker run, outdoor food trucks with distant gathering for staff and tenants.
- Morale was never an issue. We maintained most normalcy as a GA Airport.
- I really wish I could think of something, the people with morale issues were the ones that had to stay here and work onsite. Remote workers seemed ok.
- Used Kahoot! to play games virtually and maintain a learning environment.

Please share lessons learned from your airport's remote work experience.

- It proved to be difficult to just jump into a remote work environment without much advanced planning. Hats off to our small IT section who was able to make it possible. I come from a paper driven airport system, many processes were manual, files are still kept in a filing cabinet. This was the most difficult part, having to pack up files that I was planning to work with and transport them back and forth to the office.
- A hybrid schedule can work.
- The biggest lesson learned is that communication is key in an organization and that our essential staff are the heart of the organization.
- It is hard on everyone when there is even one person out and working remote is not easy.
- Remote work is not ideal at all for the airport industry. The work being produced at home with other distractions abound are not conducive for a steady output of work. That coupled with the true need to have that face to face interaction with the many tenants and all of their facilities make me believe there is not a place for remote work in this industry.
- Certain functions can work that way, others can't.

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- It is much easier to handle paperwork, projects, and personnel being on-site.
- Not all employees desire to work remotely. It can be accomplished with a little planning and employers that are flexible will prevail in the competitive labor market.
- We were able to attain a workable model.
- Our remote work experience was limited. the biggest lessons learned would be the need for increased communications, and more emphasis on prioritizing tasks and assignments. There is always the underlying question as to whether it is a productive method of work.
- It took some adjustments in thinking, but overall, it was/is a successful endeavor.
- It took a pandemic for us to realize that work can get done in a different way. Employees feel like they get part of their lives back, not having to spend as much time and money commuting to/from work every weekday. If the organization remains intentional to connect and communicate with the employees, we expect to see continued positive results.
- Although some employees enjoyed the WFH initially, the majority of our staff are pleased to be back in the office and interacting with their teams.
- I didn't like it because all my work files etc, was in my office.
- The remote workers were not extensive (in numbers) nor time working remotely, so I would say it is workable, but I'm not sure we had any profound lessons learned.
- The practice results in reduced production and, occasionally, morale issues. It is not a sustainable model despite the allure.
- It worked better than expected.
- For the most part, no impact to operations. as travel picked up, operations staff recalled earlier than other departments. operations and other staff who could not work remotely were compensated with additional paid time off (on call, required to remain within specified distance).
- Be prepared IT wise for the next time or when necessary for other reasons.
- I don't feel that as much work got completed as when the employee worked on site.
- Realized that maintaining productivity during remote work is possible, and that it can be a method of allowing people to be more flexible in their work schedule. Flexibility in where and how you work can be a quality of life booster and an employment perk while still maintaining productivity expectations.
- Trust employees to be productive, give them the tools to be, and let the rest fall into place.
- Flexibility.
- Efficiency and overall work product increased.
- We learned it is possible, and that some are better at it than others.
- It has a place in the future of work, but most of us, myself included, enjoy the airport environment too much to work from home more than 1 or 2 days per week.
- "Worst thing possible for an institution. We all need ""human"" contact and seeing people in pajamas on a computer screen does not work well.—Airports are about customer service and we need to be there for the customer and trying to do that remotely does not fill the need of the users.
- Remote work may ""work"" for the lazy people, for any that feels uncomfortable next to other humans.
- Hope we don't have to ever do that again.
- It is not a good idea. People don't work as hard and communication breaks down.
- Providing flexibility to staff to work remotely from time to time has been very popular with staff.
- It was very difficult for those that were still working on-site to get info as quickly from the at home workers.
- Zoom was a big hit at the time, would rather have personal meeting now.
- None that come to mind. Business as usual for administrative functions at home as opposed to office work. Technicians were still required to be onsite

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- Being prepared from an IT perspective (increased tech support), helping managers "manage" remotely, how to effectively manage poor performers, give reviews and take care of the non-remote staff.
- Whether someone enjoyed a remote work experience is highly individual. However, it did decrease the spread of COVID. We mandated the vaccine and lost approximately 70 employees due to this requirement (termination/resignation). Managing a remote work environment is difficult and can be socially isolating. At the same time, it allows for more work/life balance in some cases. In other cases, employees are working longer hours - just tied to their home desk. Much work can be achieved in a remote work environment, however, that was in-person in the past. It forced us to be innovative and think outside of the box.

Please describe how employees were trained during the pandemic.

- Virtual meeting.
- Employees were trained virtually on a number of subjects utilizing Microsoft Teams.
- There were no changes to how personnel were trained during COVID.
- A few AAAE online courses.
- NATA fuel training.
- Teams for local and other webinars.
- City-required IT & HR training has been done online for years so that continued with remote workers. Required airport training (e.g. recurrent driver training) must be accomplished in-house.
- We continued to have the same Part 139 requirements and TSA training requirements so existing staff could be trained in person or remotely as available. New SIDA training was developed through an online tool so candidates could take the training independently.
- Processes were discussed on actions that needed attention when employees worked remotely. For example: incoming phone calls or in-person visits needed to be routed to the applicable person.
- It varied, but typically through zoom meetings and online courses.
- NATA Safety 1st as well as other online sources.
- Employees watched video, signed sheet and sign in sheet was sent in electronically.
- Utilized Teams for Company training or other vendor sites.
- Webinar and various virtual meeting formats.
- Badging training/tests were done remotely via personal computers. During the height of the lockdown we did as much training individually on personal computers as we could.
- Online with follow-up online testing (Class Marker)
- Virtual meetings, recorded webinars
- We used in-house training software to push out training and used teams to train remotely.
- Online training.
- Webinars and other virtual options.
- In-person training with social distancing in place, limited number of attendees. Some Zoom and web-based training.
- Training was assigned via our daily ops managing program Veoci.
- We trained as we normally trained, as a group in our conference room. South Dakota never had a shutdown or masking mandate.
- LMS was already in place at the city level for all training events. The airport was not a 139 airport and thus didn't need any 139 training.
- Online.
- Ppe usage.
- One.
- On line/zoom fuel provider training certifications.
- Outside taught training was done on line. In service training was conducted by airport staff.

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- Online training was accomplished.
- Online courses.
- Online.
- Mostly on-line training, with some in person training with social distancing in place.
- Memos were regularly sent out to employees. These memos had guidance from the CDC regarding safe work environments.
- Zoom and video.
- AvFuel Hub Training Portal online.
- All of the safety training we use is and has been computer based.
- Virtual training was used when possible.
- Employees were set up with technical tools and we used a lot of online training. Department meetings/trainings were held via Google and some people just had to come in and be trained onsite.
- Virtually. We used MSFT Teams and shared our screen. - This was for remote workers. On-site continued to be trained in-person.

Do you have any other thoughts to share on this topic?

- I come from a very small office environment. Our size, however, should not discount how critical our function is to our airport. I have always had a deep concern about infection and spread of COVID throughout our office and how crippling that would be for continued operation. Though teamwork and being able to work face to face with each other is very important to me, I believe that a staggered work shift and staying in "bubbles" to lessen the operational impact should there be an outbreak in our office be something considered.
- More content became remotely available and because the cost of travel wasn't there more training occurred. 100% remote training isn't sustainable as networking and other human interaction is very important.
- For a GA airport with two employees, it works for us. Everyone is happy!
- I prefer the ability to remotely work. I find that my productivity can increase due to the increased attention to my tasks during working remotely. There are less distractions that when in office, with people stopping in to chit-chat. Customer service and responsiveness have stayed at a high level and unless that changes, a valid reason of going back to full time, in-person work isn't found.
- Both zoom training and online training courses worked well.
- There is always a way to stay at work and I believe allowing employees to work from home that are not sick causes problems in the workforce.
- Being a lowed manned airport, we could do the work of the airport while keeping distance and being safe.
- I hope we never have another one
- To help reduce expenses, we implemented an "All-Hands Furlough" meaning all employees took one day off without pay per bi-weekly pay period. Furlough days were staggered to allow continued operations. No employees were laid off.
- Employees were not in close proximity for long duration of time.
- I feel like most airport positions require some on site presence. There's a certain level of team building that helps to be done in person. Allowing people to work from home when dealing with personal issues or during times of low activity can not only demonstrate a level of value the authority has for that employee, but also can be quality of life boon. This applies strongly to admin/office staff.
- Positions like operations, custodial, maintenance (boots on the ground require heavy on site presence) have to be handled on a case by case basis and cover demands of each airport.
- As we all know: when you've been to an airport; you've been to just one airport!

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- Work from home can be a valuable alternative for most white collar workers, however it will not be valuable for blue-collar workers, or for workers whom the public relies on for information or direct f2f interaction.
- One thing we haven't seen, but expected, was complaints by employees who cannot work from home. I don't totally know why, but we have done our best to take good care of employees throughout the pandemic, including generous merit and cost of living raises each year, even a mid-year cost of living raise in April 2022.
- Limited staff
- Airport manager felt it was very important to service in any way possible medical and military aircraft during COVID.
- I have a small staff with direct contact with the public. I have always had a procedure for mitigating the Flu season. I used the same procedure to handle the COVID outbreak, with increased surface sanitization.
- It was a difficult time but with good practices and attention to our work environment none of the Airport staff had or has COVID.
- Go back to pre-COVID practices and your airport will thrive again . . . we are.
- Use common sense.
- We monitored body temperature every morning prior to work and later during the day if any possible symptoms manifested themselves during the day. Employees were sent home if anything was amiss. Also, employees were encouraged to be self-aware of their health prior to even coming to work, and I don't remember having to send anyone home due to symptoms appearing.
- In a perfect world it would be nice to work from home, but an airport environment is constantly changing and to service passengers and keep the airfield safe, you need to have humans at work.
- It is my hope that we will continue with a hybrid workplace BUT allow managers to best decide who should work remotely and how often. I could tell by the goals and objectives that were met how well my staff was working remotely. I think it's still a BIG discussion for management at our airport.

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APPENDIX F

Verbatim Comments, Airport Employee Survey

If [you were trained via distance learning at any time during the pandemic (March 2020–March 2022)], what is your perception of the quality of distance learning training versus in-person training?

- It's good, although highly reliant on the quality of the presentation material and energy of the presenter while being highly susceptible to being ineffective based on the trainee's degree of attention.
- Gradually got better as time went on.
- For annual training, providing distance learning was more efficient.
- We had some teleconference training, and it was hard to stay focused and not work on other items with the activity running in the background.
- It all depends on the trainer, but this is also true of in person learning. I did find that hybrid attempts at trainings or meetings were very poorly done most of the time.
- In person can be better
- Not the same level of interaction – can't build off other participants as well - but still effective
- Acceptable
- In-person is superior training. There is value in distance learning, however, the collaboration and learning discussions that take place in-person are critical to me for internalization of the knowledge gained.
- I liked self-paced courses, but all-day courses done remotely were very tiring.
- Not as good. Hard to make connections when not in person.
- In-person is always better, but I felt the virtual training was of good quality and valuable.
- 40% of in-person training value was experienced
- Rough at first due to learning curve on new computer software including MS Teams, Zoom, and One Drive, but no problem now!
- Mostly good except it cannot replace some in person training like some portions of driver's training on the airfield.
- Quality of presentations seems to remain the same, class engagement and ability to monitor who is actually paying attention is significantly decreased
- Good, but less interactive, less engaging than in-person
- Not as good.
- The same engagement with the materials and trainer cannot be achieved virtually. It's much easier to listen to a training distracted than in person. But provides a good option.
- Distance learning can be as effective as in-person learning, but is not necessarily effective. I found interactive and break-out elements particularly helpful, actually increasing ability to participate where in-person would have been more challenging.
- Equally as good
- Good, but scripted and lesson plans didn't allow trainees to apply the new techniques during role-playing.

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- I believe in person is always better.
- Not as good as in person
- While instructors tried to be engaging, I felt that the "human element" that was missing from the training was a determent to my overall learning. I felt it was very difficult to remain focused and engaged.
- The training was adequate.
- In-person is more inclusive and more impactful to gauge what is being taught. With distance learning, it seems as if you just go through the motions to get it done, little to no interaction.
- In general, less effective but really depends on the topic
- I think the quality is there for some trainings, others cannot be duplicated in a virtual environment.
- Distance learning worked very well.
- Quality of distance learning was excellent due to the training provider.
- The perception of the computer-based training with video demonstrations provided awareness and was presented as a guide. Yet, it was effective.
- Slightly worse in general, though for learning how to do something detailed a little better due to screen sharing.
- I took mandatory CLE classes. Same.
- With proper equipment (cameras/mics, large enough screens) and motivation to pay attention (trainees WANT to learn, subject matter is interesting/presented in an engaging manner) virtual works just fine. All parties must know HOW to participate and be comfortable with it. New rules/guidelines sometimes need to be implemented and/or adapted to the individual audience.
- Good as long as my internet service was working OK.
- It is not the same experience as in-person training. When attending distance learning training, it is too easy to be distracted and to lose focus on the training.
- The online computer-based training was just as easy to complete at home as in the office at the airport. However, some training subjects for our team requires in person on site interaction.
- Mixed—some training was easier via distance while some was significantly more difficult

What are the benefits to working remotely?

- I get more done as I usually start when I wake up. I get to spend a lot more time with my family and mix my hobbies in my daily routine. I save money on gas and wear on my vehicle.
- More time for self-care like working out, walking dog.
- No commute. Greater flexibility of work hours. Lower stress environment.
- Less travel time, more efficient due to fewer interruptions, better work/life balance
- Short Commute, less expensive,
- Shorter commute, better quality of life, similar productivity as in office work.
- I was able to stay at home with my kids a day or two a week during the Fall 2020. But for the most part I worked very little remotely.
- Quiet for reading if no others around.
- Work life balance and flexibility/saving money and time without the commute/quiet when needed for concentration
- There were days I only had online meetings so working from home allowed me to save gas.
- Save on gas, can attend more back-to-back meetings, more time to do core work tasks, ability to attend to health easily/over lunch break, comfortable and free to work however you want to in a way that is most productive for you and your team.
- I can work longer during the day with breaks to eat or take care of minor household tasks. Cutting down on my commute was great for my time and the environment.
- No commute, quiet office time.

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- I have the ability to use my time effectively, and with minimal distraction. I do not support working completely remotely, but do support a hybrid.
- No commute, more time with kids, less effort to get ready
- I enjoyed not having to commute. Being able to do small tasks around the house.
- I love the hybrid model. I get more "quiet work" done more quickly at home due to no interruptions and no "get ready for work" time.
- Tend to be more productive with less interruptions or distractions.
- Saves time and money commuting, meals, work attire. Reduced stress levels, comfortable environment, greater flexibility
- My position requires being on site, can't inspect a runway from my couch. Other office duties are transferable
- Fewer interruptions, time savings with no commute, gas cost savings, less stress
- Not hardly any benefits
- No commute means more time for activities before/after work. Allows for short breaks to do various short chores like switch a load of laundry. Meals are better prepared and healthier - have more equipment than the office breakroom (ie a toaster oven, stove top, oven).
- The ability to get more sleep and take care of personal and work tasks simultaneously. No commuting to work that resulted in less fuel expenses and risk of getting into a vehicle accident.
- Personal location options, reduced commute. Fewer benefits to Airport, more for personnel. However, we worked remotely very rarely, and much work must occur at the Airport. On-site staff carried more workload than remote workers because of the limited options for tasks which can be successfully completed off-site.
- More flexibility with schedule and when work is performed
- flexible work schedule, ability to handle family/personal items, better work/life balance, no stressful commute, saving on gas/tolls to/from work, saving personal vehicle usage, ability to work longer hours instead of having a pre and post work commute.
- Personal expense reduction.
- Less interruptions with officers walking into my office.
- Flexibility in day to get work done
- No commute (reduce carbon footprint), more time with family, save money/time
- No driving expenses.
- No after-school care expenses.
- No rushing to get to extra curricular activities.
- Flexibility, more productive, reduced stress (during COVID), more focused (less non-work related interruptions), reduced commuting (better for the environment, better for my sanity), learn and use new technology (i.e. Teams, Zoom, Skype, etc.)
- I do have some flexibility if I need to take care of things at my house (e.g., being there when a contractor needs access).
- More personal time for myself. Gained 5 hours from not having to commute. I would sign-off at 5 and get an hour of exercise in by 6 when I normally get home from work. Less gas and wear and tear on the car. Lower car insurance since it was not longer used for commuting. Gained 1 hour in the morning not getting ready for work. Ate better-I was home able to put dinner in the crock pot and do better meal prep. No distractions from the office water cooler conversations.
- Flexibility.
- Independence/Autonomy.
- Productivity.
- Multi-tasking with home/family responsibilities.
- Greater flexibility, save on commute time and gas, fewer in-office distractions (water cooler conversations)

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- Improved work-life balance, comfortable environment, cost savings
- Having the flexibility to stay home when you only have virtual meetings that particular day.
- Able to do chores around the house.
- Less interruptions, more efficient, able to work longer hours, no commute, more precise meetings, save money on gas and clothing, more time with family
- No interruptions during workday; save money on commute and meals; work harder remotely taking fewer breaks/lunches and the time commuting I am working.
- Less time with my commute and more time to see my family.
- Flexibility.
- You should really offer a choice of hybrid, work remotely a few days and in office a few days. In person interaction is helpful but not necessary every day. However the savings in personnel time of not having to drive in and out allows for more work, and better quality of life.
- More efficient.
- No commute time/more personal time. Somewhat more flexible hours.
- Work flexibility. no commute, eating lunch at home instead of eating take-out.
- While working remotely, it eliminated travel time which is approximately 2 hours daily.
- Less stressful. More productive as there were less interruptions.
- Fewer distractions, cost savings on transportation to and from work.
- Lack of commute; ease of completing individual work.

What are the challenges to working remotely?

- Must have direct contact with the team and the facilities. Impossible to see the frontline through the proper perspective from that great a distance.
- Staying focused; stopping work at regular time.
- More distractions. Less sense of community and connection to those you work with. Less access to technology (better printer, larger computer monitors, etc).
- Less time for collaboration with peers, limited interactions with tenants, businesses, etc.
- Miss the small things you notice on-site when walking around an airport. Many times, minor maintenance issues will become major before they are seen.
- Face to face interaction and quick questions are much more formal in a remote setting.
- Not feeling productive.
- I need to see the airfield and a lot of resources available at work.
- Team interaction/general information sharing in less formal settings is lost, which is a good way to keep silos of information from forming/some elements of the job have to be done in person (checking concessions, etc.) Recommend hybrid work solution.
- Having family in the house that think you can just get up and do household things during the work day.
- Disconnection from the operation, loss of face time, relationships suffer.
- No personal connections with employee, tenants, etc.
- The challenge is that we operate a campus that has physical and operational challenges. You need to be on site to see, know, and ultimately manage what's going on. The personal relationship effects are also significant. You need positive relationships to be effective and you can't that through virtual meetings.
- Less contact with team, strained work relationships, home responsibilities encroaching on work.
- Communication can be difficult via email. Sometimes a simple question takes a while to answer, when stopping by someone's office is easier.
- Sometimes, it is best for me to meet with people face to face, thus my affinity for the hybrid model. I also have 2 monitors at work, so it's easier for me to do detailed spreadsheet work.
- Not interacting with co-workers for little conversations that normally take place around the office that helps with getting stuff done.
- Lose participating in the team environment and spontaneous collaborations.

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- Position requires being on site
- Isolation, inability to reach people, tools, equipment, resources left at office not available at home
- Computer problems at home, computer support
- Engagement with employees. Especially for those newer to the organization. Airport wasn't prepared for remote work (i.e., all had desktops and no laptops). Far less active throughout the day (refilling coffee is 5 steps away in the kitchen instead of 100+ to the breakroom, same with restrooms, etc.). Fear of colleagues and supervisors questioning if you're being productive and always feeling the need to be available.
- Distractions from family members at home. Not able to monitor staff productivity or access to non-digital files.
- Staff tend to "multi-task" during meetings, reducing effectiveness and engagement. Many tasks require on-site personnel, or are completed more quickly through on-site equipment and systems. Increased security concerns and required mitigation with remote workers.
- Staying on task when there things at home that also need to be done. Distractions from family and pets.
- In-person management of employees and operations.
- Staying connected to the team; pulse of the airport.
- The face to face contact, you need that in life.
- Collaboration with other work groups
- Lack of personal connection with colleagues, balancing child's needs with work (lack of child care).
- No dedicated office space.
- Always "connected" to work.
- Technology challenges (wifi connectivity and Internet speed).
- No face-to-face, personal interaction. Virtual connections are not the same.
- Less "water cooler" knowledge. Less camaraderie. No "dedicated space" to work (when your home is also your office, the lines between work-time and non-work time blurb sometimes).
- I don't get to interact with members of my department, I am unfocused, I don't have my full work station capabilities, I feel out of touch with my airport(s).
- Computer issues. We didn't start working remotely as fast as others because our IT department set it could not be done. Once we did get home, they provided us all the equipment but if you ran into issues the answer was always "bring your computer in".
- Collaboration.
- Interaction with co-workers.
- Lack of interaction with co-workers and project management. "Hallway conversations" are very good opportunities for interaction and quick responses if needed.
- Communication - less face-to-face communication causes more misunderstandings, more impersonal, less connected.
- Limit to some resources usually found in offices, personal IT or connectivity issues, power outages
- Projects tend to move slowly across the finish line and sometimes feeling isolated from the team.
- Operations and maintenance are difficult to perform remotely. Its hard to repair runway lights from your house.
- Access to files.
- Longer hours of work and more deadlines to meet.
- Communication + trust.
- Not being able to walk in and see someone about something urgent, not seeing the condition of the airport or what is changing in terms of people flow, construction changes, etc.
- More difficult to engage with others. Loss of informal work discussions. Harder to monitor the work of subordinates.

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- Some work physically can't be done remotely. More self-discipline is needed to stay on-task. Proper environment is needed to reduce distractions.
- Internet connectivity, hardware compatibility, Zoom fatigue.
- Losing the person to person exchange of information. Not being able to reach someone else that was working remotely by just walking over to them for information or a signature. Requesting information that was not accessible instantly when needed.
- Passenger interaction.
- Less interaction with the team, feeling out of touch with the operation at times.
- Work/life separation is non-existent; any work that requires any type of collaboration is significantly more difficult; team building almost impossible.

How were you able to maintain work-life balance while working remotely?

- More easily than normal. I went in to work 2-3 days a week and remained in my office or drove alone on/around the airfield.
- Taking breaks, working out, end work at 5pm.
- By making myself adhere to a predictable work schedule that somewhat mimicked my typical work day while in the office (such as taking lunches and "leaving" work at a certain time).
- I set firm start/stop times to ensure I wasn't working just because my laptop was sitting out
- I had a dedicated office that I would work in and shut the door when I was done for the day.
- I was able to dedicate a specific space away from family for remote work.
- Getting outside/getting activity in every day.
- Made it work.
- Set up separate room for work and leave it and your computer at the end of the work day. Felt comfortable taking extra time during the day for family situations because I could easily finish working later.
- Setting boundaries with family that while I'm working they are not to come into the work area/office without permission.
- Isolated myself from the rest of the house.
- Setting scheduled lunch and knock off times in attempt not to overwork.
- I was only fully remote for a couple of months. During that time, making frequent use of phone calls and virtual meetings helped.
- More time at home = better work life balance.
- I was more productive for work and had the time with my family that I normally don't get (school events, medical appointments, etc.).
- I didn't
- I kind of liked it. I was able to walk the dogs.
- Easily. When it was time to work, I opened my laptop and worked at the kitchen table. When it was dinner time, I closed my laptop and cleaned off the table. It was a routine I easily kept.
- Having a separate area set up for just doing work that is away from the normal living space in my home. Also, sticking to a normal work schedule and routine for getting ready for work.
- Sticking to a schedule.
- Hard to stay truly focused and engaged with work. Hard to separate work time from home time, doing work at odd hours of the day.
- I split my time and went in to the office 2–3 days a week for part of the day because I needed access to equipment at the office.
- I worked from beginning to ending of each shift at my desk and then only after my shift was over, did I do home/personal stuff.
- As a consultant, I was not as they continued piling on projects so I transitioned to the airport and it's been a hybrid situation. Airport work/life balance far and away exceeds the consultant work-life balance.
- Setting certain hours assigned to work and the other time for personal.

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- Same as in-office. I actually worked remotely very rarely, and my household is well set up for remote work. I ran my office from my house for some years before working for the Airport.
- I tried to plan my day so that I was working before my kids were awake and after they were finished with remote school for the day so I could help them while also completing my work.
- Handle work issues as they arise, while dealing with personal life matters as opportunities arise.
- A little difficult even with a segregated office area.
- I would only work minimal from home. I was told by HR that I had to.
- Strict times for when is work and when is non work.
- I wasn't able to balance work-life! we didn't have any good/safe child care options - we weren't willing to put toddler back in daycare so I quit my job to stay home with my daughter.
- Get out and exercise during lunch. Use lunch break to do something constructive around the house.
- Setting a strict schedule. Setting up my home office to mimic, as much as possible my actual office.
- I cut off the work day at the time I did when I was in the office. I worked in an empty bedroom. I treated my days like I was in the office meaning I go through my usual pre-work routine of exercising, showering, eating breakfast, etc.
- Yes and No. I live alone and in an condo. I had alot of time to work during the pandemic and working in finance we had our challenges keeping cash flow going while major projects continued. I worked more because I was up in the morning and would make coffee and start my day early. Many nights I worked late because I had everything at home. Yes, the airport got alot of hours from me during the pandemic.
- Stick to schedule as if I were in the office.
- In the office 4-days a week and home 1-day a week, not including weekends.
- Fairly well. Set up office in spare room and was able to walk away at end of work day on most days.
- Eliminated time wasted by driving or commuting, set work/life boundaries and monitor working hours closely.
- Setting boundaries and specific work day hours. I initially found myself checking and responding to emails from 6am to 7 or 8pm. I realized I wasn't able to disconnect like I would if I went into the office.
- You don't.
- Yes.
- By working remote, it was the best work-life balance in years.
- Better balance working remotely.
- It's still in progress.
- Better.
- Yes.
- Exercise, calendaring.
- For me, it is/was pretty easy because I have an office that is mostly separated from the rest of my house/family and I have plenty of work to do on the computer and via phone. We use personal phones for work and I'm used to getting work-related calls, so nothing really changed there.
- Plenty of exercise, quality time with family, stepping away from the computer, and sometimes things have to wait.
- I would keep my work hours to be the same and had an office to which I could close the door and remain focused. I would make calls only during working hours and would shut it down at the regularly scheduled time.
- Set-up a separate office away from home life.
- Designate the same location at home to work each day and the same hours. Walk away from that location when the work is done for the day and focus on family.
- I didn't.

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If you were in charge of the organization/department during the pandemic, what would you have done differently, if anything?

- Maybe tried to create more opportunities for employees to determine their own schedules so they could benefit from the work from home culture similarly to the non-essential personnel. This may be difficult, but I believe it would be considered more fair to our essential workers who gave so much of themselves to their jobs while many leaders worked remotely in relative safety and comfort.
- Allowed more remote work after the six weeks Admin worked from home.
- I would have continued to allow a hybrid schedule so that people could continue to have some limited work from home options if they desired.
- Required folks to return to the office.
- I would have increased interaction and made sure that all team members had the tools and equipment available at home for remote work, including ergonomic chairs, desks, etc.
- Although I was able to work remotely a day or two here or there it was not allowed long-term. I'd like to see more hybrid options for working from home.
- 100% remote not partial. It's less risky especially for people who are high risk. Every time we make our way to work praying we don't catch Covid even while vaccinated as that has proven to not exempt catching a severe bout of Covid.
- Slower return to work transition.
- Nothing. I think the hybrid model worked for us. People had flexibility, we were more productive (through additional accountability efforts), and we had scheduled happy hours (alcohol not required) during the work day to keep engagement.
- Provided more and consistent communication about health and wellness opportunities. Though I did not experience this need, many did, and I feel were unaware of others feeling the same or of services offered.
- I'm not sure if there was anything that could have been done better since it's been so unpredictable. We did have staff cuts on consultant lead projects that we shouldn't have done as quickly as we did.
- Nothing, everyone did the best they could under the circumstances.
- Maintain some level of fairness. On site people returned to work months before office workers and this caused resentment.
- Daily team Zoom meetings at a designated time.
- Have people work different shifts.
- Allowed staff more time to work remotely especially if they had children still attending school virtually. I would have created a hybrid work schedule that would stagger and alternate days that staff went to work at least 1-2 days a week.
- I feel our Airport leadership managed reasonably identifying personnel who could work remotely (a small percentage of overall personnel), yet encouraging those to be in the office with mitigation elements in place. I was in the office almost the entire time, except for short rare moments. I have not ever tested positive for COVID-19, though I've been tested reasonably frequently for various cause. We did some things right. :)
- Implemented more software solutions.
- Been more open to change.
- I think we handled remote work fairly well, but I would have liked individual zoom check-ins with my manager occasionally instead of always having the whole senior leadership team on the zoom.
- Encouraged MORE remote work.
- Not worked from home. I am in charge of the emergency services, but HR required all managers to work from home.
- Provided technology to better help working remotely.

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- While I think the flexibility of schedules was necessary for those with children, I would have asked employees come in at least 1-2 days a week to be with their departments.
- Communication - we would our normally once a month meeting via video. If I was in charge, I would have had at least a weekly team meeting if not more often and I would meet individually with each employee. I also would do water cooler video meeting weekly. Taking no more than 15 minutes to just chat as a team. No work-related talk, just talk. We had to keep signing these thrown together ridiculous remote contracts as we continued to stay home. HR did not know what they were doing.
- Hybrid schedules and mandatory team meetings weekly.
- Occasional group activity like maybe lunch - have lunch delivered to each employee in department and have lunch gathering on Zoom/Teams, etc.
- During the height of the pandemic, I believe all necessary steps and measures were taken appropriately. As people come back to work, travel, school, etc., it's time for people to get back in the office and move forward with work, life and family.
- I don't know if anything could be done differently. I would just continue to make sure those that could not be remote felt valued and compensated appropriately for their time.
- Better communication from upper management.
- Listen to employees individually rather than adhere to a blanket policy for everyone. Obviously more is known now than in the beginning but some decisions were very detrimental.
- Offered a session to give tips on working remotely.
- More individual return to work conversations and planning.
- Accepted the feedback received from surveys and had an informal dialogue to find a common ground to encourage and upbuild the morale first and then promote new and innovative opportunities for the organization in the future.
- I think the organization did the best it could.
- Nothing that I can think of. We responded as best we could given the ever-changing guidance and circumstances. It did lead to the organization creating a part-time remote-work policy.
- Pay for employees high-MPS home internet since they are saving on office space.
- I would have worked towards having a more effective way of making sure people were actually working. I would have also made sure everyone had the proper equipment to have online meetings periodically just to socialize with the people in the Department and assure the personal connection was maintained.
- Supervisors and managers worked 2 days at home and 3 days at the airport each week. The front-line team worked three 12-hour days and 4 hours work from home each week to limit exposure. I would not change that plan.

Morale took a hit during the pandemic, what "fun" thing did your airport do that you can share?

- We held more meetings and tried to energize the team a bit on the call. We spoke plainly and recognized the extenuating circumstances we were all in. Following the return to work, when the pandemic was in a lull, I held a "family day" for my departments (with safety measures) where I hopped on a dunk tank in 40 degree weather and got dunked many times over. It helped lighten everyone's moods and remind everyone why we work and who we work for.
- Department lunches, food trucks.
- We had a get together for coffee in the parking lot so we could meet but still be outside and 6′ apart.
- No-weekly catchup sessions which sometimes involved an engaging game for the department.
- Staff in other divisions set up weekly games over Zoom, like trivia or pictionary.
- Kept the lights on!
- We would have a weekly happy hour where we wouldn't talk about work but just have fun and chat. It was not scripted or structured. Just good people having fun.

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- We did get together for staff birthdays. Order take out.
- Nothing. That is something else I would do. Virtual happy hours, support groups, etc.
- I lead a team and made it a point to try to either have a virtual happy hour or in-person happy hour on a monthly basis. This has been good for our team.
- Not much. All fun events were canceled. There were attempts but with covid precautions it didn't have the intended results
- Offered lunch to staff more often.
- Send photos of successes to others.
- We continued our weekly team meetings virtually to discuss work projects, but also gave us some time to speak to each other about any non-work related topics.
- That's tough. We couldn't share food or snacks, we couldn't meet together face-to-face, except in specific scenarios and with social distancing and masks. It wasn't actually organized, but our section, mostly being on-site much of the time, encouraged one another. We had no increased turn-over during the pandemic.
- Holiday goodie bags I suppose.
- We had a company-wide employee recognition Zoom. Once we had all returned to the office we had a socially distanced, outdoor event with a taco truck and ice cream truck.
- Friday afternoon "happy hours."
- Not much. Many employees still had to work 100% at the airport.
- We had an all airport cookout. We invited tenants, contractors, vendor, and airline personnel.
- We did nothing during the pandemic. We have this Spring started to have some social events. We recently had a big picnic out in airfield in our Snow equipment building.
- Occasional virtual happy hour.
- Our group would have virtual "Happy Hour" where we would just talk and catch up on a monthly basis.
- Virtual scavenger hunt . . . goosechase.
- Customer appreciation day and Quarterly reward and recognition for top employees.
- Virtual Christmas party with a V-Jay and games and we shared videos of decorations for a contest.
- Nothing, but we've been very busy and short-handed.
- Foreign Object Debris (FOD) Walk.
- Employee picnic's/bbq's.
- When it became possible, we brought in food for in-person team meetings and gatherings. Managers and supervisors spent more time with the team in the field when at the airport.

Do you have any other thoughts to share on this topic?

- I believe the modern-day culture is anything but modern. From schedules, to work/life balance, to how people are compensated, including employer's flexibility in exchanging insurance and vacation benefits for wages at the individual level. We have a long way to go on modernizing our business staffing models!
- It seems to me that individual producers were able to be as productive or more productive while working remotely. Managers and senior leaders seem to add value by making decisions at meetings, so there seems to be a bias towards in office work the higher in the organization you go.
- I actually worked at a small hub during 2020–2022 and just recently moved to a large hub. But both airports didn't/don't really allow work from home, it was just available occasionally when I was sick or had a kid that needed to be at home. But I do believe long-term airports are going to have to allow some time to work from home to be competitive in the employee marketplace.
- Obviously some jobs require being physically present at the airport. Each department should be ruled separately based on the type of work and allow them to establish remote vs on-site vs hybrid.

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- Health over the need to see employees daily when it's unnecessary, should be key. That is what you would call caring for your employees and higher productivity/employee satisfaction.
- There is room for hybrid schedules (WFH once or twice a week) that I think are sustainable long term and confer a lot of quality of life benefits, without a person's productivity suffering.
- I am in full support of a hybrid model of working from home and office.
- Every department has different needs for working remotely. I'd personally like it if I could do 1 or 2 days remote.
- I do see an improvement in our ability to communicate via video conferencing and sharing of information electronically has increased substantially from the pandemic.
- I feel the ability to work a hybrid schedule would allow me to get more work done and personal tasks completed simultaneously. I still would need to go into the office at least once a week to review my staff's performance and productivity. Most meetings could be virtual, unless it required visual inspection that would be difficult to perform remotely.
- I feel a hybrid work experience for specific positions is possible. I feel 100% remote work for any position is ultimately detrimental to the Airport, and the community.
- I think remote work is sustainable for some job functions within our organization but not all.
- Airports should continue recognizing the many benefits to remote work and continue supporting their employees working remotely as much as possible, as it will INCREASE morale since it allows a better work/life balance.
- I believe that moving to a hybrid model was probably appropriate, but the standards are extremely low. For instance, airport employees in most departments (there are obvious exceptions) only have to come in for 2 days a week. I often see that isn't really adhered to consistently depending on the department, job, etc. I also have a hybrid agreement (1 day a week from home), but have found that it is much more productive and important for me to be on site. I work across several departments who are here no matter what and I think it's important to be here in person. I also learned very early on in the pandemic that working from home was not a sustainable solution for me. I now have a manager (only within the last 4 months) that is on site five days a week. I haven't had a manager on site for 5 years and I feel it is important for our very small, fledgling department that I am here consistently. I do believe it also sets the tone for future opportunities at the organization because I firmly believe that being engaged in-person is key to department success.
- We are still working remotely some. I currently come into the office Tuesdays, Thursdays, and Friday mornings. I think we continue unless something goes wrong. I would add we had a perfect audit in 2020. The first in my 7 years at the airport. So we as a team were focused and got the job done.
- I think there should be flexibility where employees can have the option to work remotely on certain days of the week. In addition, I think there should be flexible work hours with core hours (e.g. 9am–3pm).
- I think best long term solution would be a hybrid schedule that combines working remotely and in office time, maybe 2 days remote and 3 days in office.
- Remote work is a valuable option necessary to attract talent. Employers should embrace this option or create hybrid work options when necessary or dictated by the job. Airports should not deny remote work for a department because another dept. cannot work remote (i.e. maintenance, ops, public safety). Generally, remote work would also make additional terminal space available to rent to tenants.
- I think employees enjoy the flexibility and has improved employee moral for those that can take advantage of the "hybrid" work schedule. Those positions that cannot take advantage of remote work should receive an additional 3 - 5 percent bump in pay to help offset.
- Hybrid work.
- If you hire the right employees, trust your employees to do their work well, and base their work on results and not time in a chair, everyone wins.

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- Flexibility is key.
- It was and is still a difficult time. Due to lost staffing, budgets, and other resources, those that did have to come to work had to work harder with less and they are the ones that had their morale lowered knowing that some of their counterparts were working from home (or wherever). As part of the Maintenance Division, the Plumbers, Electricians, Custodians, etc. did not have the luxury of working from home, nor did their Supervisors and Managers, and had to come to work and expose themselves and their families to the virus.
- I only worked a couple of days remotely, once after receiving the booster and feeling bad and a couple of times when there was an exposure in the office.
- Now that people are returning to the workplace and having the audacity to complain, it is very disheartening. I believe that remote working has disrupted how the organization functions and it will take time to get people back into a smooth operation again.
- Most members of the team contracted COVID at some point during the pandemic. Our team's mission involves face-to-face contact with colleagues, airport partners, and the traveling public all day every day. Providing support for the team as a whole, as well as for each team member personally, was key during the pandemic and remains a key part of our success today.

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Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI–NA	Airports Council International–North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
GHSA	Governors Highway Safety Association
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S. DOT	United States Department of Transportation

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